

Lenovo

Moto G9 Plus



Lenovo Moto G9 Plus Level 2 – Service and Repair Manual

REVISION HISTORY

<i>Revision</i>	<i>Date</i>	<i>Notes</i>
1	08/13/2020	Initial Release.



TABLE OF CONTENTS

Revision History	2
List of Tables	7
Table of Figures	7
Safety Information	8
Cautions and Warnings	8
Battery Safety Guidelines	8
Introduction	9
External Views	9
Main Exploded Views	11
Disassembly and Recovery	14
Disassembly and Recovery Map	14
Tools Required for Disassembly	15
SIM Tray Removal	17
IMEI Label Removal	18
Water Detection Label Inspection	19
Battery Cover Removal	20
FPS Connector Bracket Removal	23
Top Bracket Removal	24
Water Detection Label Inspection	25
Speaker Box Heat Spreader Removal	26
Speaker Box Removal	27
Water Detection Label Inspection	28
Fingerprint Sensor Flex Removal	29
2M Macro Rear-Facing Camera Removal	30
Rear-Facing Camera Conductive Cloth Removal	31
64M Rear-Facing Camera Removal	32
Rear-Facing Camera Conductive Foam Removal	33
2M Depth Rear-Facing Camera Removal	34
Rear-Facing Camera Conductive Foam Removal	35
Rear-Facing Camera Support Pad Removal	36
Rear-Facing Camera Conductive Sponge Removal	37
Front-Facing Camera Copper Foil Removal	38
Front-Facing Camera Removal	39
LCD Connector Conductive Cloth Removal	40
Main PCB Removal	41
8M Rear-Facing Camera Removal	43
8M Rear-Facing Camera Grommet Removal	44

Lenovo Moto G9 Plus

Level 2 – Service and Repair Manual

Audio Jack Grommet Removal	45
PMU Copper Foil Removal	46
Receiver Removal	47
Sub Board Removal	48
USB Grommet Removal	50
Vibrator Removal	51
Vibrator Grounding Foam Removal	52
Vibrator Adhesive Removal	53
Coax Cable Removal	54
Battery Heat Spreader Removal	55
Battery Removal	56
Battery Foam Removal	58
Main FPC Removal	59
Rear-Facing Camera Lens Removal	60
NFC Antenna Removal	61
5G WIFI Antenna Removal	62
Battery Cover Heat Spreader Removal	63
Volume Key FPC Removal	64
Google Key FPC Removal	65
Sim Tray Hole Grommet Removal	66
Volume Key Removal	67
Google Key Removal	68
Parts Refresh	69
Tools Required for Parts Refresh	69
Battery Cover Cleaning	70
Front Housing Cleaning	71
Top Bracket Cleaning	72
Main PCB Cleaning	73
Front-Housing Thermal Gel Cleaning	74
Assembly	75
Assembly Map	75
Tools Required for Assembly	76
Water Detection Label Assembly	77
Sim Tray Hole Grommet Assembly	78
Volume Key FPC Assembly	79
Google Key FPC Assembly	80
Receiver Assembly	81
Vibrator Grounding Foam Assembly	82
Vibrator Adhesive Assembly	83

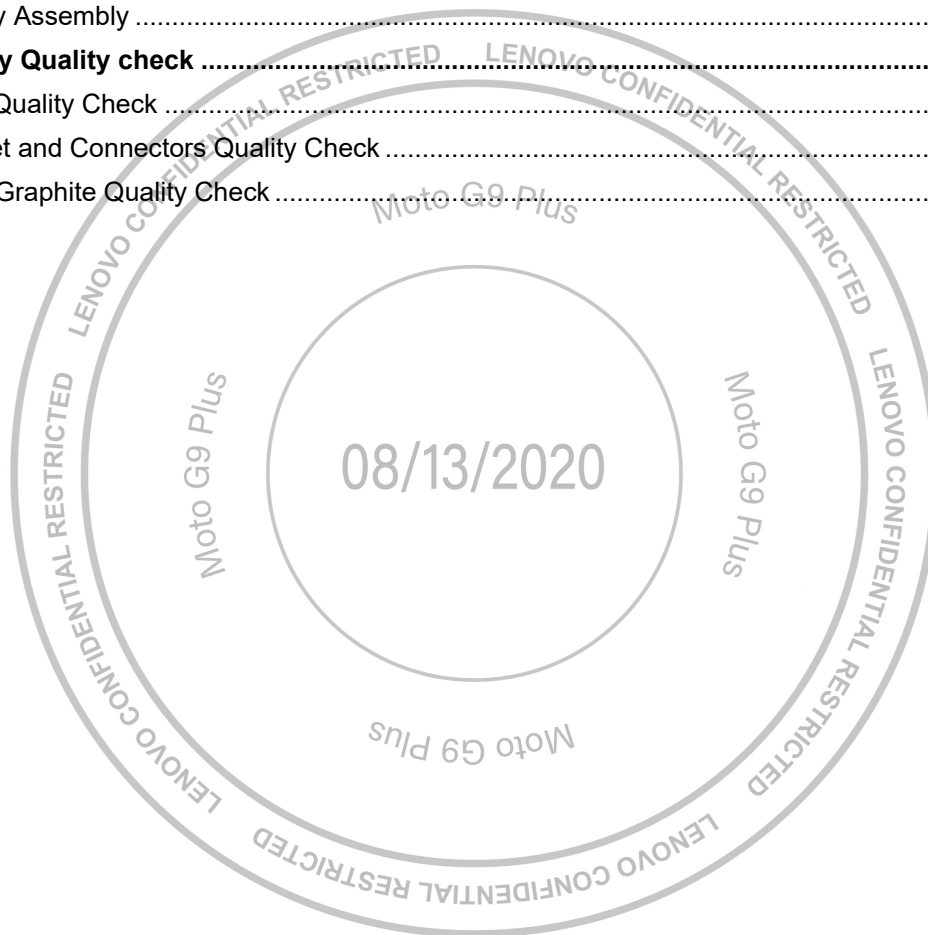
Lenovo Moto G9 Plus

Level 2 – Service and Repair Manual

Vibrator Assembly	84
Rear-Facing Camera Conductive Foam Assembly.....	85
8M Rear-Facing Camera Assembly.....	86
Audio Jack Assembly	87
Thermal Gel Assembly	88
Main PCB Assembly	94
Front-Facing Camera Assembly	95
Front-Facing Camera Copper Foil Assembly.....	96
USB Grommet Assembly	97
Water Detection Label Assembly.....	98
Sub Board Adhesive Assembly.....	99
Sub Board Assembly.....	100
Main FPC Assembly.....	101
Coax Cable Assembly.....	103
Rear-Facing Camera Conductive Cloth Assembly	105
2M Macro Rear-Facing Camera Assembly.....	106
64M Rear-Facing Camera Assembly.....	107
Rear-Facing Camera Conductive Foam Assembly.....	108
PMU Copper Foil Assembly	109
Rear-Facing Camera Conductive Sponge Assembly.....	110
Rear-Facing Camera Support Pad Assembly.....	111
2M Depth Rear-Facing Camera Assembly	112
Rear-Facing Camera Conductive Foam Assembly.....	113
LCD Connector Conductive Cloth Assembly	114
Speaker Box Assembly	115
Water Detection Label Assembly.....	116
Battery Foam Assembly.....	117
Battery Adhesive 1 Assembly	118
Battery Adhesive 2 Assembly	119
Battery Adhesive 3 Assembly	120
Battery Assembly	121
Battery Heat Spreader Assembly.....	123
Battery Cover Heat Spreader Assembly.....	124
Rear-Facing Camera Lens Adhesive Assembly	125
Rear-Facing Camera Lens Assembly	126
5G WIFI Antenna Assembly.....	127
NFC Antenna Assembly.....	128
8M Rear-Facing Camera Grommet Assembly.....	129
Speaker Box Heat Spreader Assembly	130

Lenovo Moto G9 Plus
Level 2 – Service and Repair Manual

Top Bracket Assembly	131
Volume key Adhesive Assembly	132
Volume Key Assembly	133
Google Key Adhesive Assembly	134
Google Key Assembly	135
FPS Flex Assembly	136
FPS Connector Bracket Assembly.....	137
Battery Inlay Assembly.....	138
IMEI Label Assembly	139
Sim Tray Assembly	140
Assembly Quality check	141
Battery Quality Check	141
Grommet and Connectors Quality Check	142
Screws Graphite Quality Check	143



LIST OF TABLES

Table 1. External Views Parts List..... 10

Table 2. Assembly Exploded View Parts List 11

TABLE OF FIGURES

Figure 1. External Views..... 9

Figure 2. Assembly Exploded View 11



SAFETY INFORMATION

Cautions and Warnings

Electrostatic Discharge

The phone components may be damaged by electrostatic discharge (ESD). Always use an ESD mat and ground strap when working with internal components.

Battery Safety Guidelines

Handle Battery with care. Ensure Battery edges and surfaces are not dented or deformed. If the Battery Pack is dropped to the floor, it may be internally damaged and must be scrapped.

Ensure all surfaces, fixtures, and phone components contacting the Battery are smooth and clean.

Ensure Battery and its insulation are not damaged (e.g. scratched, dented, punctured) prior to and throughout assembly.

Prior to assembly, ensure Battery edges and surfaces are not dented or deformed, and that fixtures and parts that will contact the Battery are free of foreign material.

Ensure screws and screwdrivers do not contact the Battery.

Failure to adhere to Safety Critical Note(s) may increase risk of rupture, burning, or failure to function safely when used by the customer.

INTRODUCTION

External Views



Figure 1. External Views

Table 1. External Views Parts List

Reference #	Description
1	Speaker Mesh
2	Front-Facing Camera
3	Volume Key
4	Power Key/Side FPS
5	Camera Deco
6	Flash Lens
7	Camera Lens
8	Batwing Logo
9	Rear-Housing
10	Audio Holes
11	USB Port
12	MIC
13	Main Lens
14	Front-Ring
15	Google Key
16	Sim-Tray
17	Earphone Holes

Main Exploded Views

Figure 2. Assembly Exploded View

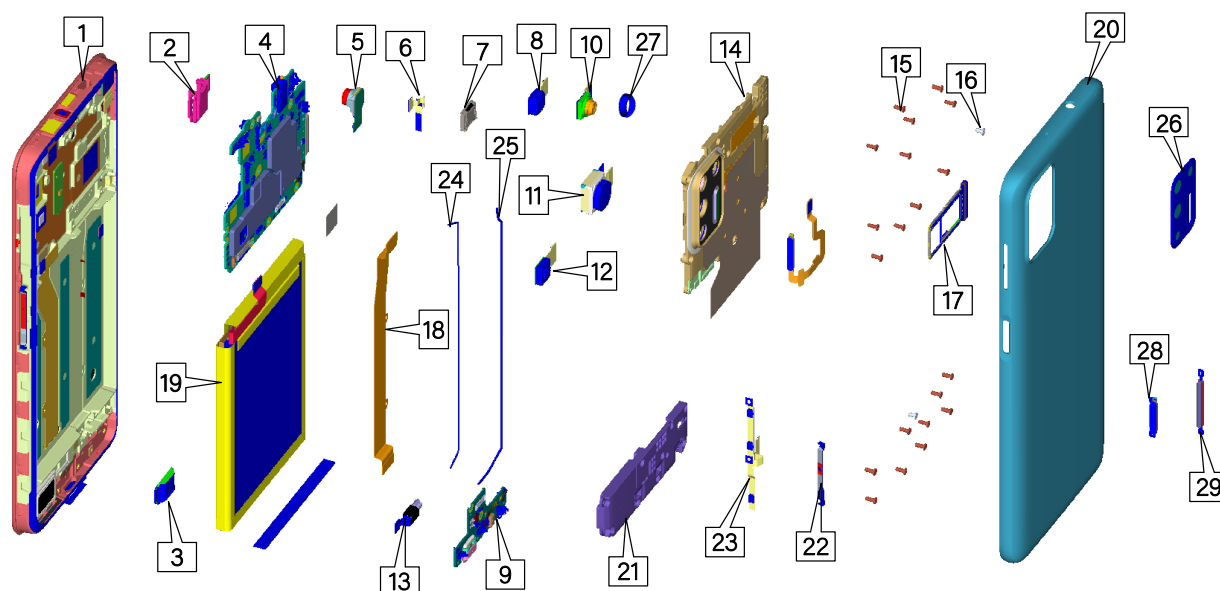


Table 2. Assembly Exploded View Parts List

Reference #	Description	Recovery Part
1	Odessa FRU TP LCM Assy CS	Yes
2	LC5061AA000486+LXF_M600_A01_Headsetseali	Yes
3	LC5061AA000487+LXF_M600_A01_USBrubbersle	Yes
4	XT2087-1 MB 4+128G DS CS XT2087-1 MB 4+128G SS CS XT2087-2 MB 4+128G DS CS XT2087-2 MB 6+128G DS CS ASSY,BD KIT,Odessa20 BR 4+128 MB PCBA	Yes
5	MDL,CAMR,16M,FRNT,;C0EF04;FF:Q Tech	Yes
6	LC6680AA000510+M600_GoogleButtonFPC	No
7	LC2626AA000035+Receiver_Normal_9mmx8mmxH	No
8	MDL,CAMR,2MAF,Guam/Borneo/Odessa,L02BA00	Yes
9	Odessa Sub Baord CS ASSY,DTR BRD,Odessa20 BR 4+128 SUB PCBA	Yes
10	MDL,CAMR,8M,rear,Nairobi;C98F11;Qtech;UW	Yes
11	MDL,CAMR,64M,AF,Liberty;AF64M;LGW1A20;Of	Yes
12	MDL,CAMR,2M,Rear,Nairobi;tacoma;FOV;Ofil	Yes
13	LC2627AA000042+MOTOR_CylinderMotor_20.37	No
14	Odessa Rear Housing DD CS Odessa Rear Housing CR CS	Yes
15	LC4610AA000282+cross screw 1.4x3.5M600	No

Lenovo Moto G9 Plus
Level 2 – Service and Repair Manual

Reference #	Description	Recovery Part
16	LC4610AA000281+Phillips screw 1.4x2M600	No
17	LC5017AA000443+LXFM600_DD_SIM Tray_DS DP LC5017AA000442+LXFM600_CR_SIM Tray_DS DP LC5017AA000474+LXFM600_DD_SIM Tray_SS DP LC5017AA000473+LXFM600_CR_SIM Tray SS DP	Yes
18	LC6680AA000511+M600_Main Sub board FPC	Yes
19	ACCY,BAT,5000mAh,MG50;ATL526582;ATL;WW	No
20	LC5011AA000893+LXF_M600_A01,Battery cove LC5011AA000778+LXF_M600_A01_Batterycover HSG,BAT CVR,Odessa BR BATCVR KIT DD LM HSG,BAT CVR,Odessa BR BATCVR KIT CR LM Odessa Etching Battery Cover LATAM DD CS Odessa Etching Battery Cover LATAM CR CS Odessa Etching Battery Cover EMEA DD CS Odessa Etching Battery Cover EMEA CR CS Odessa Etching Battery Cover APAC DD CS Odessa Etching Battery Cover APAC CR CS	Yes
21	LC5016AA000163+LXF_M600_A01_SpeakerBOXco	Yes
22	ASSY,BND,Odessa20 FPS KIT DD ASSY,BND,Odessa20 FPS KIT CR	Yes
23	LC6680AA000507+M600_Volumekey&powerkeyFP	No
24	LC5234AA000053+M600_RFmid-lowfrequencyc	Yes
25	LC5234AA000052+M600_RFcoaxialcable_coax	Yes
26	LC5030AA000168+LXF_M600_A01_Lesu_darkblu LC5030AA000166+LXF_M600_A01_Lesu_Rosegol	Yes
27	LC5061AA000488+LXF_M600_A01_Wideanglereaa	Yes
28	LC5040AA000298+LXF_M600_A01_GoogleKey_Di LC5040AA000301+LXF_M600_A01_Googlekey_tr	Yes
29	LC5040AA000297+LXF_M600_A01_Volumekey_Di LC5040AA000302+LXF_M600_A01_VolumeKey_Di	Yes
-	LC5022AA000165+LXF_M600_A01_NFCantenna	No
-	LC5022AA000168+LXF_M600_A01_5G_WIFlanten	No
-	Odessa Battery Adhesive1&5420AA000820	No
-	Odessa Battery Adhesive2&5420AA000825	No
-	Odessa Battery Adhesive3&5420AA000824	No
-	Odessa Main FPC Adhesive3&5420AA000839	No
-	Odessa Motor Adhesive3&5420AA000840	No
-	Odessa Sub BD Adhesive3&5420AA000822	No
-	Odessa Cameralens Adhesive&5420AA000821	No
-	Odessa SideKey Adhesive&5420AA000819	No

Lenovo Moto G9 Plus
Level 2 – Service and Repair Manual

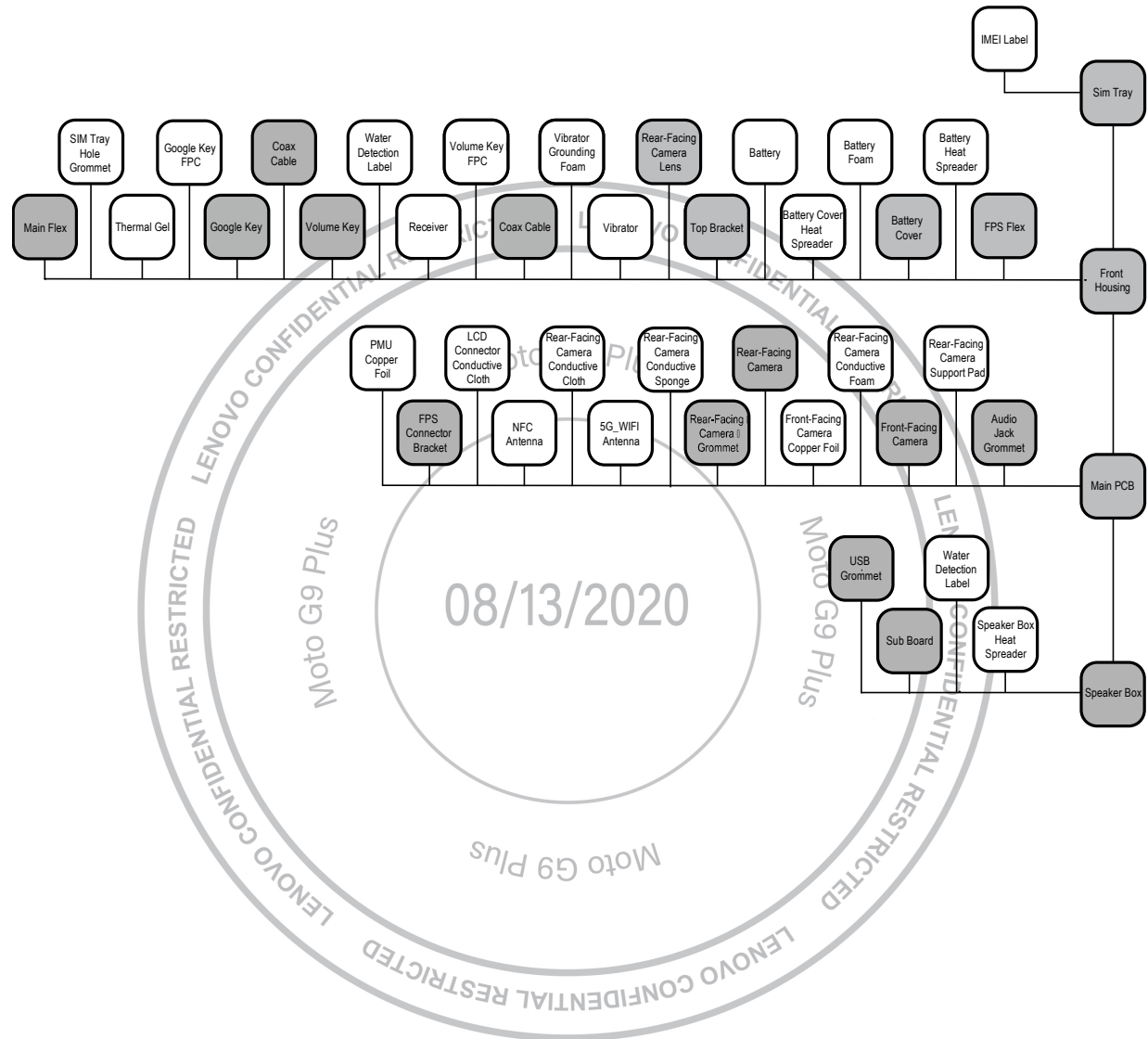
<i>Reference #</i>	<i>Description</i>	<i>Recovery Part</i>
-	LC5061AA000490+LXF_M600_A01_Cardpinrubbe	Yes
-	LC5450AA000615+LXF_M600_A01_Depthoffield	No
-	LC5430AA000837+LXF_M600_A01_Batter foam	No
-	LC5450AA000613+LXF_M600_A01_Motorgroundi	No
-	LC5470AA000155+LXF_M600_A01_Batterycooli	No
-	LC5470AA000154+LXF_M600_A01_SpeakerBOXco	No
-	LC5450AA000614+LXF_M600_A01_LCDconnector	No
-	LC5470AA000151+LXF_M600_A01_Proactivegro	No
-	LC5450AA000616+LXF_M600_A01_macroreargro	No
-	LC7015AA001336_Waterproof labeling_DGJL	No
-	LC5430AA000840+LXF_M600_A01_Depth-of-fie	No
-	LC5470AA000160+LXF_M600_A01_Rearshellhea	No
-	LC5430AA000835+LXF_M600_A01_rearconnecto	No
-	LC7015AA000352+SIM Card Tray Label	No
-	LC5470AA000163+LXF_M600_A01,PMUshielding	No
-	LC5090AA000261+LXF_M600_A01_Fingerprintc	Yes

08/13/2020

DISASSEMBLY AND RECOVERY

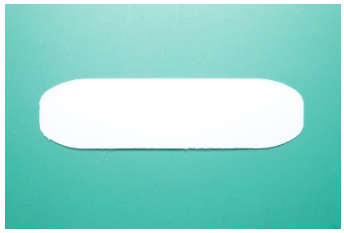
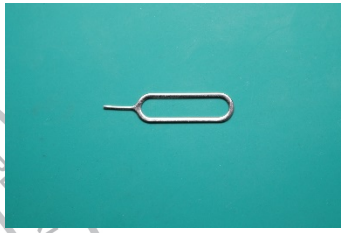
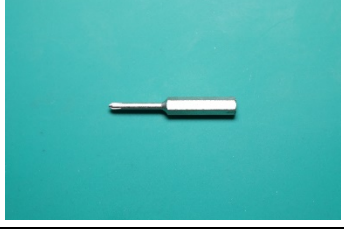
Disassembly and Recovery Map

*Shaded parts may be eligible for parts recovery.



Tools Required for Disassembly

The following tools are required to disassemble the Moto G9 Plus phone.

<i>Description</i>	<i>Part #</i>	<i>Picture/Drawing</i>
Thin Plastic Spudger	0-00-00-40826	
SIM/SD Tray Removal Tool	--	
Black Stick	--	
Torque Driver	--	
Shim	4-00-5L-10000	
PH00 Bit	--	
Plastic Tweezers	--	--
ESD Mat	--	--

Lenovo Moto G9 Plus
Level 2 – Service and Repair Manual

<i>Description</i>	<i>Part #</i>	<i>Picture/Drawing</i>
Wrist Strap	--	--
Gloves	--	--



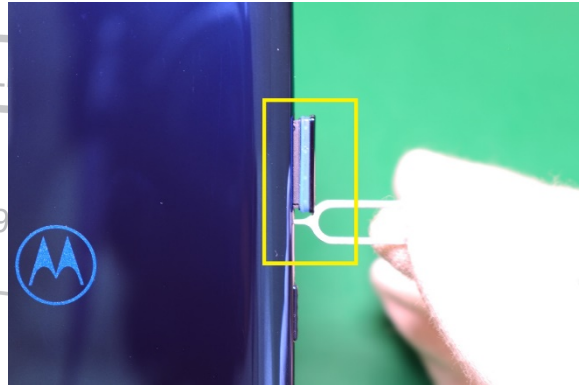
SIM Tray Removal

NOTE

The SIM Tray can be recovered and reused after removal.



1. Power off the device before beginning with the procedure. Use the SIM Tray Removal pin to push into the SIM Tray hole.
2. Press the SIM Tray removal tool to make it pop-out as shown.



3. Use fingers to take out the SIM Tray.



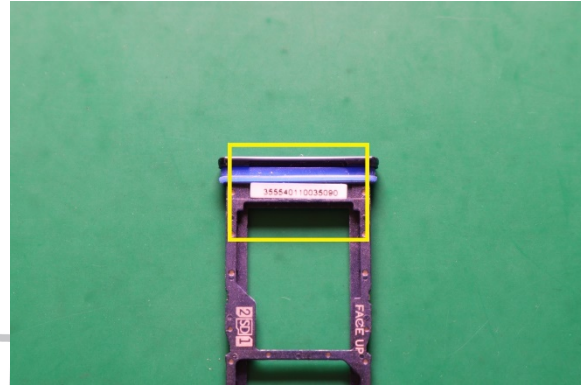
IMEI Label Removal

NOTE

Remove the IMEI Label only if necessary. After you remove it, you cannot reuse it.



Use the Plastic Tweezer to remove the IMEI Label from the SIM tray.

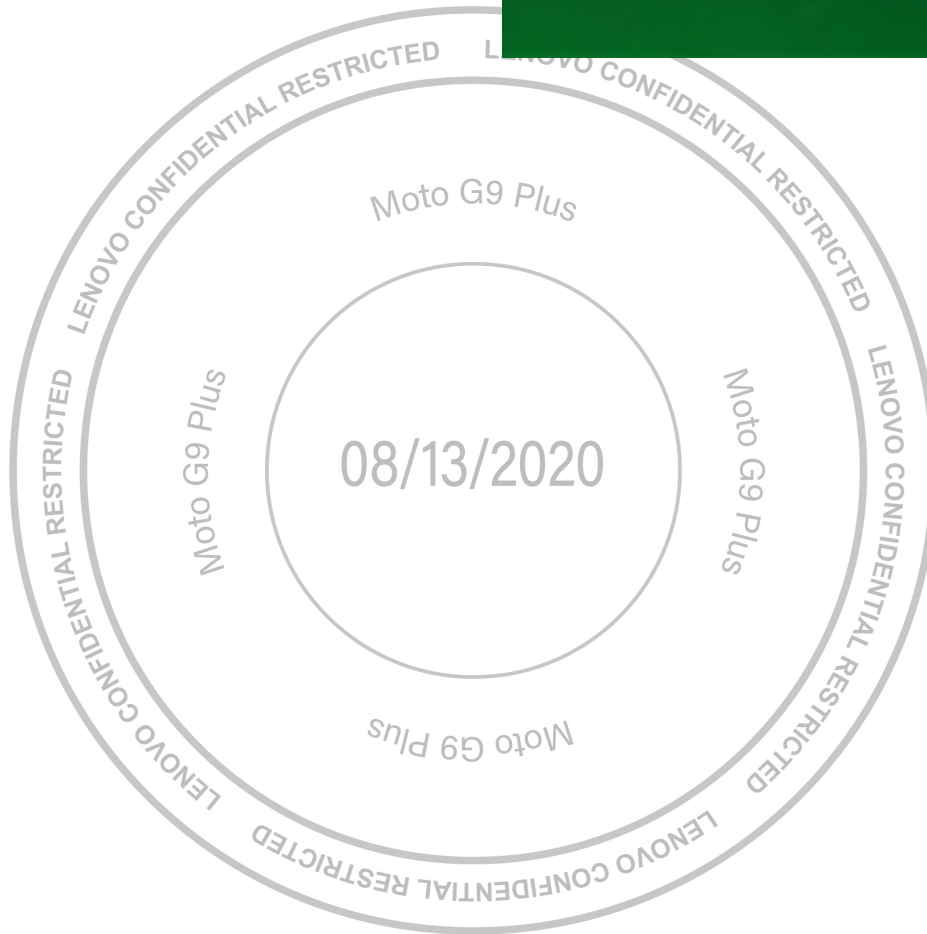
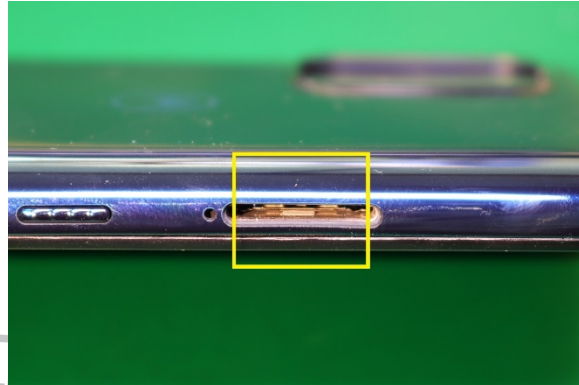


Water Detection Label Inspection

NOTE

If the Water Detection Label (WDL) has been activated, the phone's warranty no longer applies.

Inspect the WDL inside the SIM Tray Slot for signs of activation. Use external light if needed.



Battery Cover Removal

NOTE

The Battery Cover can be recovered and reused after removal if it is undamaged.



1. Use the Shim to create a gap between the Battery Cover and the Front-Housing.

NOTE

Make sure to start at the Side Key position as shown.

2. Use the Shim to create a gap between the Battery Cover and the Front-Housing.

3. Use the Shim to create a gap between the Battery Cover and the Front-Housing.



4. Use the Shim to create a gap between the Battery Cover and the Front-Housing.



5. The highlighted frame is a forbidden area. Do not disassemble at this location.

NOTE

Care not to slide the Shim to avoid damaging the Front-Housing.

NOTE

Care while removing the Battery Cover with Shim do not scratch the Battery Cover and Front rim.

NOTE

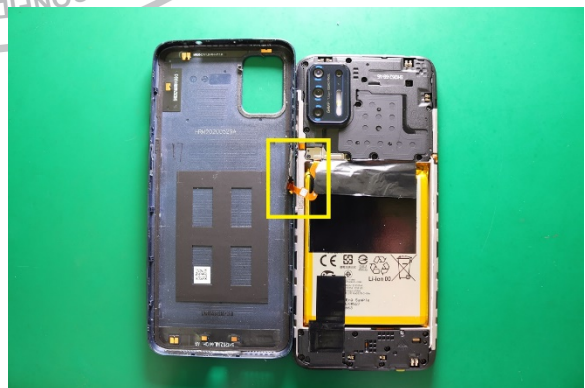
Care not to break the MIC1 Grommet on the bottom side and MIC2 Grommet on the top side.



6. Split the Battery Cover from the Front-Housing as shown.

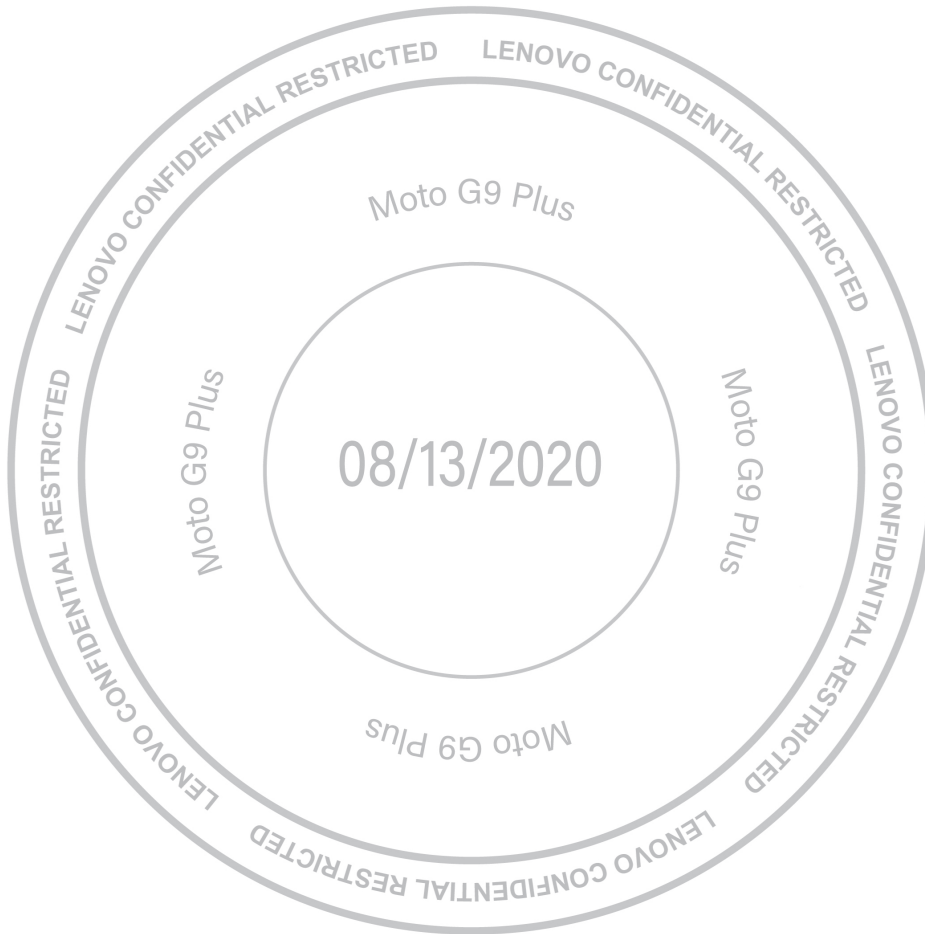
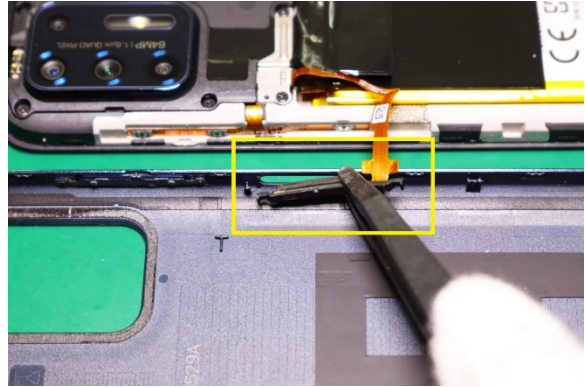
/// WARNING ///

Use care not to disconnect the Fingerprint Sensor (FPS) Flex from the Battery Cover when removing the Front Housing.



Lenovo Moto G9 Plus
Level 2 – Service and Repair Manual

7. Use Plastic Tweezer to remove FPS Key from the location as shown.



FPS Connector Bracket Removal

NOTE

The FPS Connector Bracket can be recovered and reused after removal.

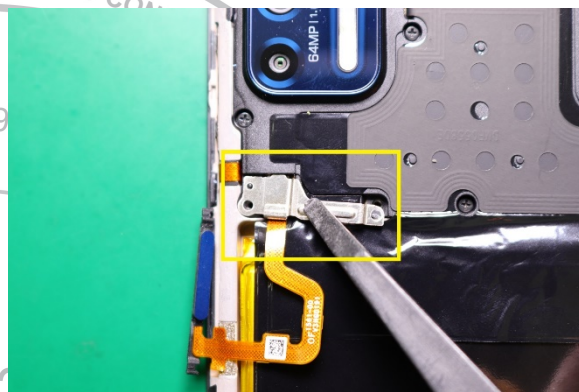
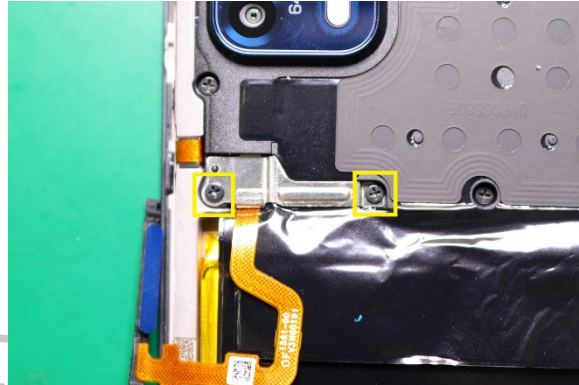


1. Use the Torque Driver with PH00 Bit to remove the two screws as shown.

/// WARNING ///

The screws cannot be reused after removal.

2. Use Plastic Tweezer to remove the FPS Connector Bracket from the location as shown.



Top Bracket Removal

NOTE

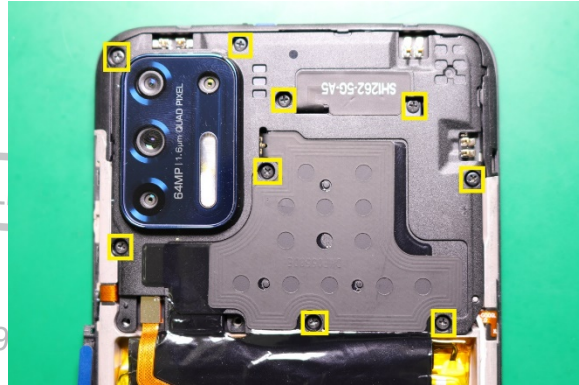
The Top Bracket can be recovered and reused after removal.



1. Use the Torque Driver with PH00 Bit to remove the nine screws as shown.

/// WARNING ///

The screws cannot be reused after removal.



2. Use the flat end of the Black Stick to lift the Top Bracket from the location as shown.

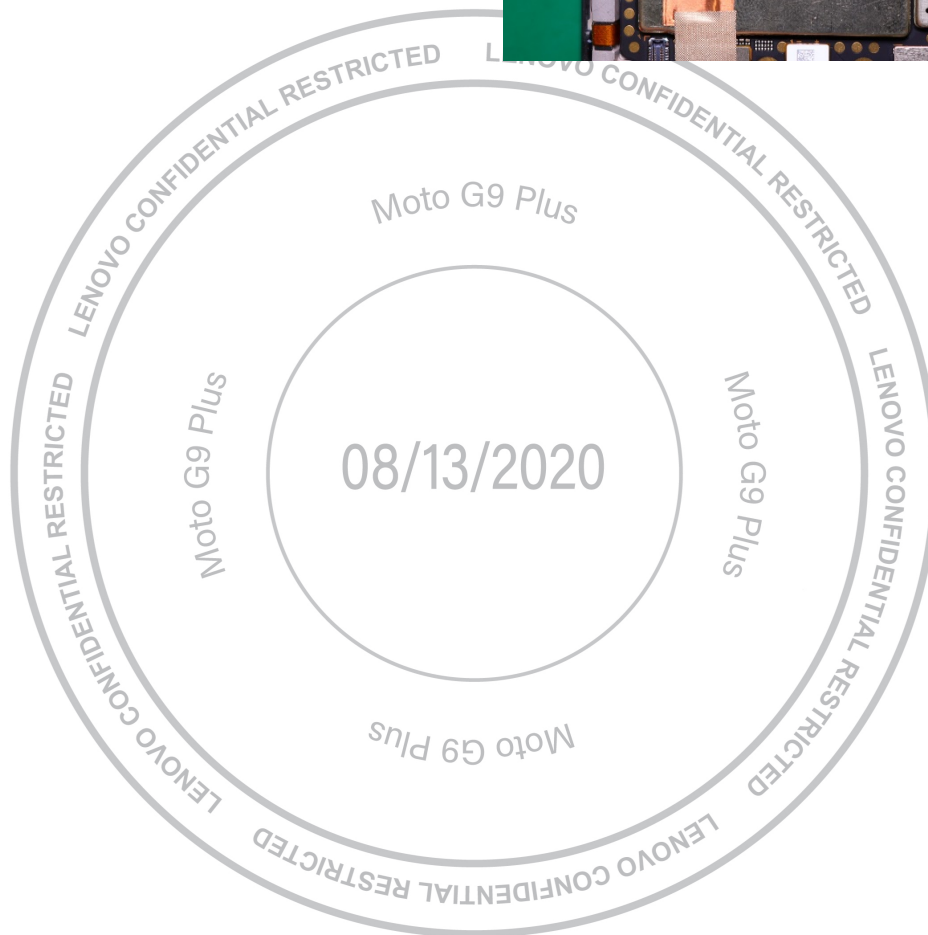
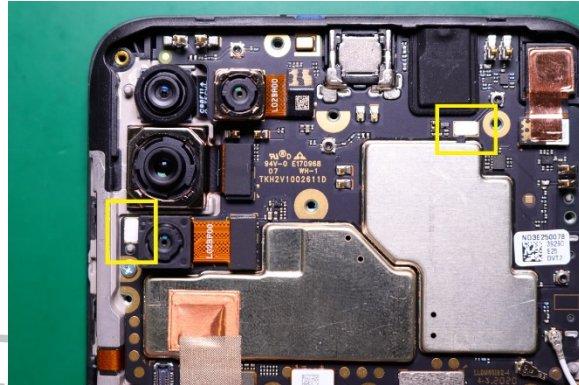


Water Detection Label Inspection

NOTE

If the Water Detection Label (WDL) has been activated, the phone's warranty no longer applies.

Inspect the WDL, which is visible on the Main PCB. Examine it for signs of activation.



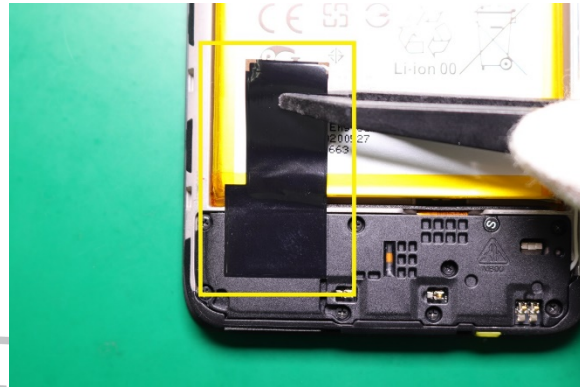
Speaker Box Heat Spreader Removal

NOTE

After you remove the Speaker Box Heat Spreader, you cannot reuse it.



Use the Plastic Tweezer to remove the Speaker Box Heat Spreader from the location as shown.



Speaker Box Removal

NOTE

The Speaker Box can be recovered and reused after removal.

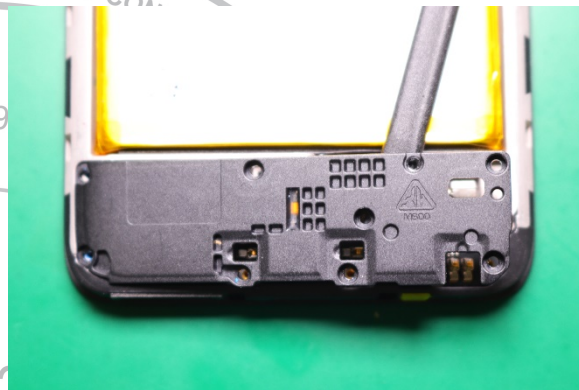
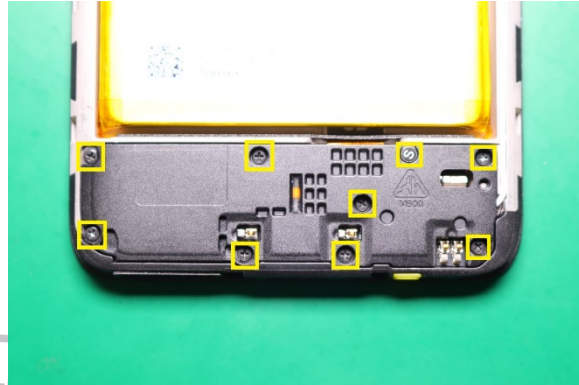


1. Use the Torque Driver with PH00 Bit to remove the nine screws as shown.

/// WARNING ///

The screws cannot be reused after removal.

2. Use the flat end of the Black Stick to lift the Speaker Box as shown.

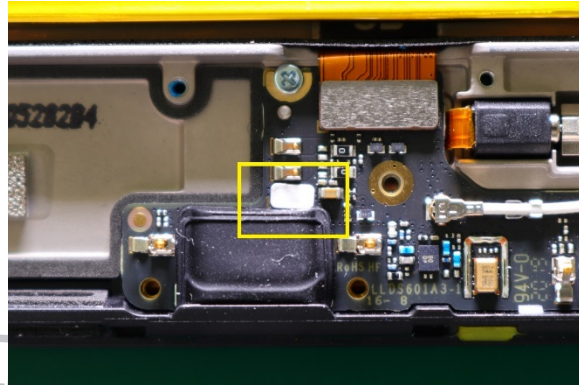


Water Detection Label Inspection

NOTE

If the Water Detection Label (WDL) has been activated, the phone's warranty no longer applies.

Inspect the WDL, which is visible on the Sub Board. Examine it for signs of activation.



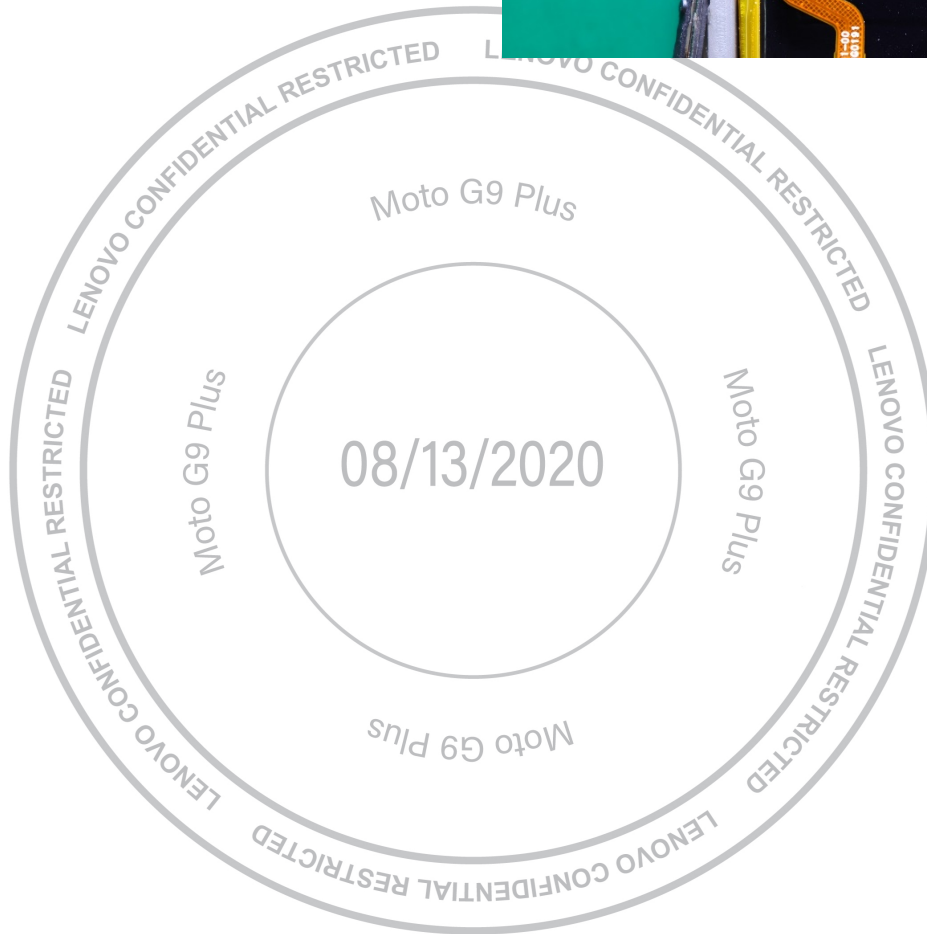
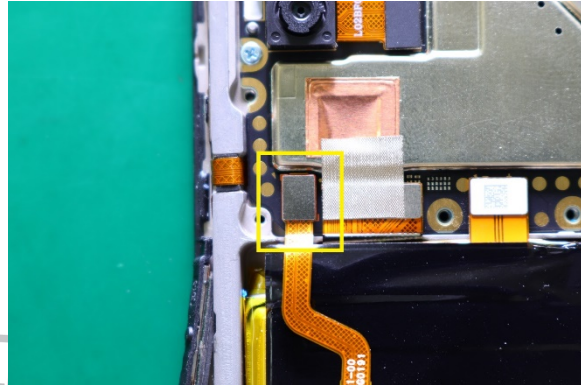
Fingerprint Sensor Flex Removal

NOTE

The Fingerprint Sensor Flex can be recovered and reused after removal.



Use fingers to remove the Fingerprint Sensor Flex from the location as shown.



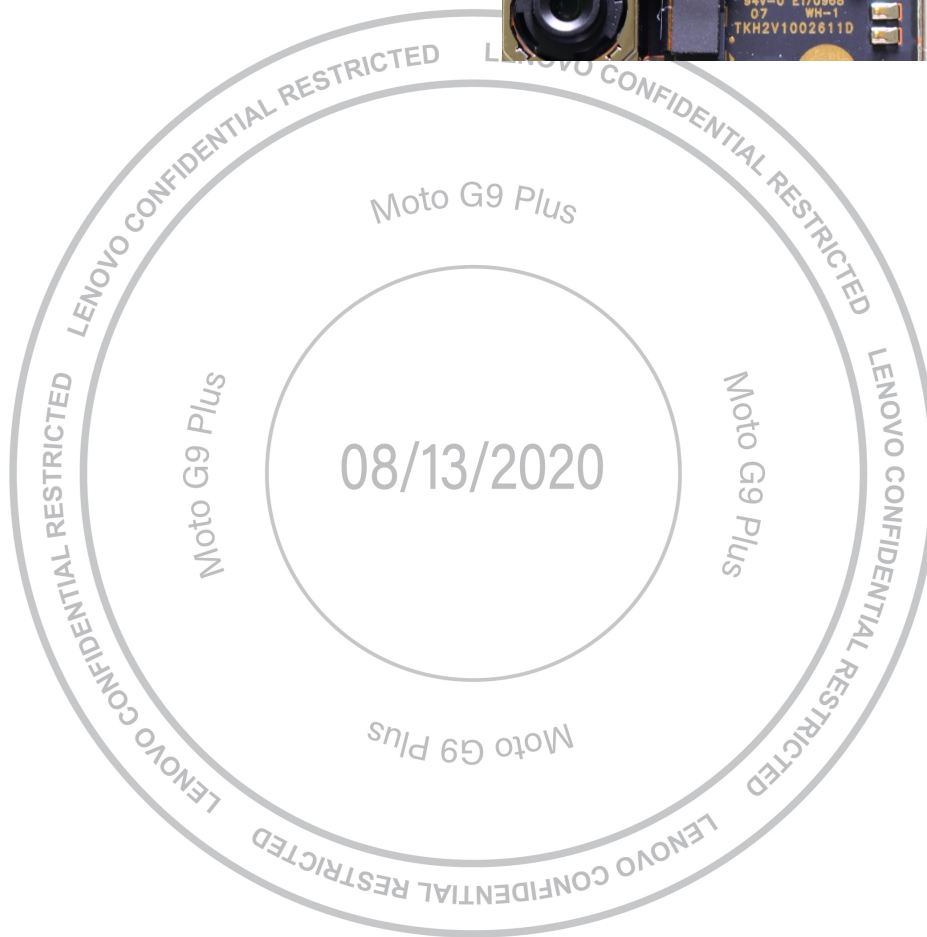
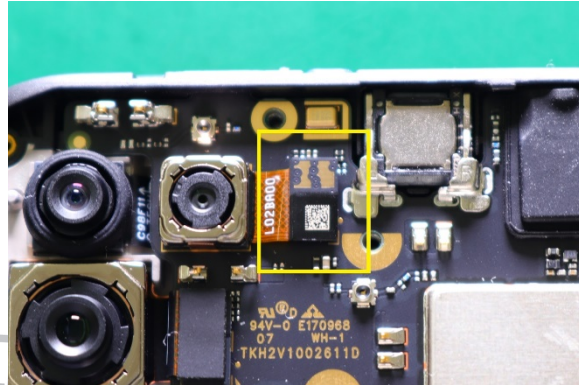
2M Macro Rear-Facing Camera Removal

NOTE

The 2M Macro Rear-Facing Camera can be recovered and reused after removal.



Use fingers to remove the 2M Macro Rear-Facing Camera from the location as shown.



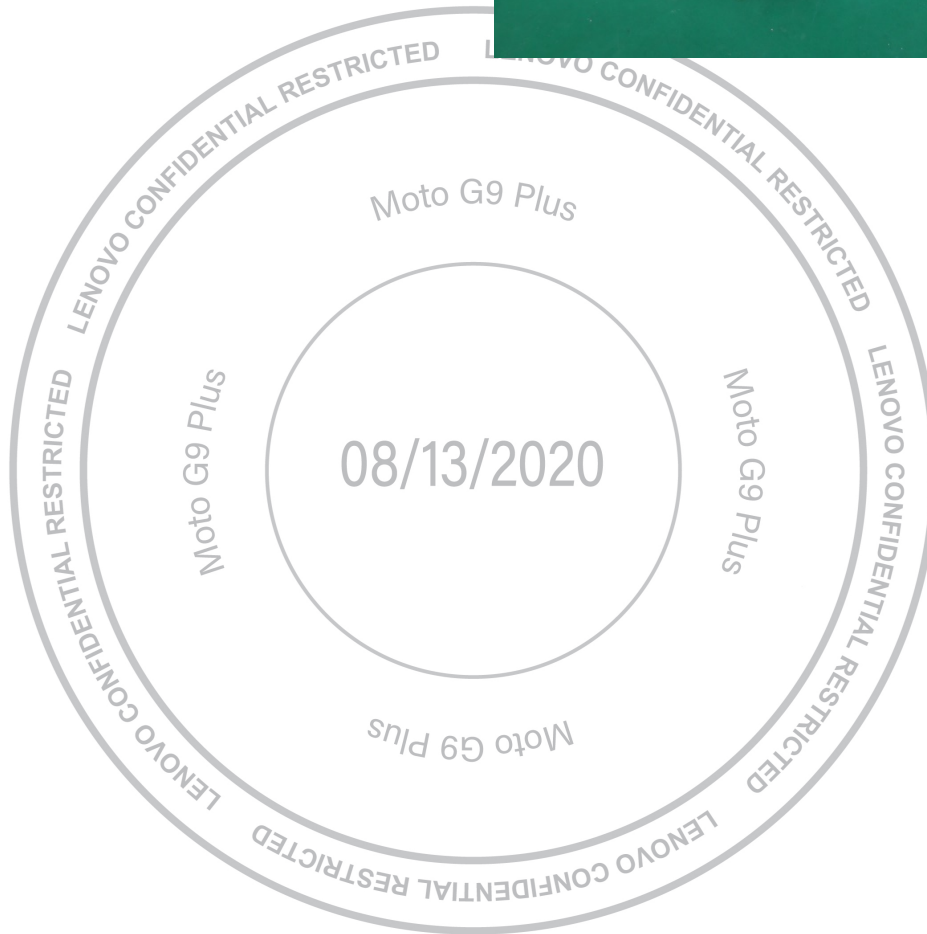
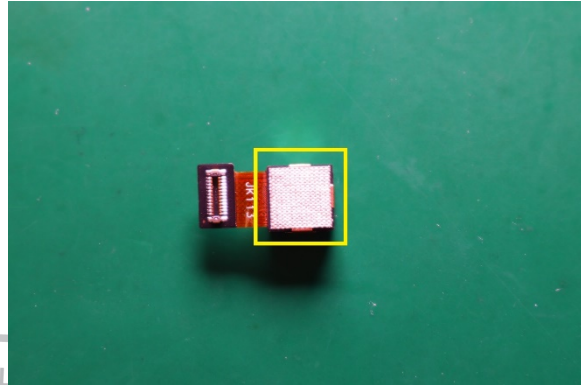
Rear-Facing Camera Conductive Cloth Removal

NOTE

After you remove the Rear-Facing Camera Conductive Cloth, you cannot reuse it.



Use fingers to remove the Rear-Facing Camera Conductive Cloth from the location as shown.



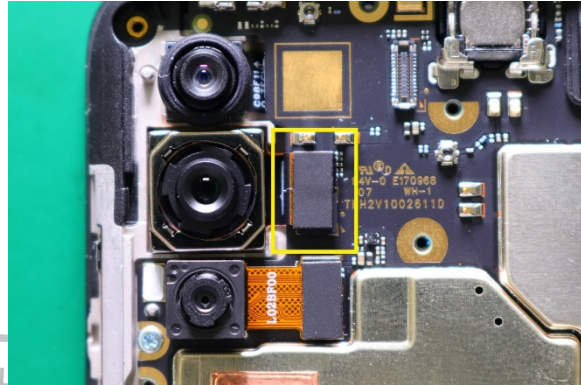
64M Rear-Facing Camera Removal

NOTE

The 64M Rear-Facing Camera can be recovered and reused after removal.



Use fingers to remove the 64M Rear-Facing Camera from the location as shown.



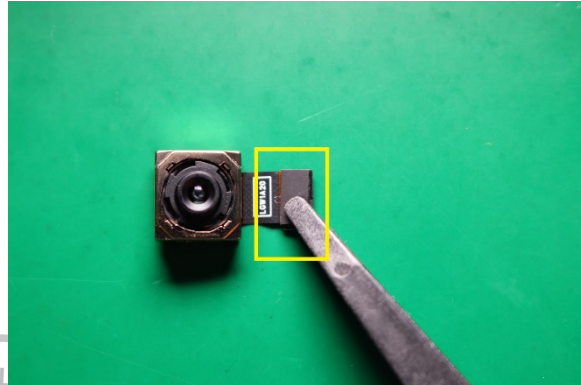
Rear-Facing Camera Conductive Foam Removal

NOTE

After you remove the Rear-Facing Camera Conductive Foam, you cannot reuse it.



Use the Plastic Tweezer to remove the Rear-Facing Camera Conductive Foam from the location as shown.



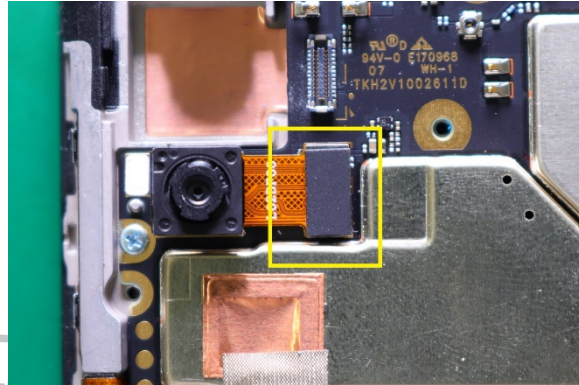
2M Depth Rear-Facing Camera Removal

NOTE

The 2M Depth Rear-Facing Camera can be recovered and reused after removal.



Use fingers to remove the 2M Depth Rear-Facing Camera from the location as shown.



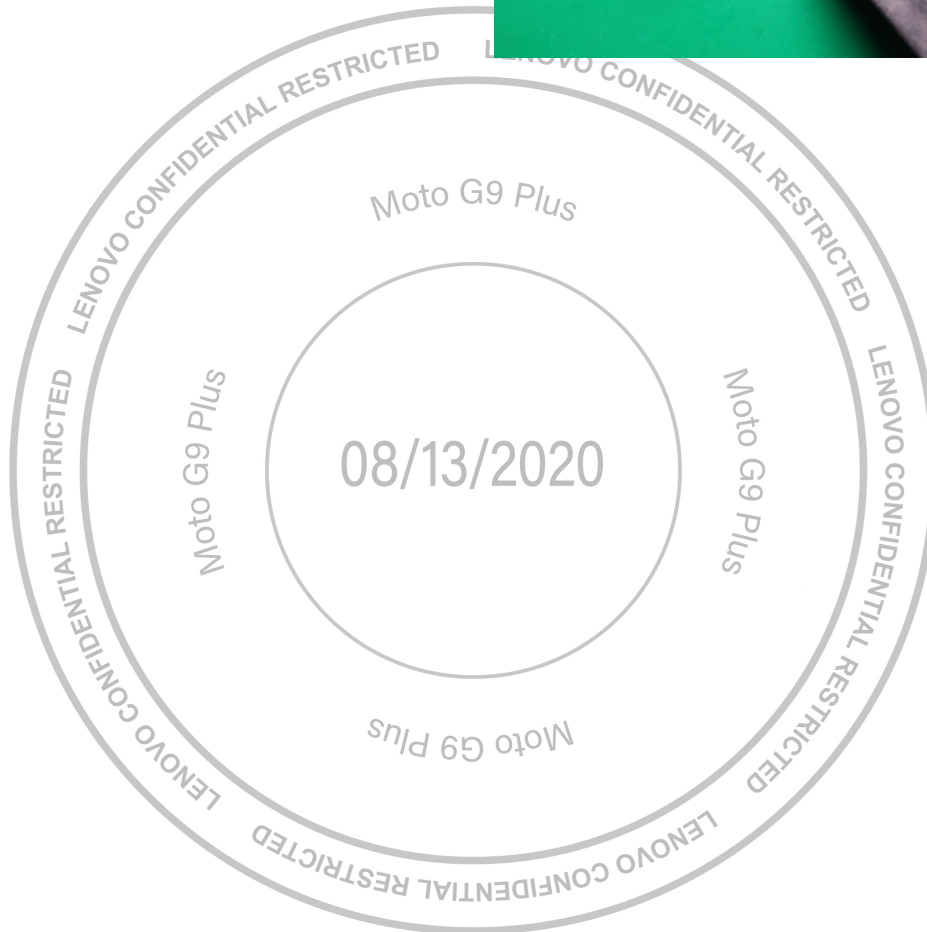
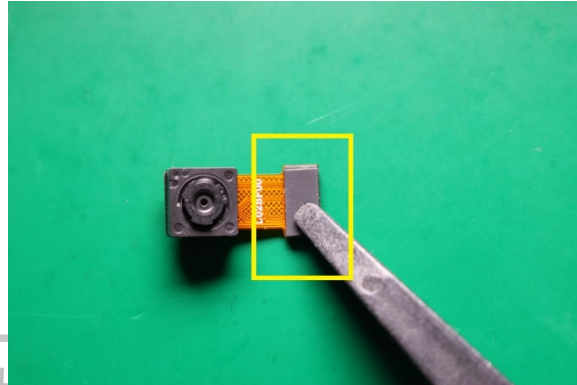
Rear-Facing Camera Conductive Foam Removal

NOTE

After you remove the Rear-Facing Camera Conductive Foam, you cannot reuse it.



Use the Plastic Tweezer to remove the Rear-Facing Camera Conductive Foam from the location as shown.



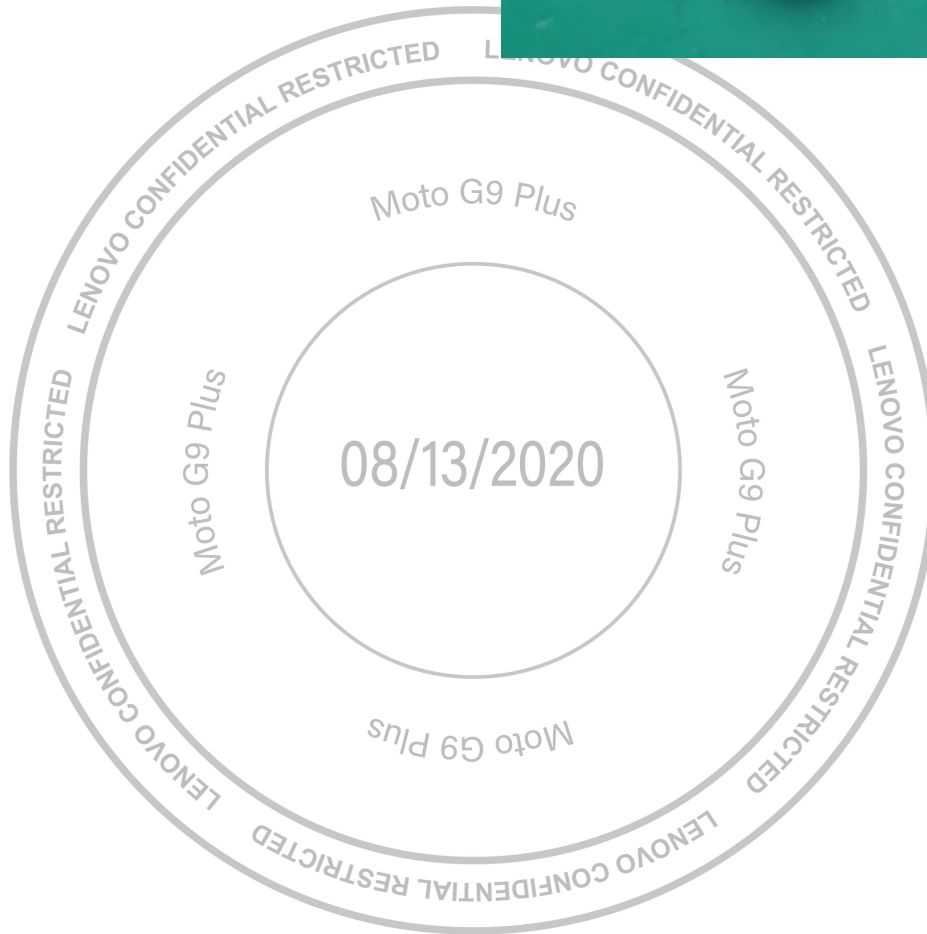
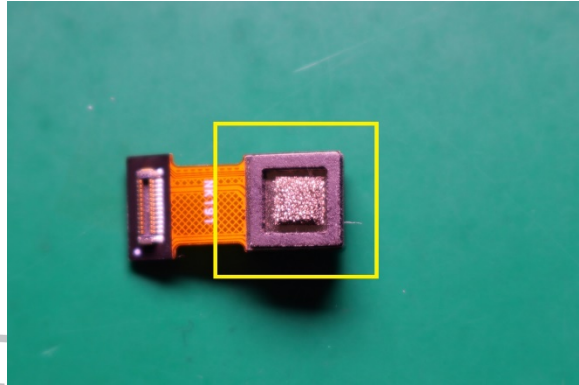
Rear-Facing Camera Support Pad Removal

NOTE

After you remove the Rear-Facing Camera Support Pad, you cannot reuse it.



Use the Plastic Tweezer to remove the Rear-Facing Camera Support Pad from the location as shown.



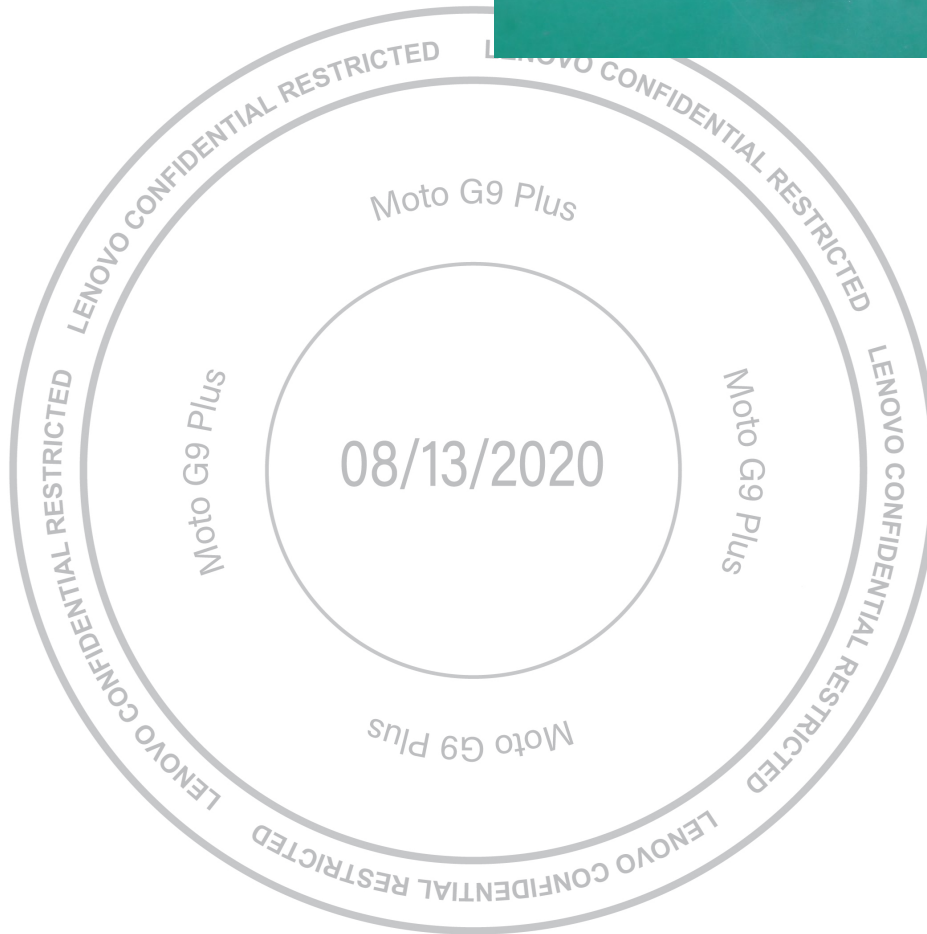
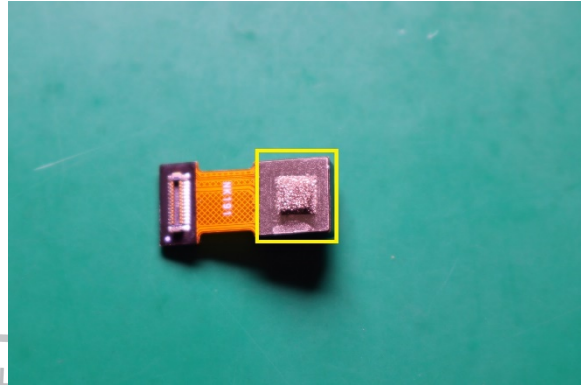
Rear-Facing Camera Conductive Sponge Removal

NOTE

After you remove the Rear-Facing Camera Conductive Sponge, you cannot reuse it.



Use the Plastic Tweezer to remove the Rear-Facing Camera Conductive Sponge from the location as shown.



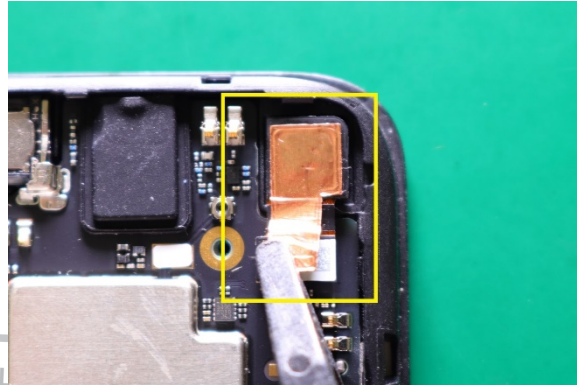
Front-Facing Camera Copper Foil Removal

NOTE

After you remove the Front-Facing Camera Copper Foil, you cannot reuse it.



Use the Plastic Tweezer to remove the Front-Facing Camera Copper Foil from the location as shown.



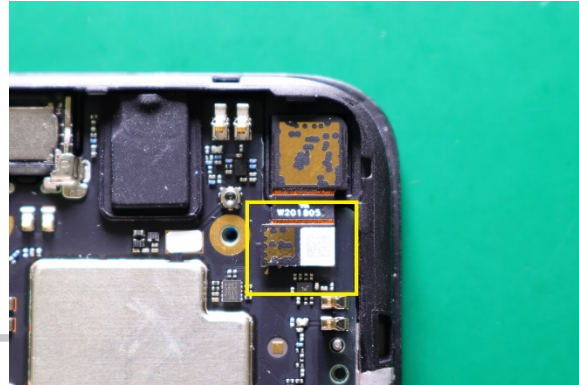
Front-Facing Camera Removal

NOTE

The Front-Facing Camera can be recovered and reused after removal.



Use fingers to remove the Front-Facing Camera from the Main PCB as shown.



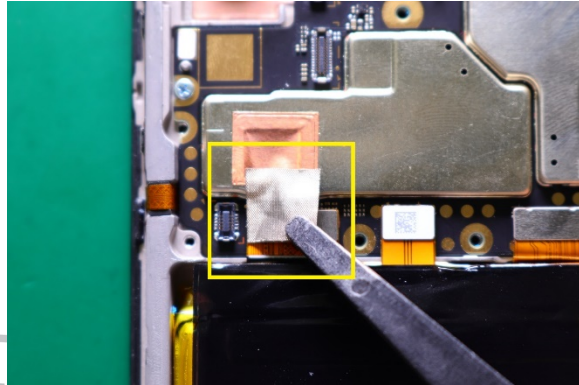
LCD Connector Conductive Cloth Removal

NOTE

After you remove the LCD Connector Conductive Cloth, you cannot reuse it.



Use the Plastic Tweezer to remove the LCD Connector Conductive Cloth from the location as shown.



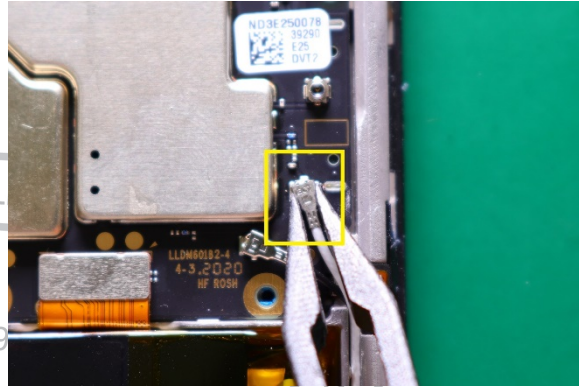
Main PCB Removal

NOTE

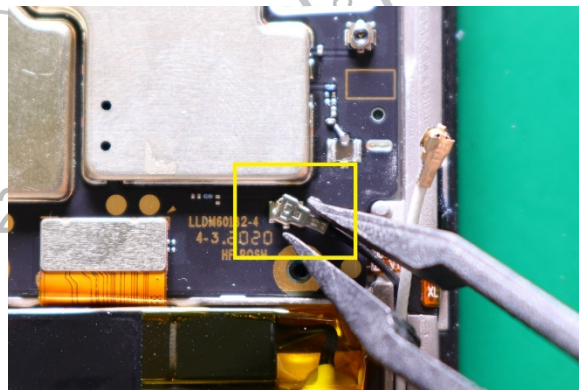
The Main PCB can be recovered and reused after removal.



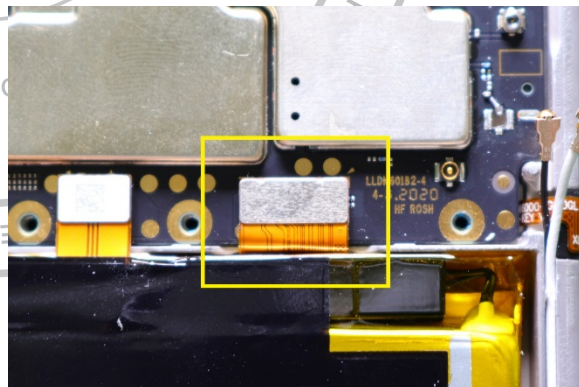
1. Use the Plastic Tweezer to remove the white Coax Cable from the location as shown.



2. Use the Plastic Tweezer to remove the black Coax Cable from the location as shown.



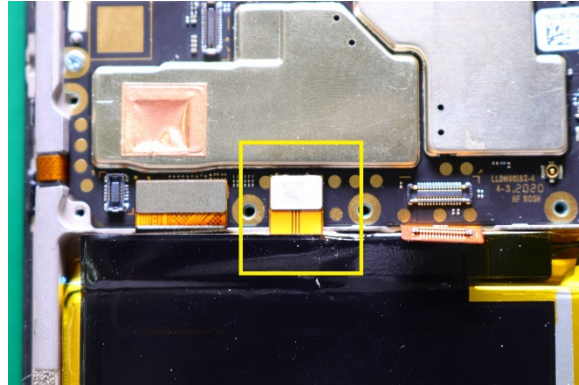
3. Use fingers to disengage Main Flex BtB Connector from the location as shown.



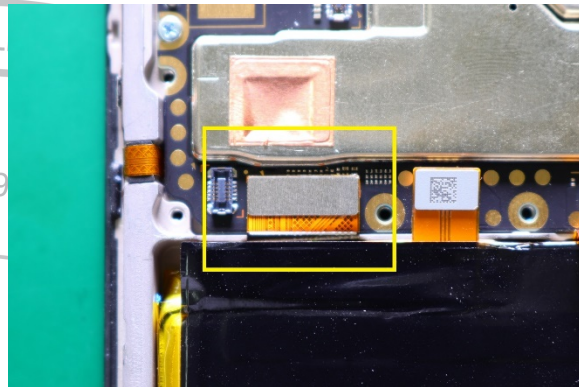
Lenovo Moto G9 Plus

Level 2 – Service and Repair Manual

4. Use fingers to disengage Battery BtB Connector from the location as shown.



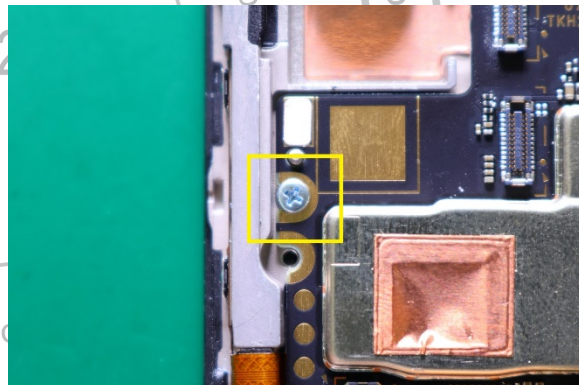
5. Use fingers to disengage Display BtB Connector from the location as shown.



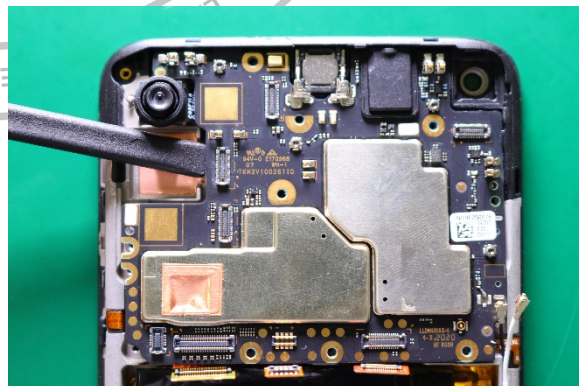
6. Use the Torque Driver with PH00 Bit to remove the screw as shown.

/// WARNING ///

The screw cannot be reused after removal.



7. Use the flat end of the Black Stick to lift the Main PCB from the location as shown.



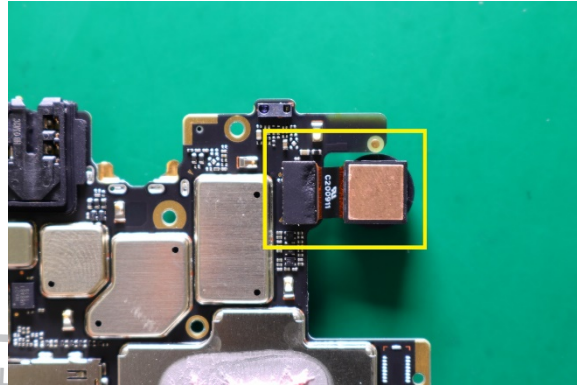
8M Rear-Facing Camera Removal

NOTE

The 8M Rear-Facing Camera can be recovered and reused after removal.



Use fingers to remove the 8M Rear-Facing Camera from the location as shown.



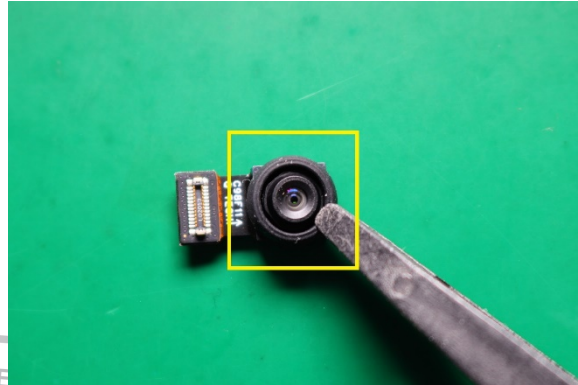
8M Rear-Facing Camera Grommet Removal

NOTE

The 8M Rear-Facing Camera Grommet can be recovered and reused after removal.



Use the Plastic Tweezer to remove the 8M Rear-Facing Camera Grommet from the location as shown.



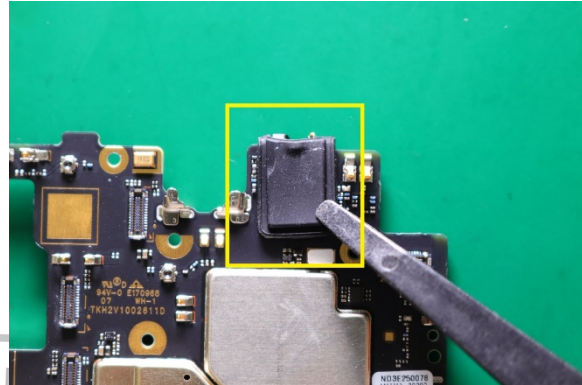
Audio Jack Grommet Removal

NOTE

The Audio Jack Grommet can be recovered and reused after removal.



Use the Plastic Tweezer to remove the Audio Jack Grommet from the location as shown.



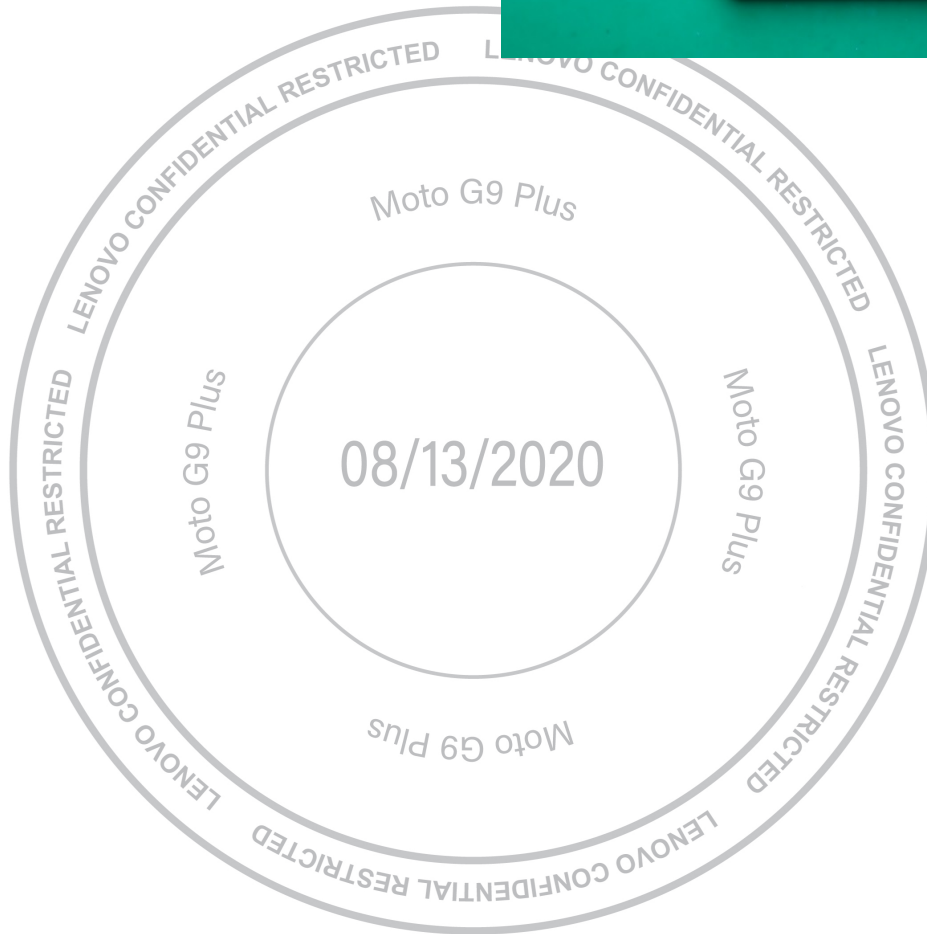
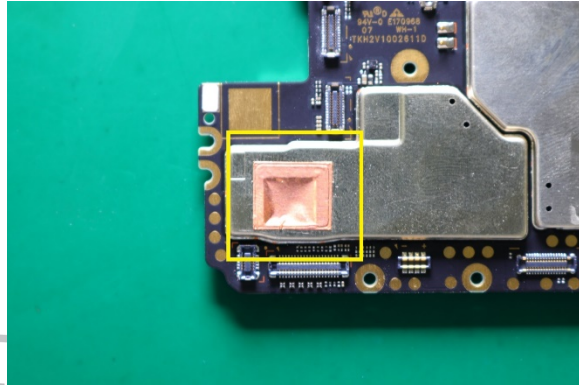
PMU Copper Foil Removal

NOTE

After you remove the PMU Copper Foil, you cannot reuse it.



Use the Plastic Tweezer to remove the PMU Copper Foil from the Main PCB as shown.



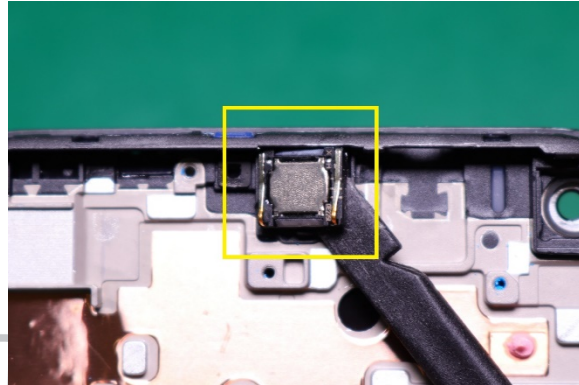
Receiver Removal

NOTE

The Receiver can be recovered and reused after removal.



1. Use the flat end of the Black Stick to lift the Receiver from the location as shown.
2. Remove the Receiver from the phone.



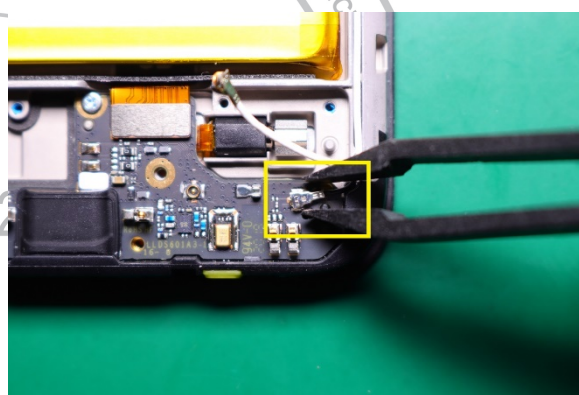
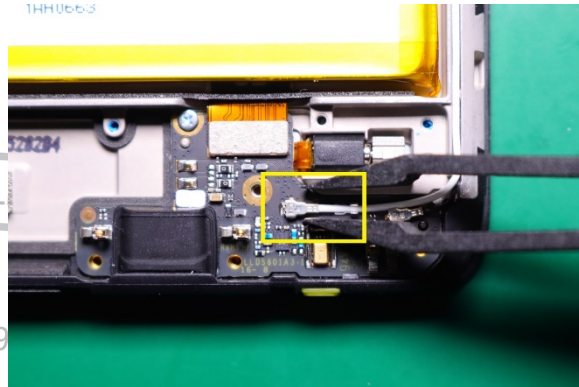
Sub Board Removal

NOTE

The Sub Board can be recovered and reused after removal.



1. Use the Plastic Tweezer to remove the white Coax Cable from the location as shown.
2. Use the Plastic Tweezer to remove the black Coax Cable from the location as shown.
3. Use fingers to carefully disengage the Main Flex BTB Connector from the location on the Sub Board as shown.



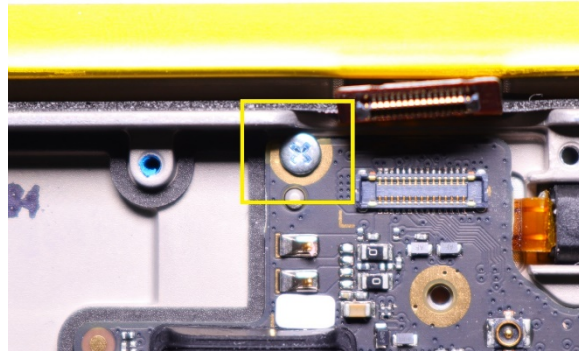
Lenovo Moto G9 Plus

Level 2 – Service and Repair Manual

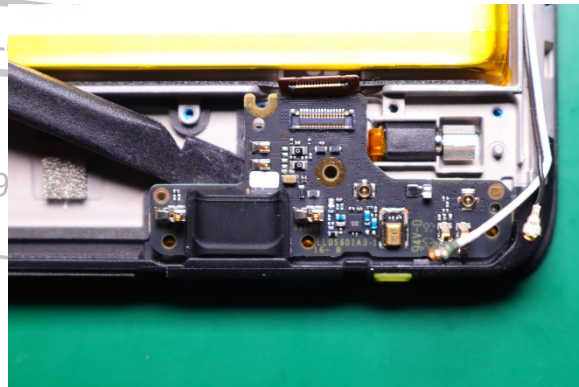
4. Use the Torque Driver with PH00 Bit to remove the screw as shown.

/// WARNING ///

The screw cannot be reused after removal.



5. Use the flat end of the Black Stick to gently lift the Sub Board from the location as shown.



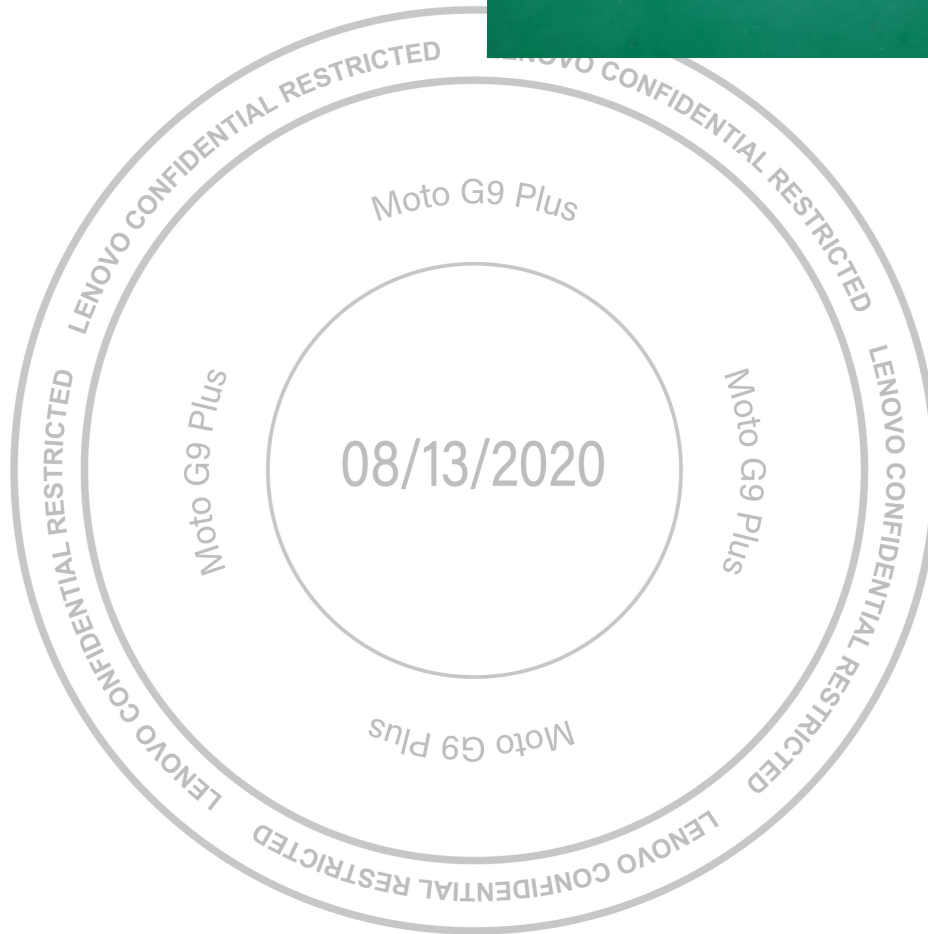
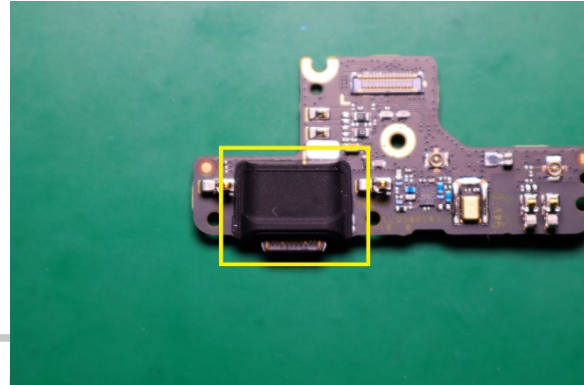
USB Grommet Removal

NOTE

The USB Grommet can be recovered and reused after removal.



Use the Plastic Tweezer to remove the USB Grommet from the Sub Board as shown.



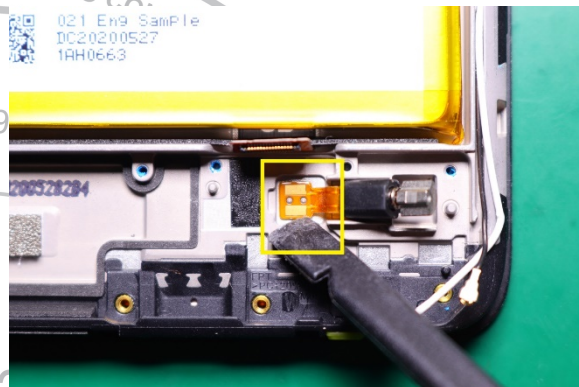
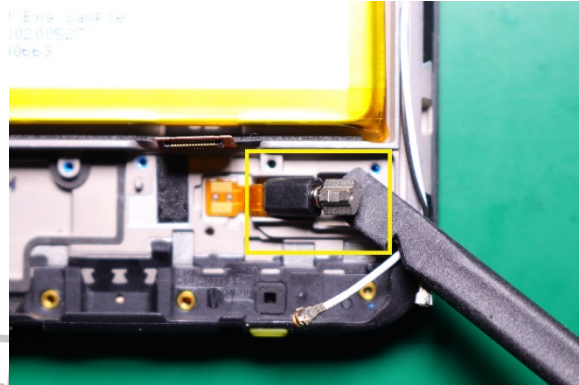
Vibrator Removal

NOTE

The Vibrator can be recovered and reused after removal.



1. Use the flat end of the Black Stick to gently disengage the Vibrator from the location as shown.
2. Use the flat end of the Black Stick to gently disengage the Vibrator Flex from the location as shown.



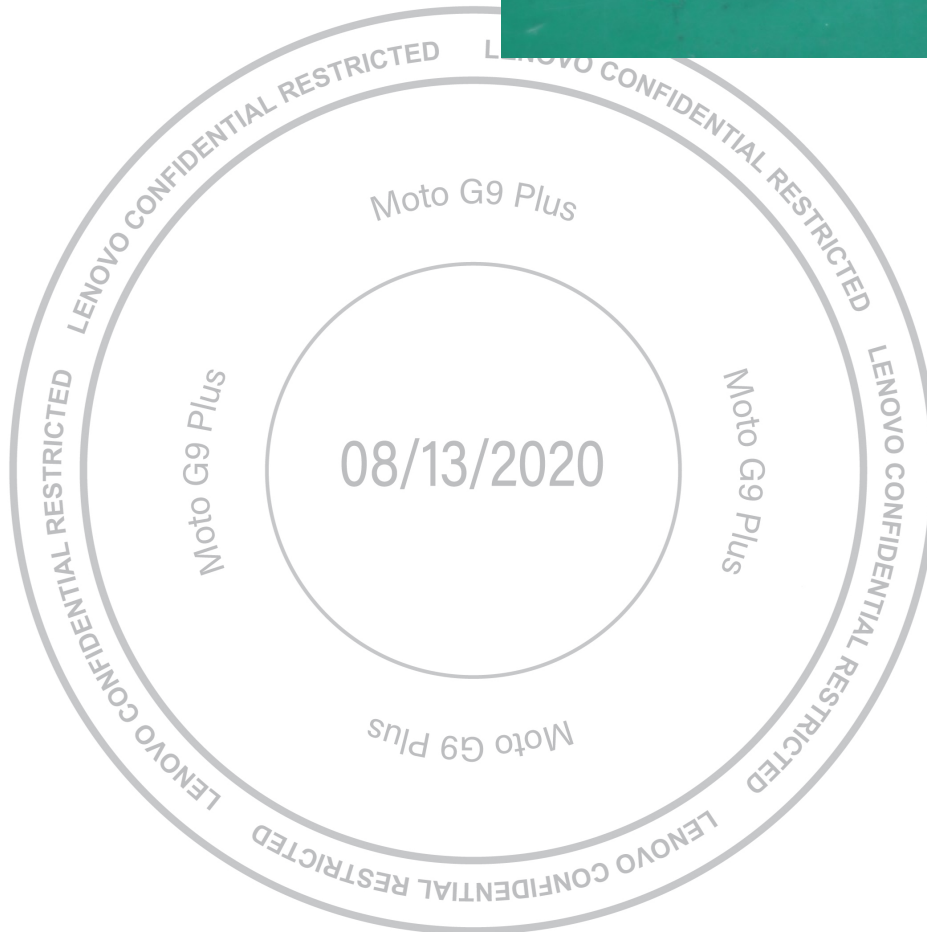
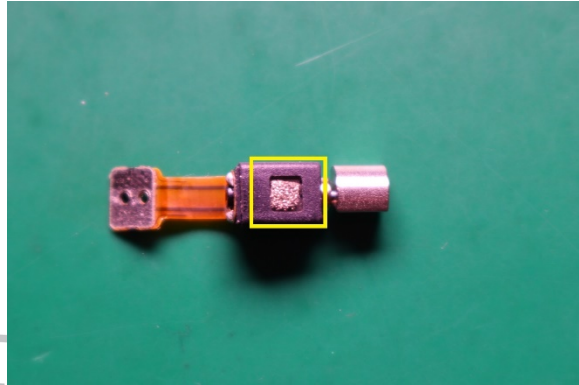
Vibrator Grounding Foam Removal

NOTE

After you remove the Vibrator Grounding Foam, you cannot reuse it.



Use the Plastic Tweezer to remove the Vibrator Grounding Foam from the location as shown.



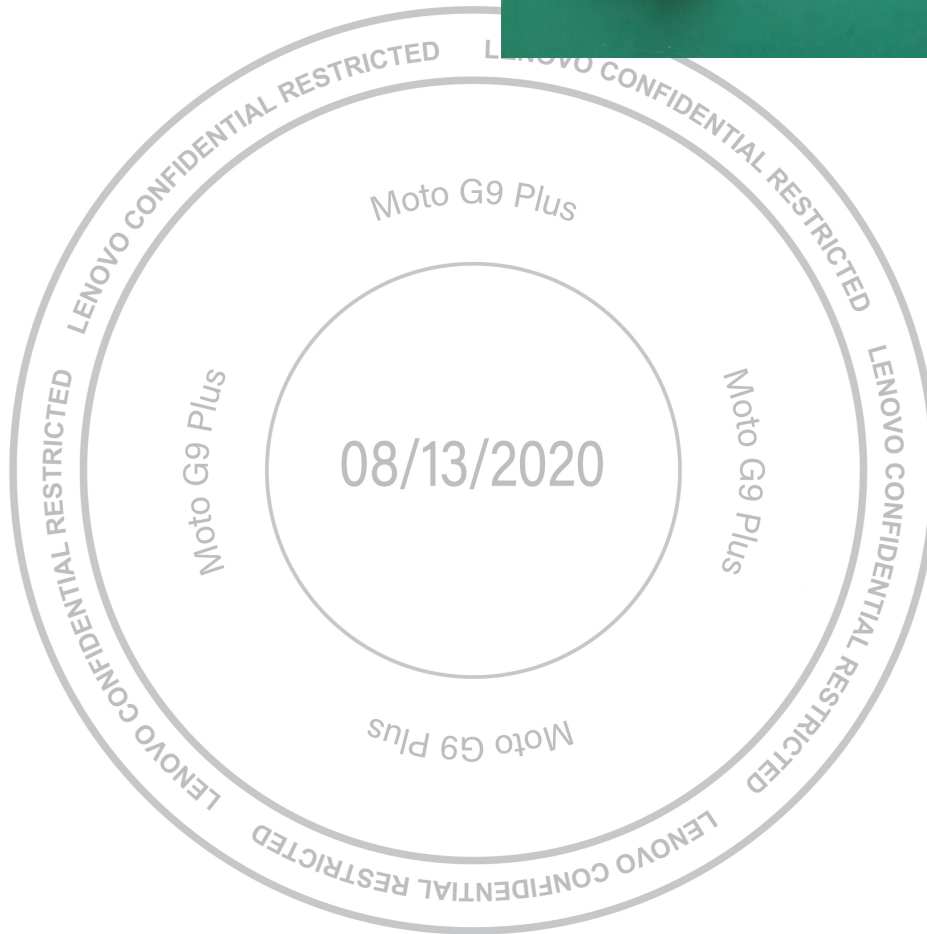
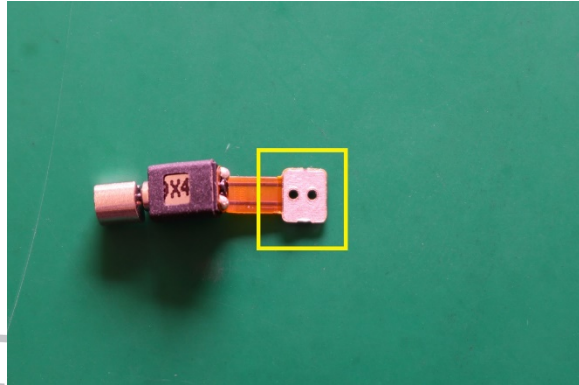
Vibrator Adhesive Removal

NOTE

After you remove the Vibrator Adhesive, you cannot reuse it.



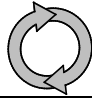
Use the Plastic Tweezer to remove the Vibrator Adhesive from the location as shown.



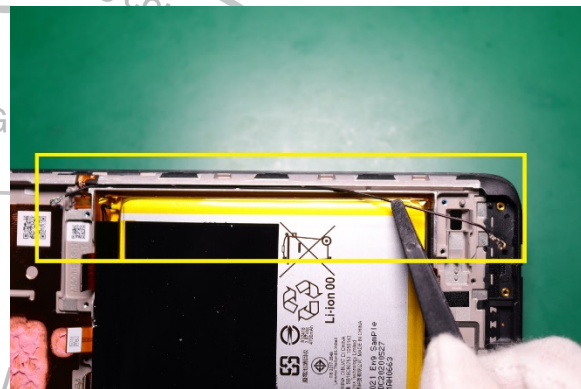
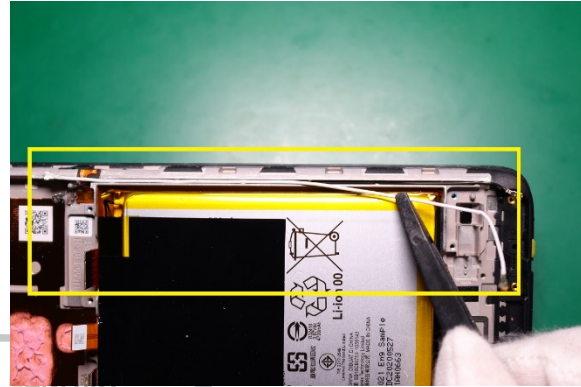
Coax Cable Removal

NOTE

The Coax Cables can be recovered and reused after removal.



1. Use the Plastic Tweezer to remove the white Coax Cable from the location as shown.
2. Use the Plastic Tweezer to remove the black Coax Cable from the location as shown.



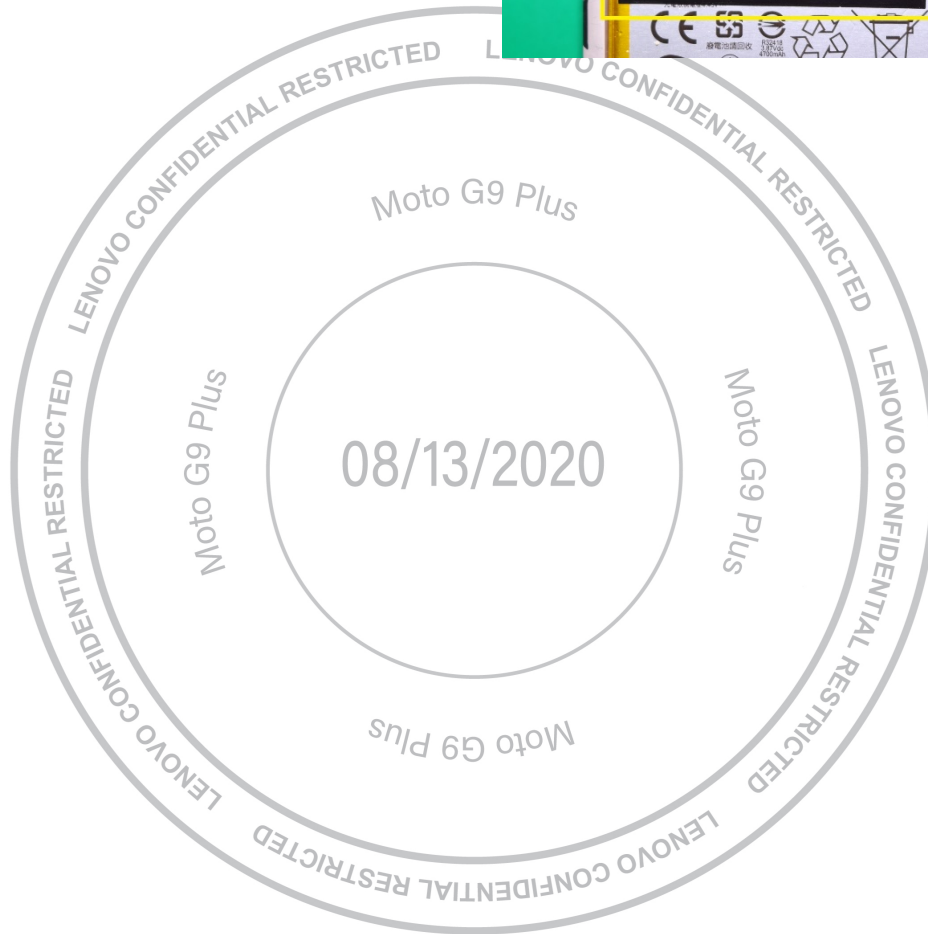
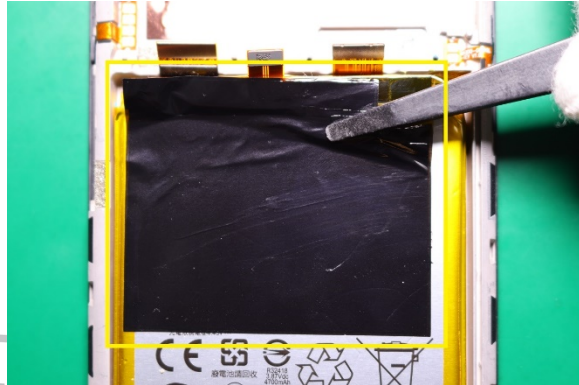
Battery Heat Spreader Removal

NOTE

After you remove the Battery Heat Spreader, you cannot reuse it.



Use the Plastic Tweezer to remove the Battery Heat Spreader from the location as shown.



Battery Removal

/// WARNING ///

Failure to adhere to Safety Critical Note(s) may increase risk of rupture, burning, or failure to function safely when used by the customer. Refer to the Battery Safety Guidelines.

NOTE

Remove the Battery only if it is defective or damaged. After you remove it, you cannot reuse it.



// CAUTION //

Handle the Battery Pack with care. If dropped on the floor, it may be internally damaged and must be scrapped.

Ensure Battery and its insulation are not damaged (e.g. scratched, dented, punctured) prior to and through assembly.

Prior to assembly, ensure Battery edges and surfaces are not dented or deformed, and that fixtures and parts that will contact the Battery are free of foreign material.

Ensure screws and screwdrivers do not contact the Battery.

1. Set the temperature of the Hot Plate to 80°C (176°F).

// CAUTION //

Do not set the Hot Plate temperature higher than 80°C (176°F). Damage to the phone may occur.

NOTE

Preheating with an oven is also acceptable if the oven is set to the same temperature.

2. Place the phone on the Hot Plate, with the Display side down.



Lenovo Moto G9 Plus

Level 2 – Service and Repair Manual

3. Use the thin Plastic Spudger to cut adhesive from the top of the Battery.



4. Use the thin Plastic Spudger to cut adhesive from the sides of the Battery.



5. Use fingers to carefully lift the Battery.



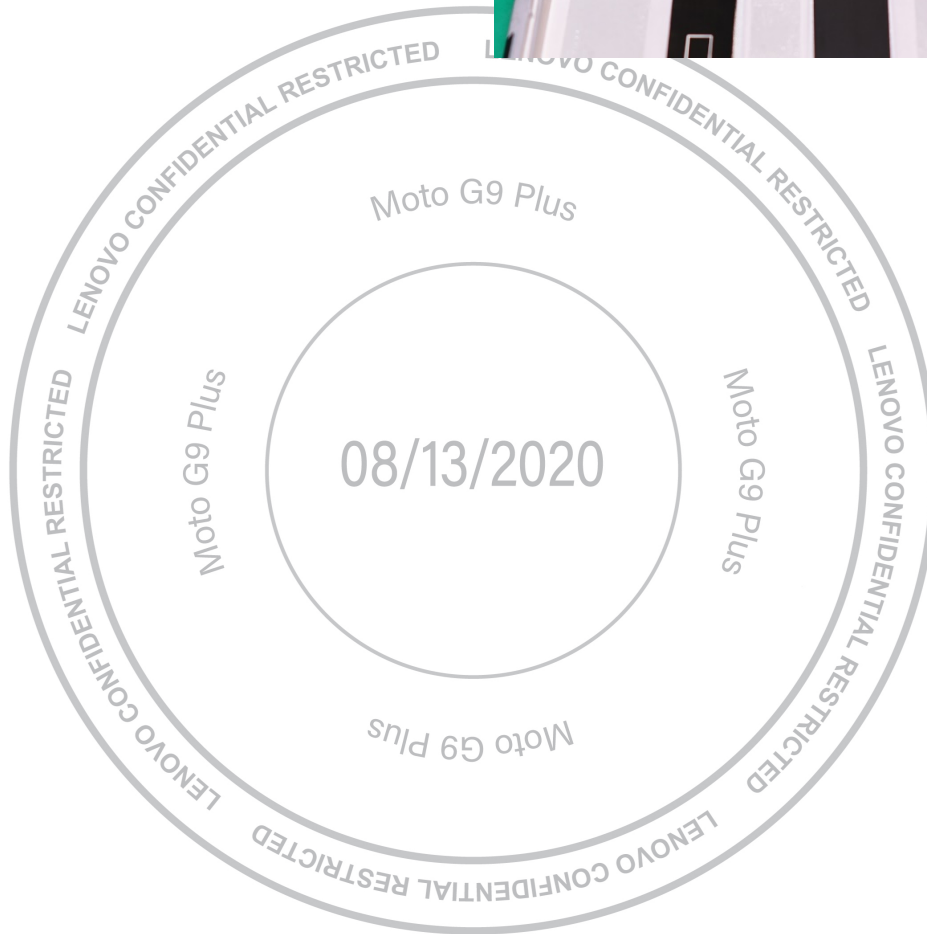
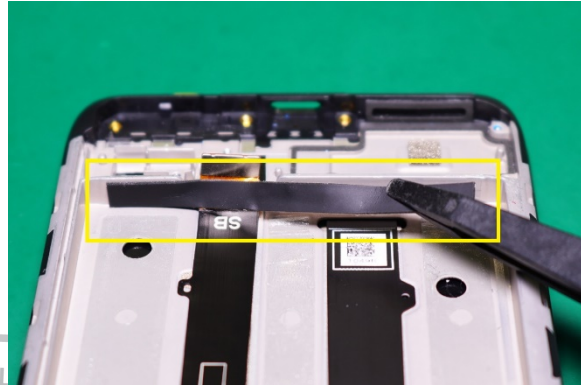
Battery Foam Removal

NOTE

After you remove the Battery Foam, you cannot reuse it.



Use the Plastic Tweezer to remove the Battery Foam from the location as shown.



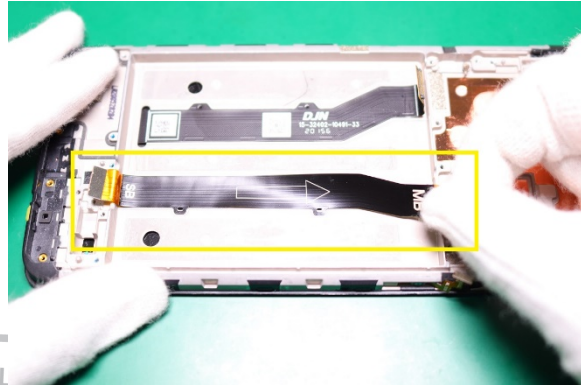
Main FPC Removal

NOTE

After you remove the Main Flex you cannot reuse it.



Use fingers to gently peel off the Main Flex as shown.



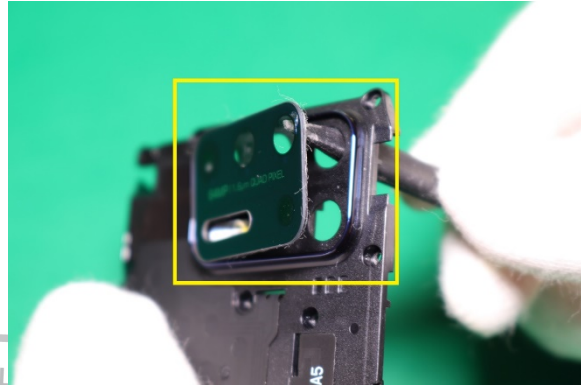
Rear-Facing Camera Lens Removal

NOTE

Remove the Rear-Facing Camera Lens only if it is defective or damaged.



Use the pointed end of the Black Stick to carefully disengage and push out the Rear-Facing Camera Lens from the location as shown.



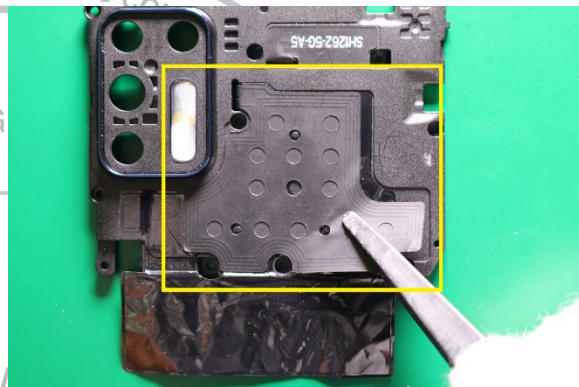
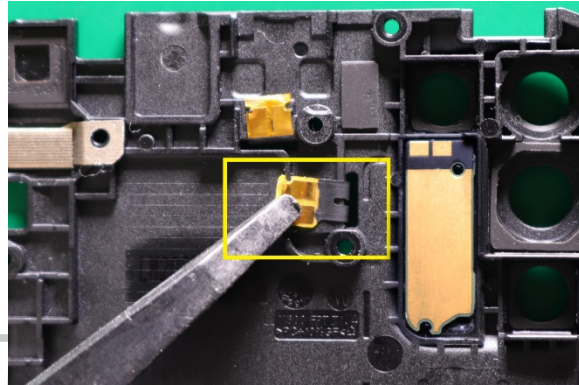
NFC Antenna Removal

NOTE

After you remove the NFC Antenna, you cannot reuse it.



1. Use the Plastic Tweezer to remove the NFC Antenna flex from the location as shown.
2. Use the Plastic Tweezer to remove the NFC from the location as shown.



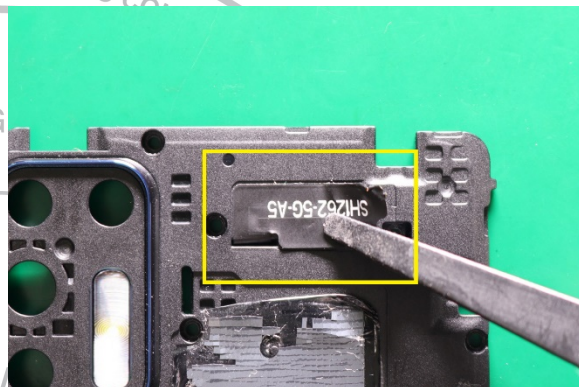
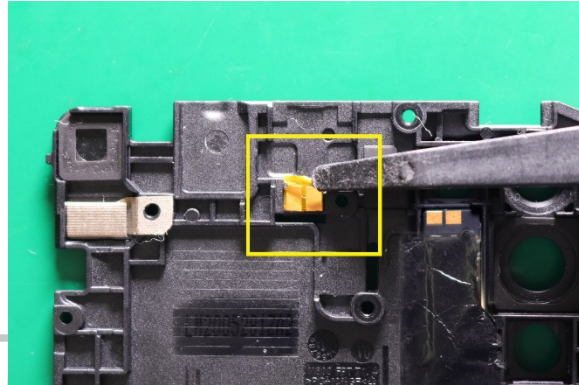
5G WIFI Antenna Removal

NOTE

After you remove the 5G WIFI Antenna, you cannot reuse it.



1. Use the Plastic Tweezer to remove the 5G WIFI Antenna flex from the location as shown.
2. Use the Plastic Tweezer to remove the 5G WIFI Antenna from the location as shown.



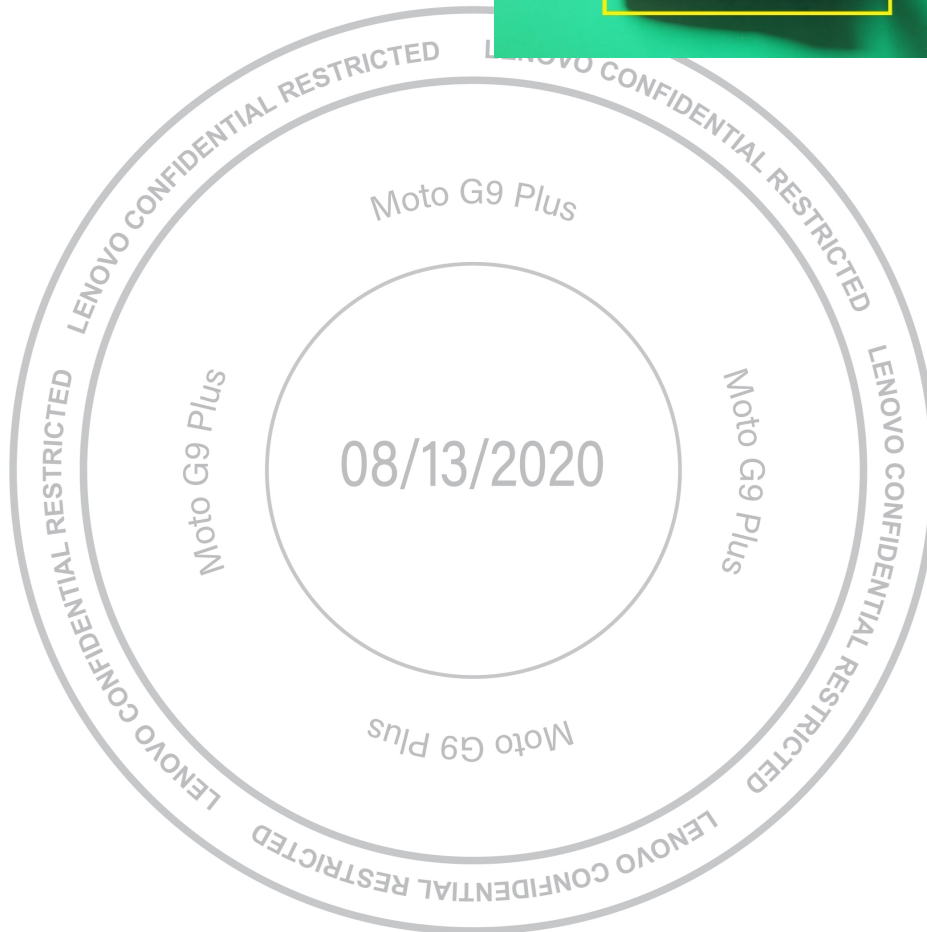
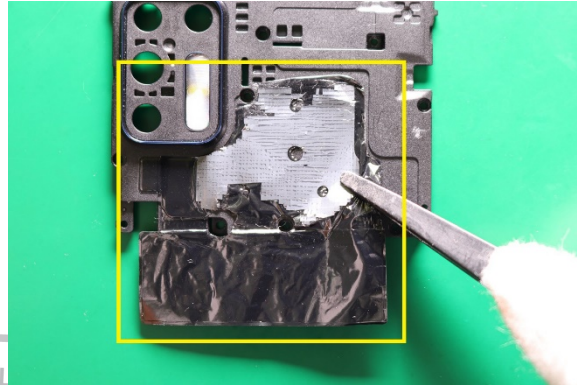
Battery Cover Heat Spreader Removal

NOTE

After you remove the Battery Cover Heat Spreader, you cannot reuse it.



Use the Plastic Tweezer to remove the Battery Cover Heat Spreader from the location as shown.



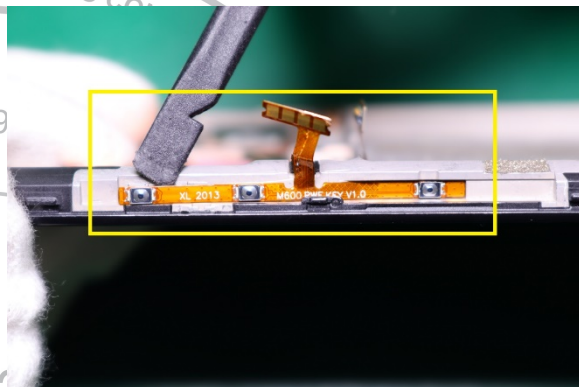
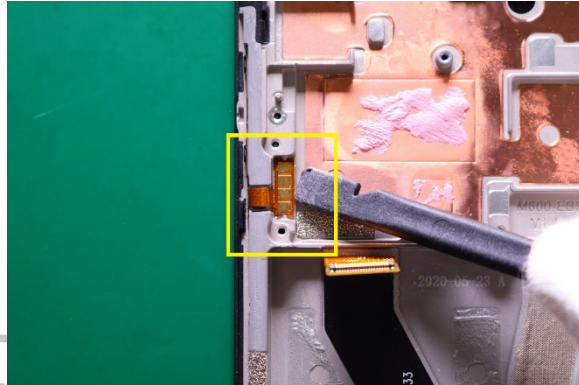
Volume Key FPC Removal

NOTE

After you remove the Side Key FPC, you cannot reuse it.



1. Use flat end of the Black Stick to disengage Volume Key Flex from the location as shown.
2. Use flat end of the Black Stick to disengage Volume Key FPC from the location as shown.



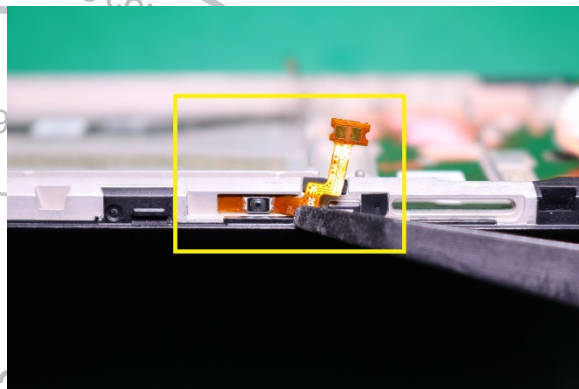
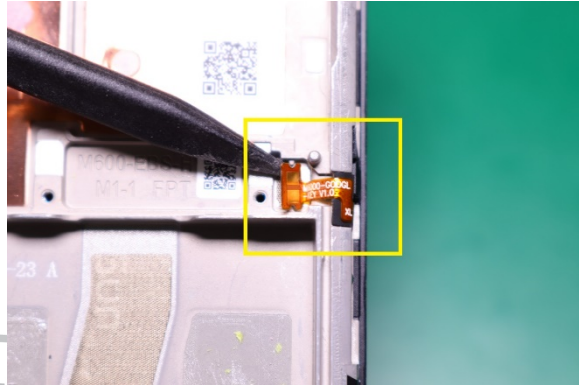
Google Key FPC Removal

NOTE

After you remove the Side Key FPC, you cannot reuse it.



1. Use pointed end of the Black Stick to disengage Google Key Flex from the location as shown.
2. Use the Plastic Tweezer to remove Google Key FPC from the location as shown.



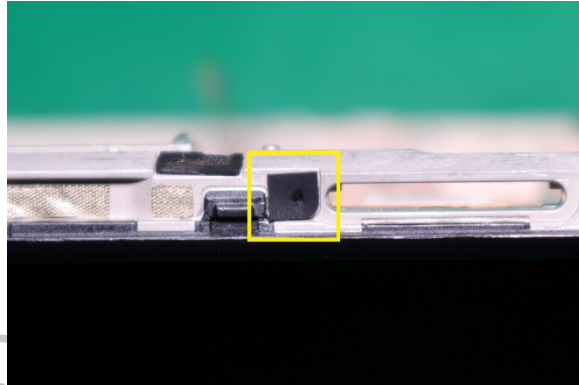
Sim Tray Hole Grommet Removal

NOTE

After you remove the Sim Tray Hole Grommet, you cannot reuse it.



Use the Plastic Tweezer to remove the Sim Tray Hole Grommet from the location as shown.



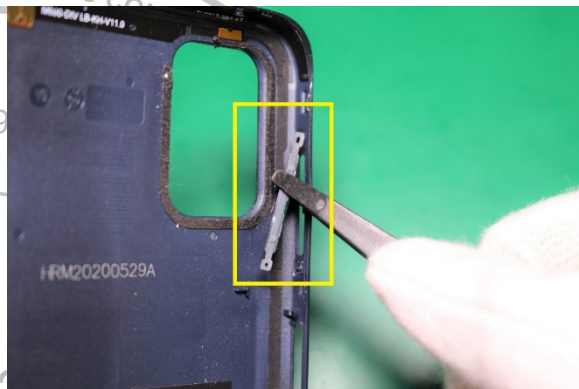
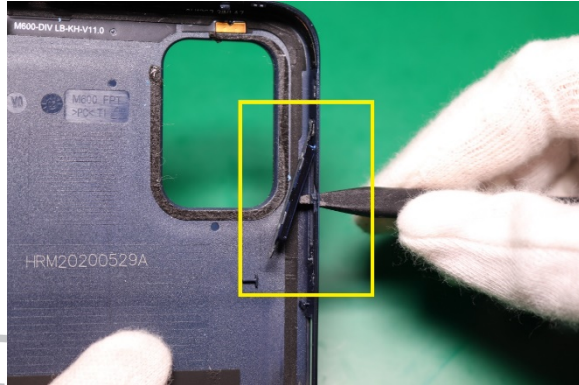
Volume Key Removal

NOTE

The Volume Key can be recovered and reused after removal.



1. Use pointed end of the Black Stick to gently push out and disengage Volume Key from the location as shown.
2. Use the Plastic Tweezer to remove Volume Key from the location as shown.



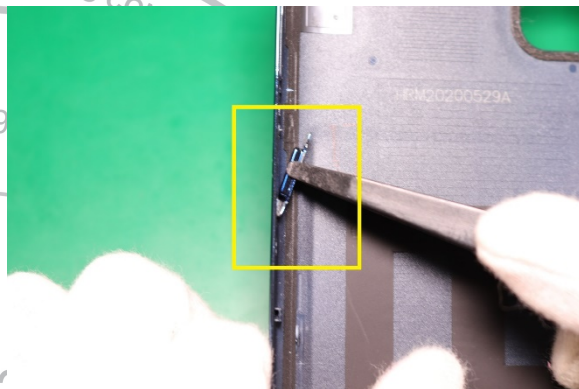
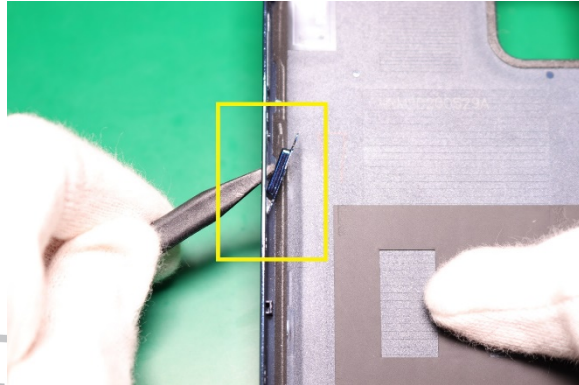
Google Key Removal

NOTE

The Google Key can be recovered and reused after removal.



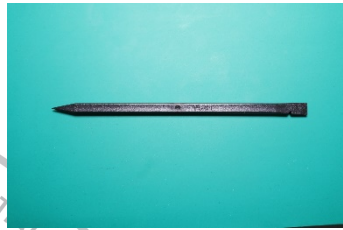
1. Use pointed end of the Black Stick to gently push out and disengage Side Key from the location as shown.
2. Use the Plastic Tweezer to remove Google Key from the location as shown.



PARTS REFRESH

Tools Required for Parts Refresh

The following tools are required for parts refresh of the Moto G9 Plus phone.

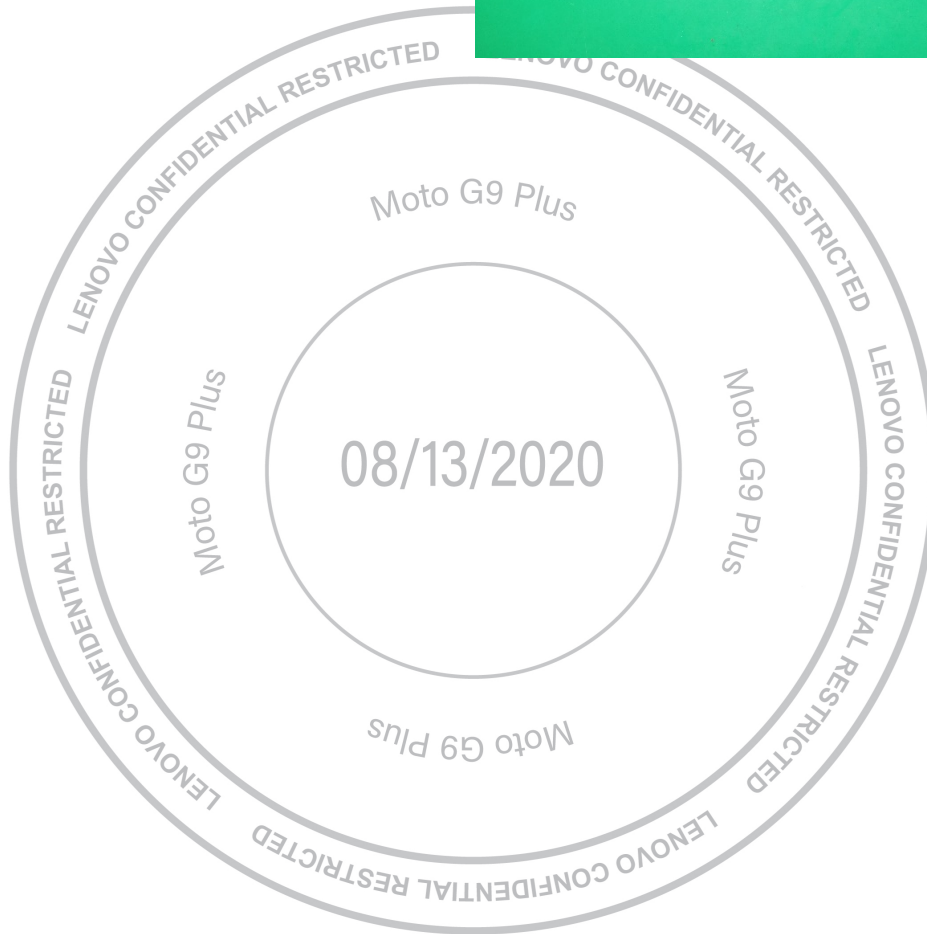
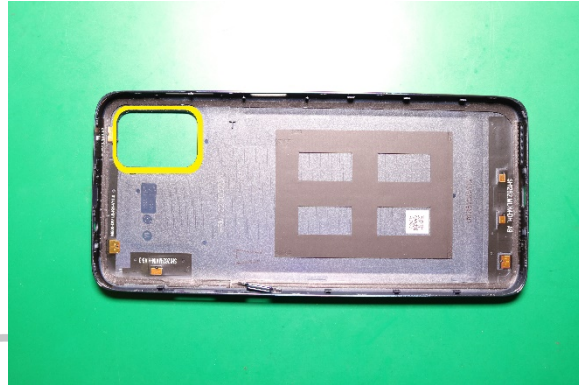
<i>Description</i>	<i>Part #</i>	<i>Image</i>
Plastic Tweezers	--	--
Black Stick	--	
ESD Mat	--	--
Wrist Strap	--	--
Isopropyl (ISP) Alcohol	--	--
Cotton Swabs	--	--
Cleaning Pads	--	--
Gloves or Finger Collets	--	--

Battery Cover Cleaning

Use the ISP Alcohol and Cotton Swabs to remove all the adhesives on the Battery Cover as shown.

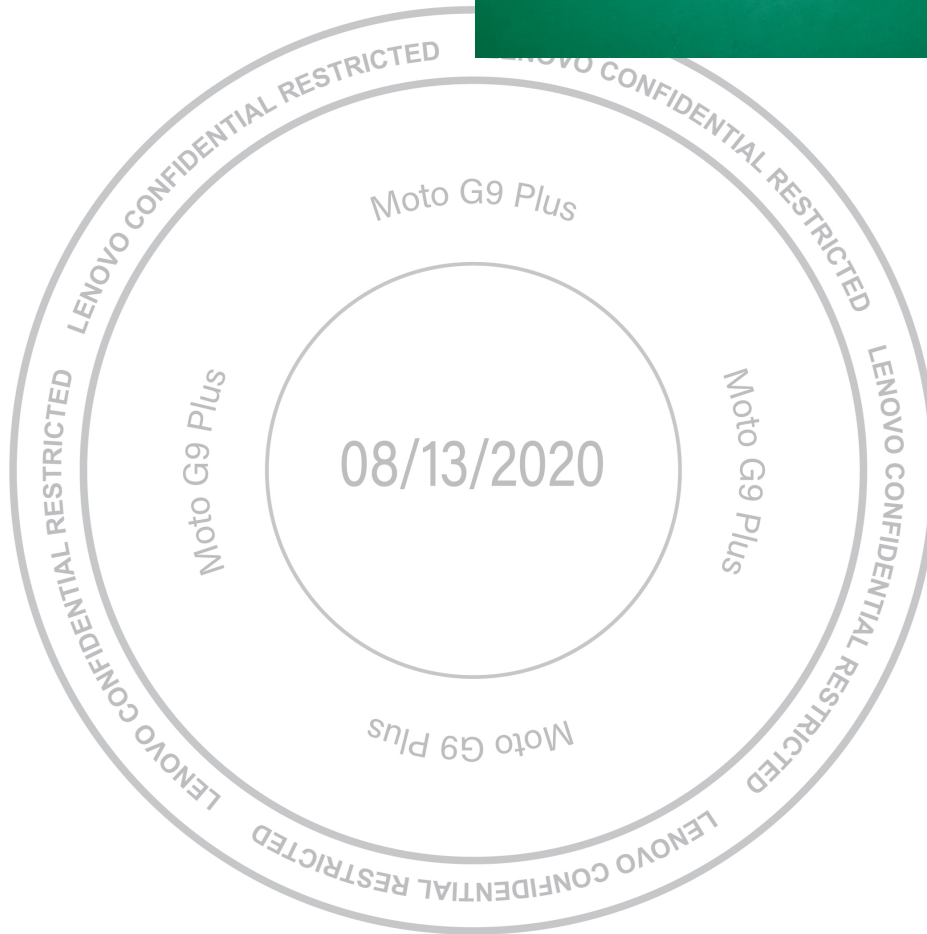
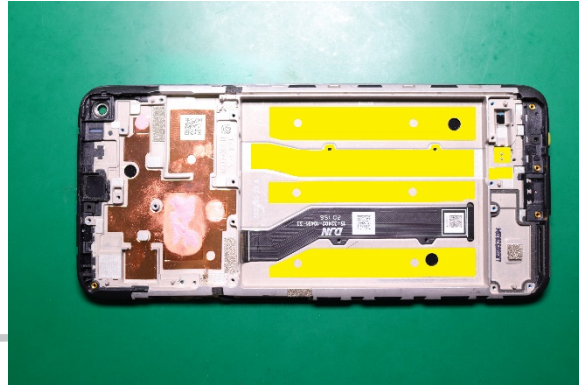
/// CAUTION ///

Remove the adhesive carefully to avoid damage to the Battery Cover.



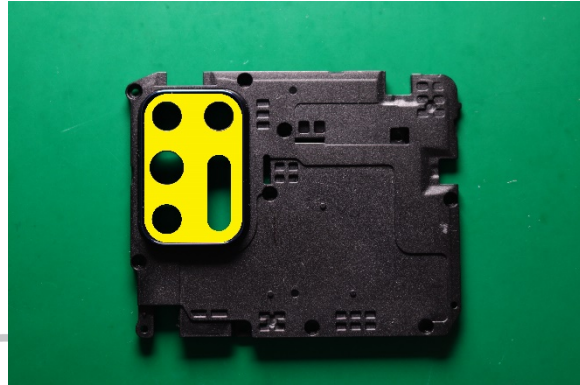
Front Housing Cleaning

Use the Cotton Swabs with ISP Alcohol to remove all the adhesives from the location as shown.



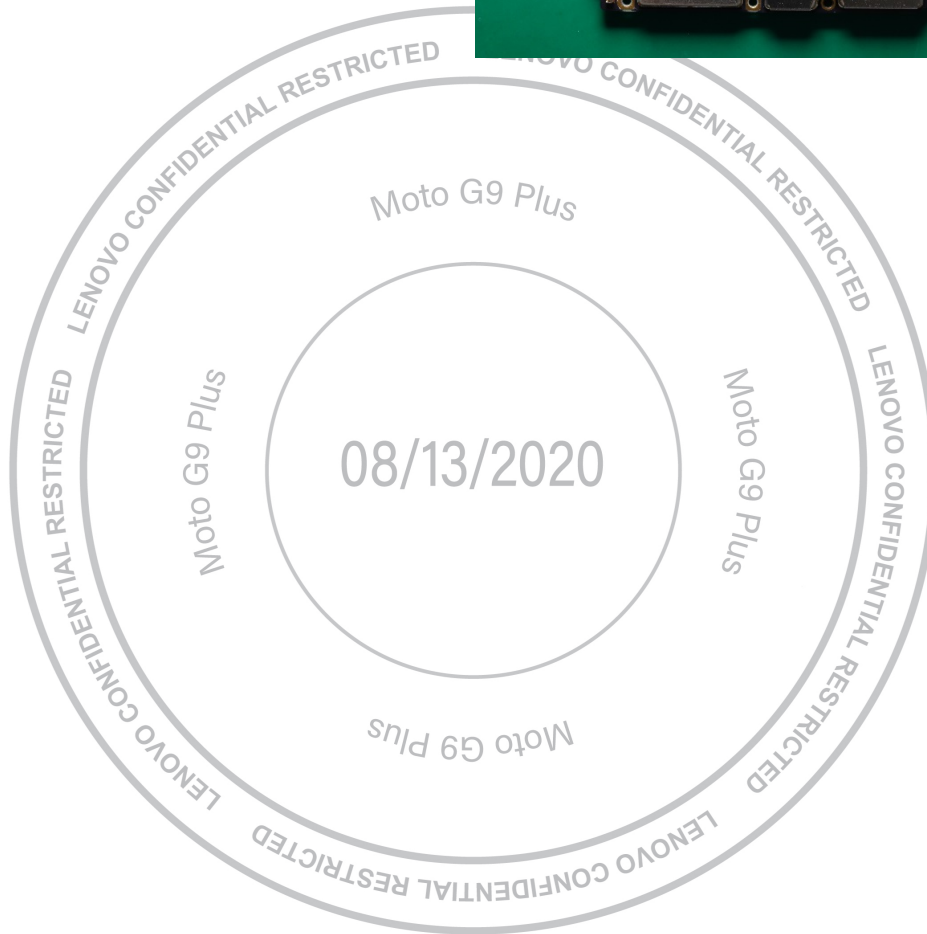
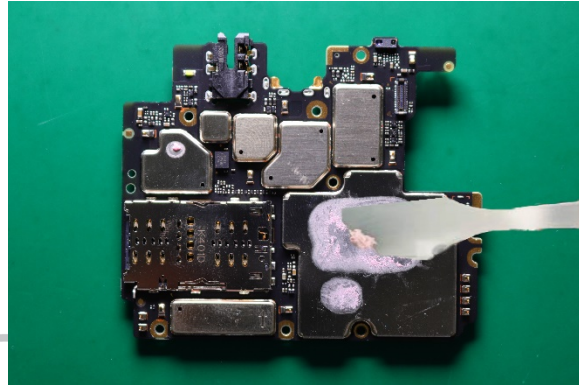
Top Bracket Cleaning

Use the Cotton Swabs with ISP Alcohol to remove all the adhesives from the location as shown.



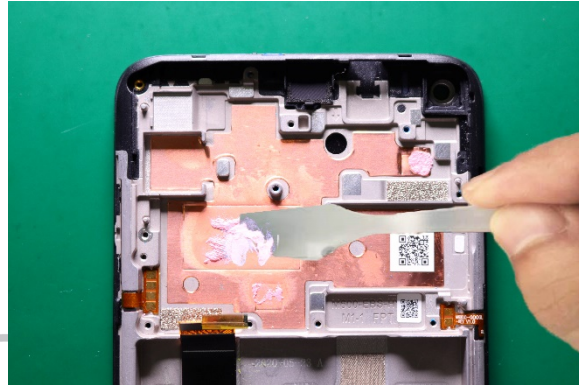
Main PCB Cleaning

Use the Cotton Swabs with ISP Alcohol to remove all the adhesives from the location as shown.



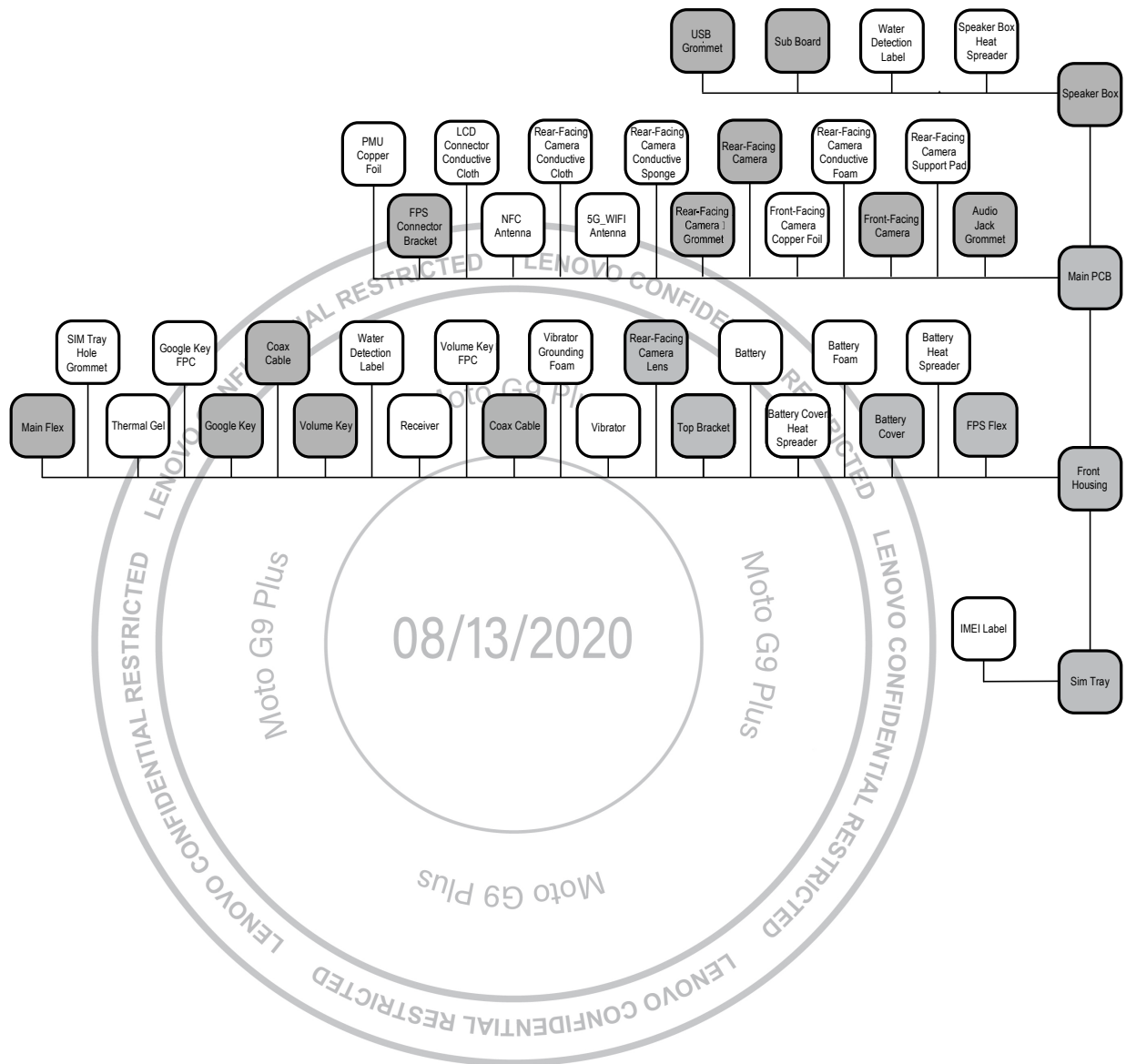
Front-Housing Thermal Gel Cleaning

Use Plastic Scraper to remove the most part of thermal putty from the Front-Housing.



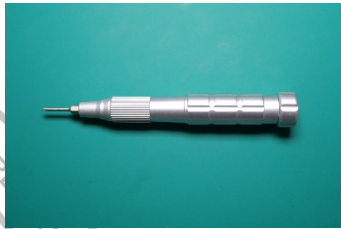
ASSEMBLY

Assembly Map



Tools Required for Assembly

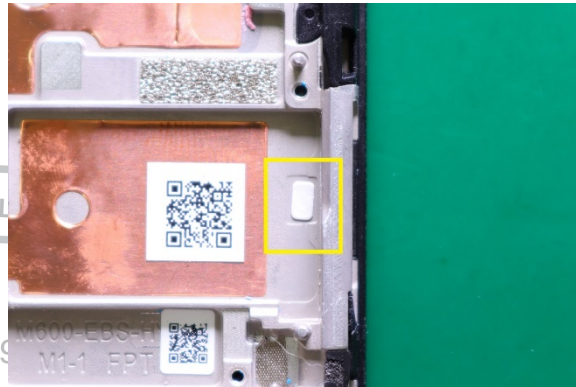
The following tools are required to assemble the Moto G9 Play phone.

<i>Description</i>	<i>Part #</i>	<i>Picture/Drawing</i>
Black Stick	--	
Torque Driver	--	
PH00 Bit	--	
Manual Roller	--	
Plastic Tweezers	--	--
Tweezers	--	--
Gloves or Finger Collets	--	--

Water Detection Label Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
LC7015AA001336_Waterproof labeling_DGJL	-

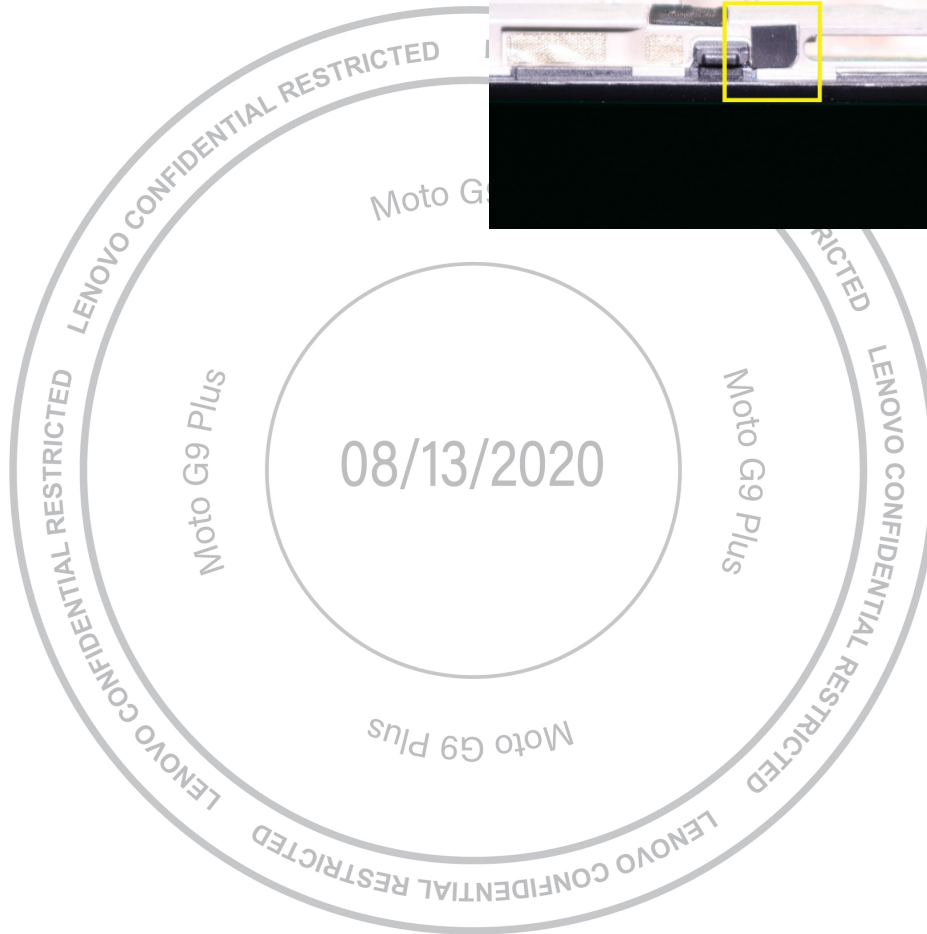
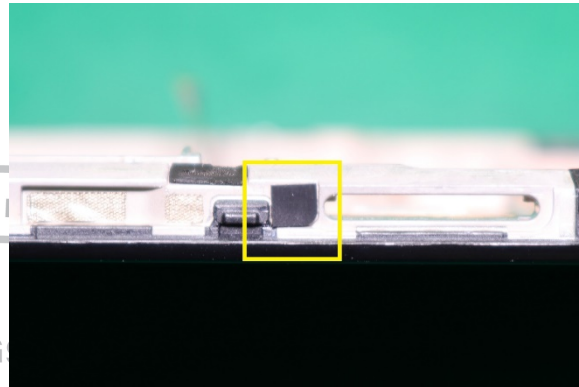
Use the Plastic Tweezer to place the Water Detection Label (WDL) on the Front-Housing as shown.



Sim Tray Hole Grommet Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
LC5061AA000490+LXF_M600_A01_Cardpinrubbe	-

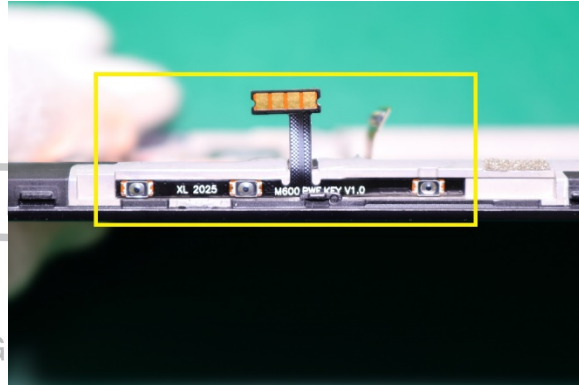
Use Plastic Tweezer to place the Sim Tray Hole Grommet on the Front Housing as shown.



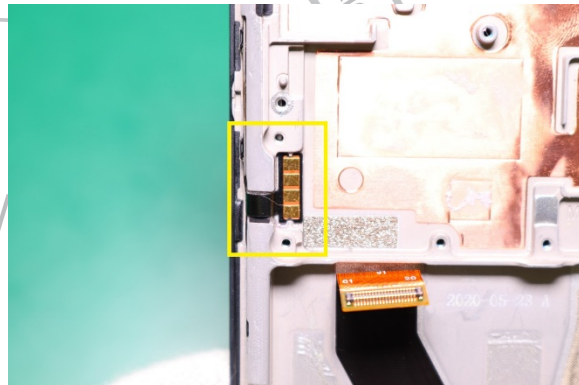
Volume Key FPC Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
LC6680AA000507+M600_Volumekey&powerkeyFP	23

1. Use fingers to place the Volume Key FPC on the Front Housing as shown.



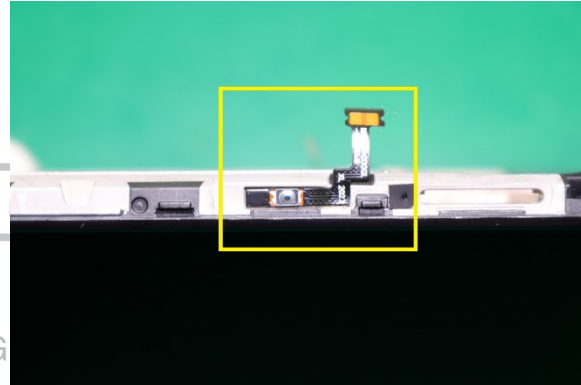
2. Use fingers to press and secure the Volume Key FPC on the Front Housing as shown.



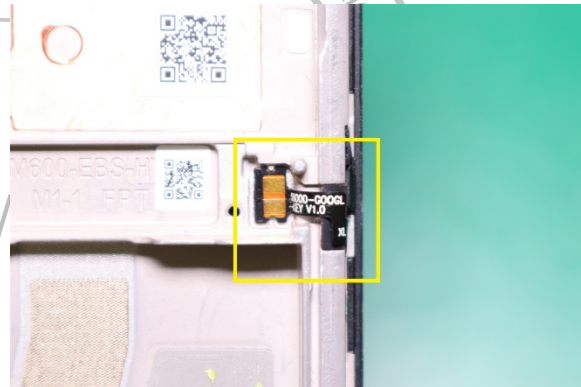
Google Key FPC Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
LC6680AA000510+M600_GoogleButtonFPC	6

1. Use fingers to place the Google Key FPC on the Front Housing as shown.



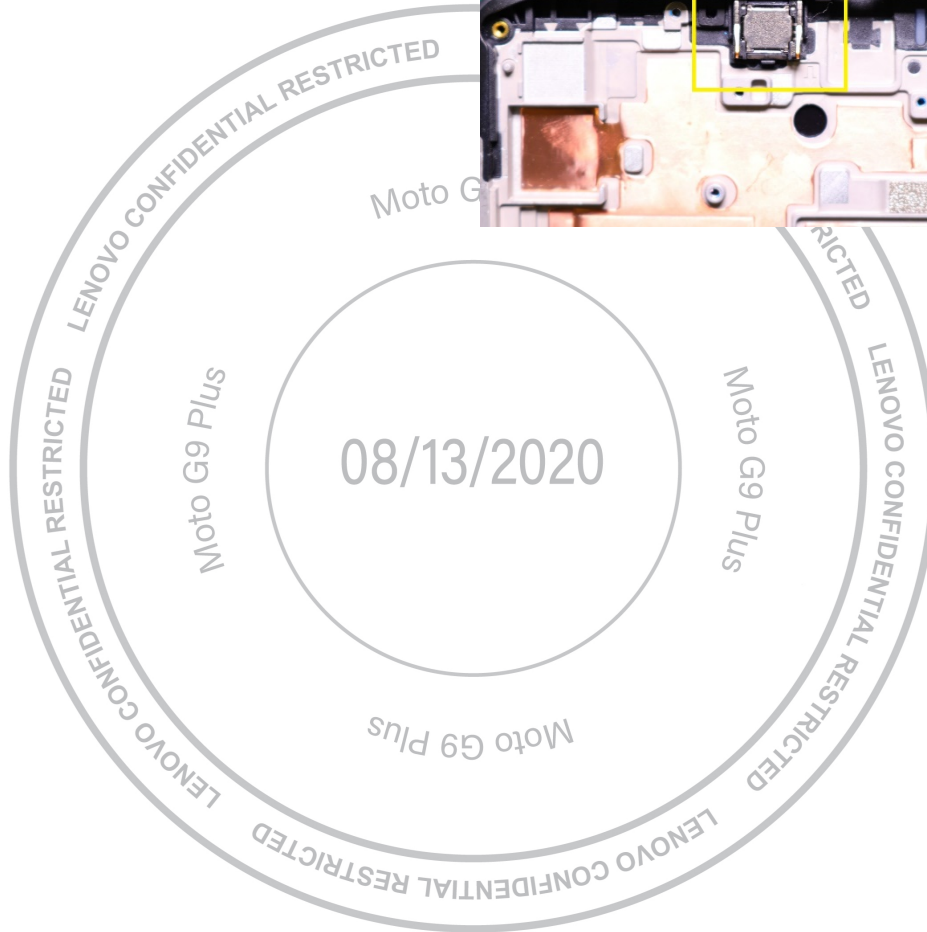
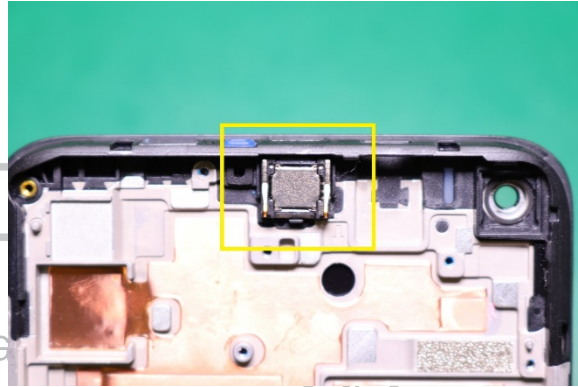
2. Use fingers to press and secure the Google Key FPC on the Front Housing as shown.



Receiver Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
LC2626AA000035+Receiver_Normal_9mmx8mmxH	7

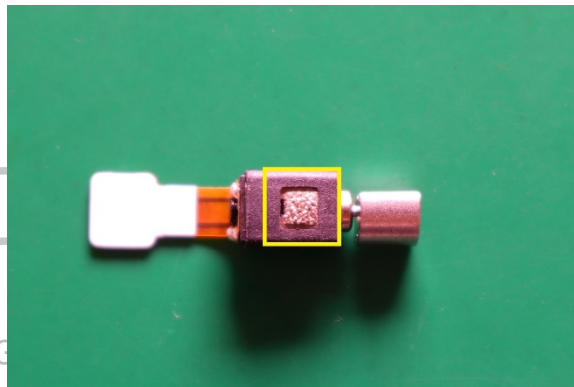
Use the Plastic Tweezer to place the Receiver on the location aligning with the pins as shown.



Vibrator Grounding Foam Assembly

Description	Reference #
LC2627AA000042+MOTOR_CylinderMotor_20.37	13
LC5450AA000613+LXF_M600_A01_Motorgroundi	-

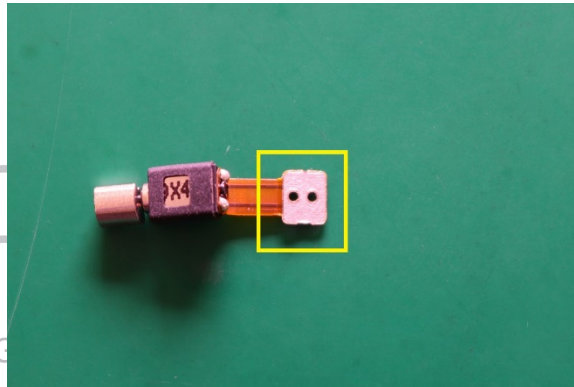
Use the Plastic Tweezer to place the Vibrator Grounding Foam on the location as shown.



Vibrator Adhesive Assembly

Description	Reference #
Odessa Motor Adhesive3&5420AA000840	-
LC2627AA000042+MOTOR_CylinderMotor_20.37	13

Use the Plastic Tweezer to place the Vibrator Adhesive on the location as shown and peel off the liner.



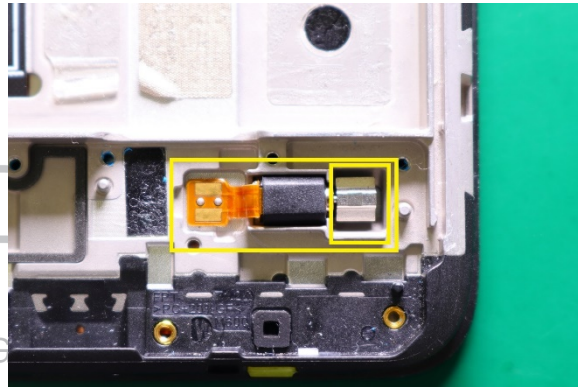
Vibrator Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
LC2627AA000042+MOTOR_CylinderMotor_20.37	13

Use the Plastic Tweezer to place and secure the Vibrator on the location as shown.

NOTE

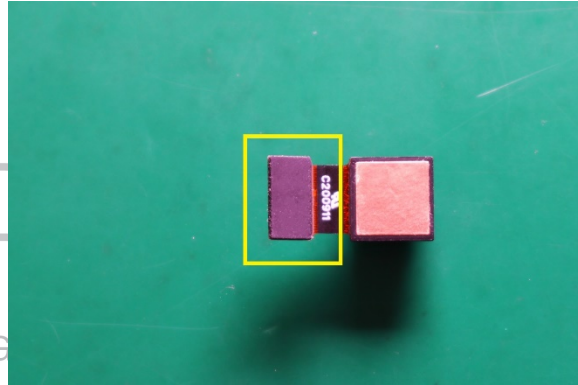
Make sure that gold finger side of the Vibrator is pointing upwards.



Rear-Facing Camera Conductive Foam Assembly

Description	Reference #
MDL,CAMR,8M,rear,Nairobi;C98F11;Qtech;UW	10
LC5430AA000835+LXF_M600_A01_rearconnecto	-

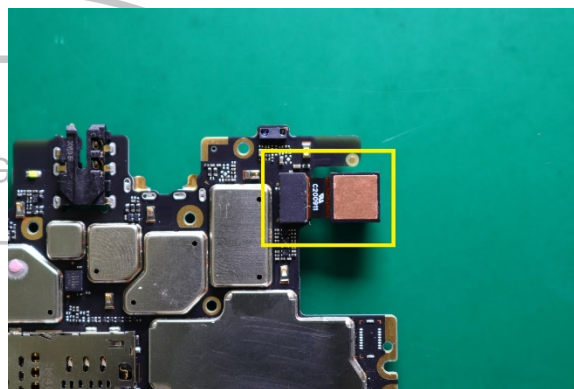
Use fingers to place and secure the Rear-Facing Camera Conductive Foam on the location as shown.



8M Rear-Facing Camera Assembly

Description	Reference #
XT2087-1 MB 4+128G DS CS XT2087-1 MB 4+128G SS CS XT2087-2 MB 4+128G DS CS XT2087-2 MB 6+128G DS CS ASSY,BD KIT,Odessa20 BR 4+128 MB PCBA	4
MDL,CAMR,8M,rear,Nairobi;C98F11;Qtech;UW	10
LC5430AA000835+LXF_M600_A01_rearconnecto	-

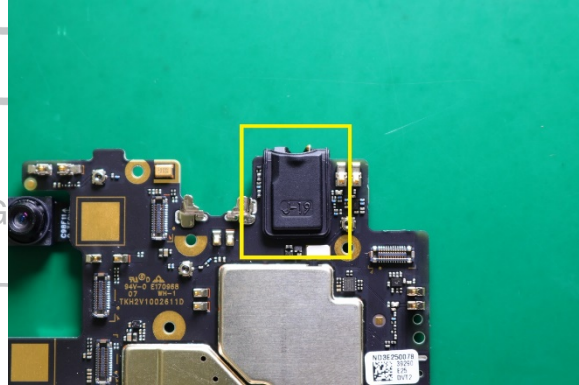
Use fingers to place and secure the 8M Rear-Facing Camera on the Main PCB as shown.



Audio Jack Assembly

Description	Reference #
XT2087-1 MB 4+128G DS CS XT2087-1 MB 4+128G SS CS XT2087-2 MB 4+128G DS CS XT2087-2 MB 6+128G DS CS ASSY,BD KIT,Odessa20 BR 4+128 MB PCBA	4
LC5061AA000486+LXF_M600_A01_Headsetseali	2

Use the Plastic Tweezer to place the Audio Jack on the location as shown.



Thermal Gel Assembly

NOTE

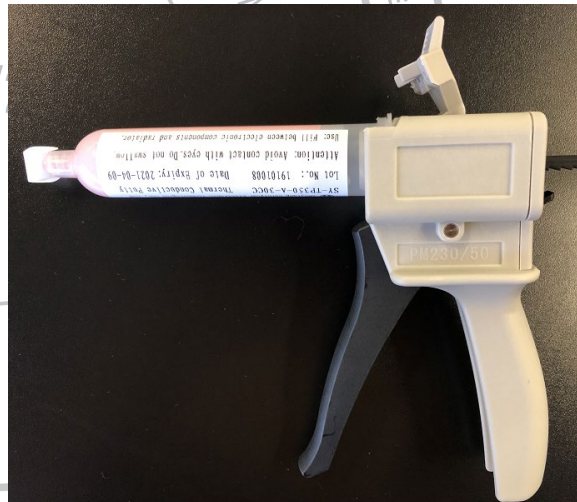
This procedure is only required when replacing the Main PCB or Front Housing.

Thermal Gel Syringe size package and Needle Gauges description

1. TDS SY-TP350 5ml can operate without dispensing tool.



2. TDS SY-TP350 30ml operate with dispensing tool.



3. Henkel LF350 30ml operate with dispensing tool.



Lenovo Moto G9 Plus

Level 2 – Service and Repair Manual

4. Cooler Master and Artic Thermal Gel syringe package.



5. When use Artic and Cooler Master 30ml package suggest using 18-gauge needle.



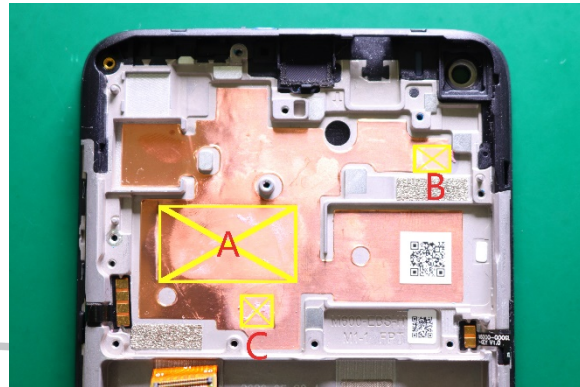
6. When use SY-TP350 or Henkel LF350 30ml package suggest using 14- or 15-gauge needle.

Needle Gauges for Injections Size Chart	
	14 GAUGE COLOR: OLIVE OUTER DIAMETER: .072IN (1.83MM)
	15 GAUGE COLOR: AMBER OUTER DIAMETER: .065IN (1.65MM)
	16 GAUGE COLOR: GRAY OUTER DIAMETER: .064IN (1.63MM)
	18 GAUGE COLOR: GREEN OUTER DIAMETER: .050IN (1.27MM)
	20 GAUGE COLOR: PINK OUTER DIAMETER: .036IN (.91MM)
	21 GAUGE COLOR: PURPLE OUTER DIAMETER: .033IN (.83MM)
	22 GAUGE COLOR: BLUE OUTER DIAMETER: .027IN (.70MM)
	23 GAUGE COLOR: ORANGE OUTER DIAMETER: .025IN (.63MM)
	25 GAUGE COLOR: RED OUTER DIAMETER: .020IN (.53MM)
	27 GAUGE COLOR: WHITE OUTER DIAMETER: .016IN (.42MM)

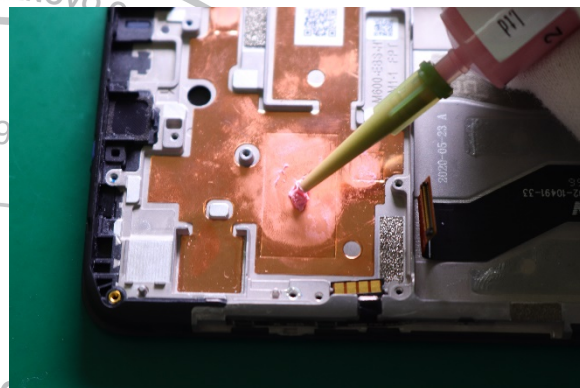
Thermal Gel Assembly

1. Use the pen to mark the three points on the Front-Housing as shown.

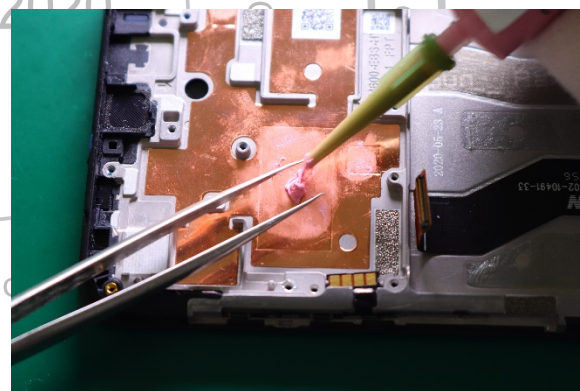
Odessa Thermal gel area	Diameter /mm
A	10
B	2
C	2



2. Use the 18-gauge needle (green color) for Artic and Cooler Master.
Use the 14/15-gauge needle for Henkel LF350.



3. Use the Tweezer to cut off the string of Thermal gel from the needle.



NOTE

The tip of the needle and Tweezers must be cleaned after each application.

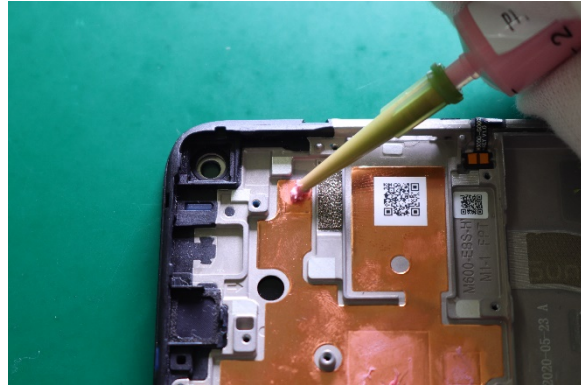
Lenovo Moto G9 Plus

Level 2 – Service and Repair Manual

- Heat conductive gel is extruded according to the amount of heat conductive gel.

NOTE

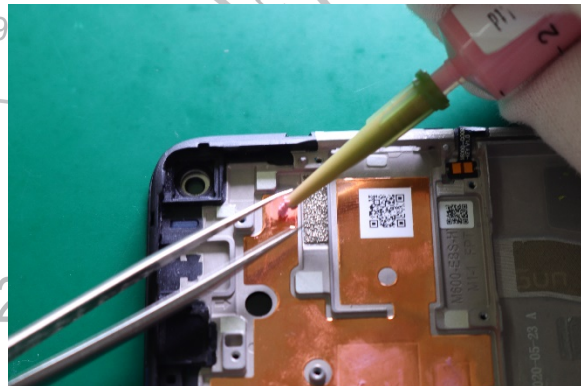
Care not to exceed the diameter while applying the Thermal gel.



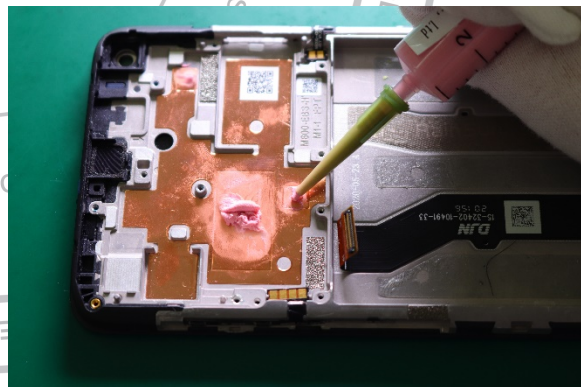
- Use the Thermal gel described below:

NOTE

Needle must be replaced daily.

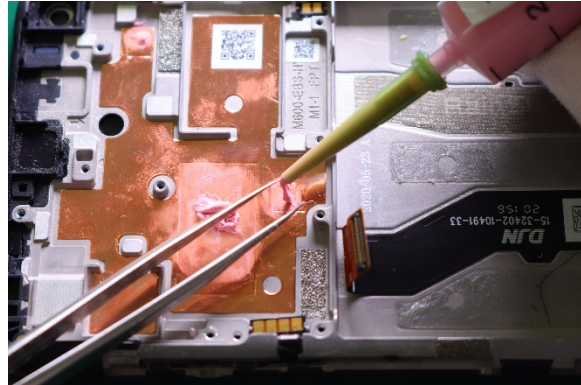


- Use the 18-gauge needle (green color) for Artic and Cooler Master.
Use the 14/15-gauge needle for Henkel LF350.



Lenovo Moto G9 Plus
Level 2 – Service and Repair Manual

7. Use the Tweezer to cut off the string of Thermal gel from the needle.



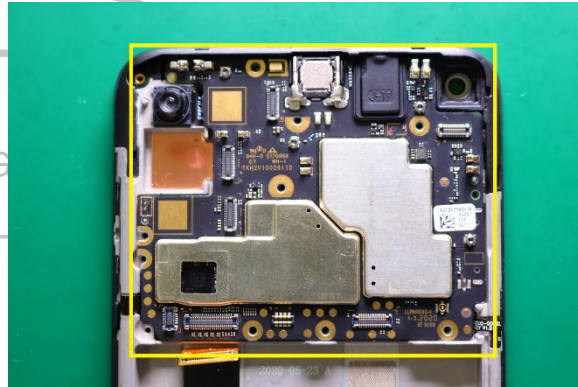
Thermal Gel list that can be acceptable substitutes

Brand	Model	URL
Cool Master	MGY-OSSG-N15M-R1	https://www.coolermaster.com/catalog/coolers/thermal-grease/mastergel-pro/
	MGZ-ND SG-N15M-R1	https://www.coolermaster.com/catalog/coolers/thermal-grease/mastergel-maker/
	RG-ICE1-TG15-R1	https://www.coolermaster.com/catalog/coolers/thermal-grease/ic-essential-e1/
	RG-ICE2-TA15-R1	https://www.coolermaster.com/catalog/coolers/thermal-grease/ic-essential-e2/
	RG-EFX1-TG15-R1	https://www.coolermaster.com/catalog/coolers/thermal-grease/extreme-fusion-x1/
Electrolube	TCP400	N/A
Henkel	LF3500	N/A

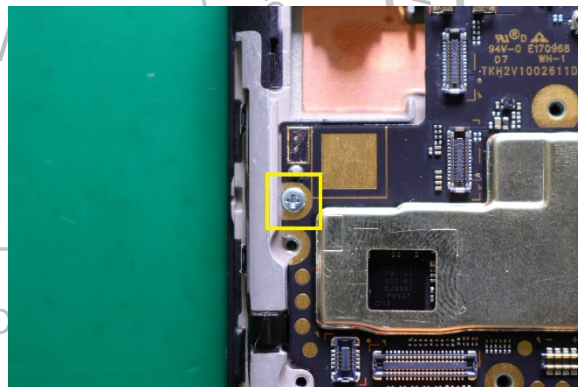
Main PCB Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
XT2087-1 MB 4+128G DS CS XT2087-1 MB 4+128G SS CS XT2087-2 MB 4+128G DS CS XT2087-2 MB 6+128G DS CS ASSY,BD KIT,Odessa20 BR 4+128 MB PCBA	4
LC4610AA000281+Phillips screw 1.4x2M600	16

1. Use fingers to place the Main PCB on the location as shown.



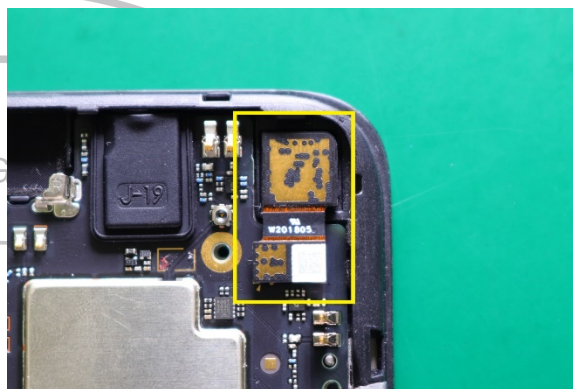
2. Use the Torque Driver with PH00 Bit to tighten the screw as shown (0.8 ± 0.05 kgf-cm).



Front-Facing Camera Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
XT2087-1 MB 4+128G DS CS XT2087-1 MB 4+128G SS CS XT2087-2 MB 4+128G DS CS XT2087-2 MB 6+128G DS CS ASSY,BD KIT,Odessa20 BR 4+128 MB PCBA	4
MDL,CAMR,16M,FRNT,;C0EF04;FF:Q Tech	5

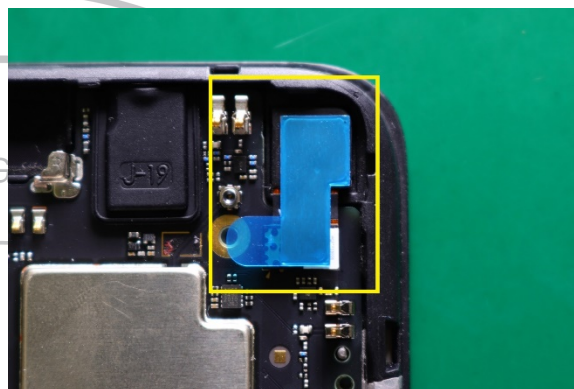
Use fingers to place and secure the Front-Facing Camera on the Main PCB as shown.



Front-Facing Camera Copper Foil Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
XT2087-1 MB 4+128G DS CS XT2087-1 MB 4+128G SS CS XT2087-2 MB 4+128G DS CS XT2087-2 MB 6+128G DS CS ASSY,BD KIT,Odessa20 BR 4+128 MB PCBA	4
LC5470AA000151+LXF_M600_A01_Proactivegro	-

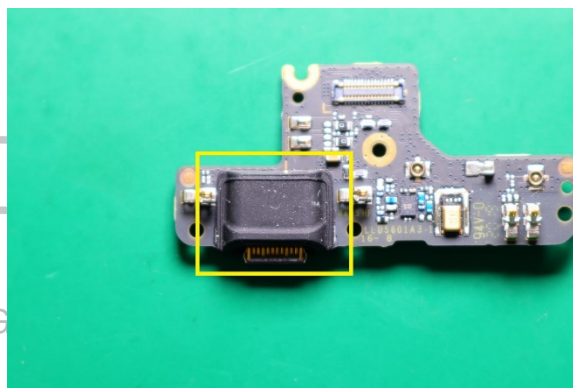
Use the Plastic Tweezer to place and secure the Front-Facing Camera Copper Foil on the location as shown.



USB Grommet Assembly

<i>Description</i>	<i>Reference #</i>
Odessa Sub Board CS ASSY,DTR BRD,Odessa20 BR 4+128 SUB PCBA	9
LC5061AA000487+LXF_M600_A01_USBrubbersle	3

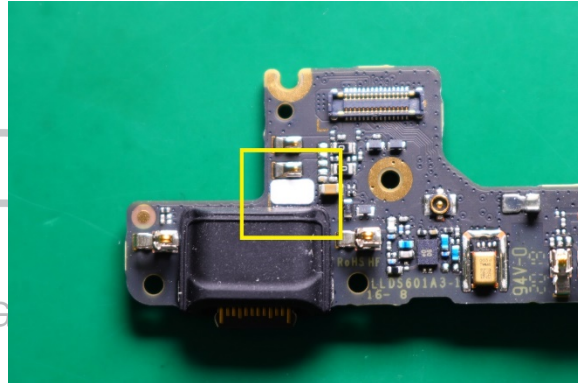
Use the Plastic Tweezer to place and secure the USB Grommet on the Sub Board as shown.



Water Detection Label Assembly

Description	Reference #
Odessa Sub Board CS ASSY,DTR BRD,Odessa20 BR 4+128 SUB PCBA	9
LC7015AA001336_Waterproof labeling_DGJL	-

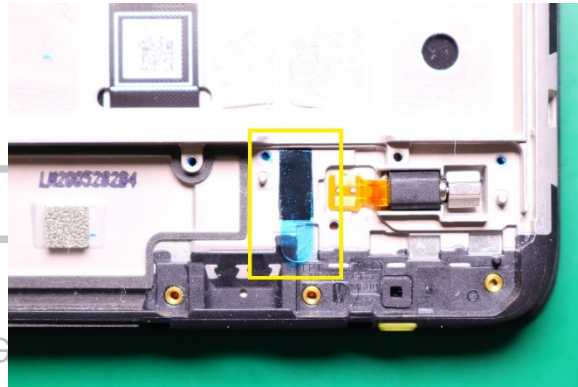
Use the Plastic Tweezer to place the Water Detection Label (WDL) on the Sub Board as shown.



Sub Board Adhesive Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
Odessa Sub BD Adhesive3&5420AA000822	-

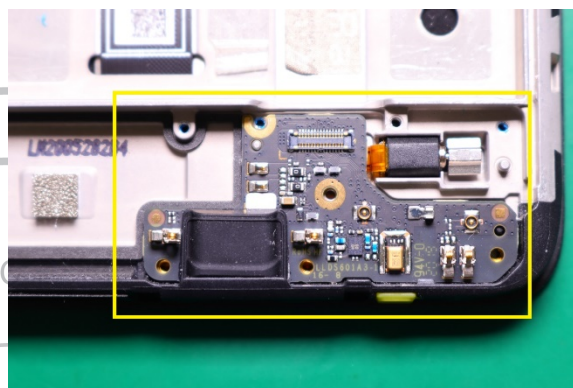
Use the Plastic Tweezer to place and secure the Sub Board Adhesive on the location as shown and peel off the liner.



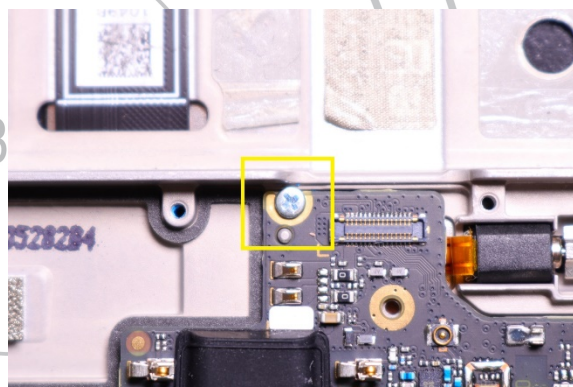
Sub Board Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
Odessa Sub Baord CS ASSY,DTR BRD,Odessa20 BR 4+128 SUB PCBA	9
LC4610AA000281+Phillips screw 1.4x2M600	16

1. Use fingers to place the Sub Board on the location as shown.



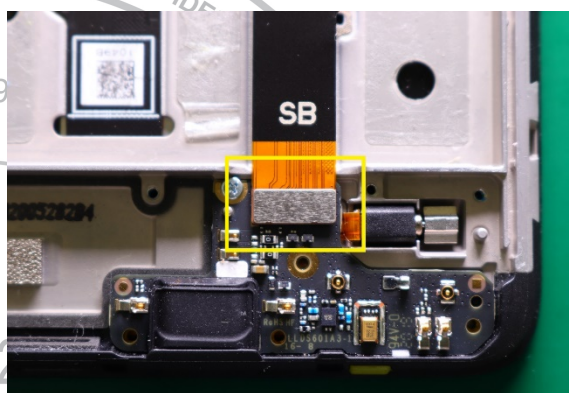
2. Use the Torque Driver with PH00 Bit to tighten the screw as shown (0.8 ± 0.05 kgf-cm).



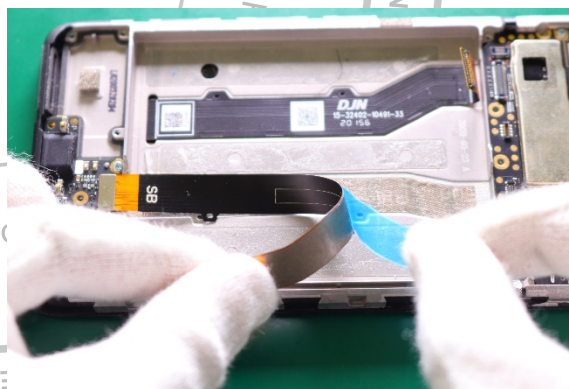
Main FPC Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
XT2087-1 MB 4+128G DS CS XT2087-1 MB 4+128G SS CS XT2087-2 MB 4+128G DS CS XT2087-2 MB 6+128G DS CS ASSY,BD KIT,Odessa20 BR 4+128 MB PCBA	4
Odessa Sub Board CS ASSY,DTR BRD,Odessa20 BR 4+128 SUB PCBA	9
Odessa Main FPC Adhesive3&5420AA000839	-

1. Use fingers to place and secure the Main Flex BtB Connector on the Sub Board as shown.



2. Use fingers to peel off the liner from the Top of the Main FPC as shown.



3. Use fingers to carefully place the Main Flex on the phone as shown, sticky side down.



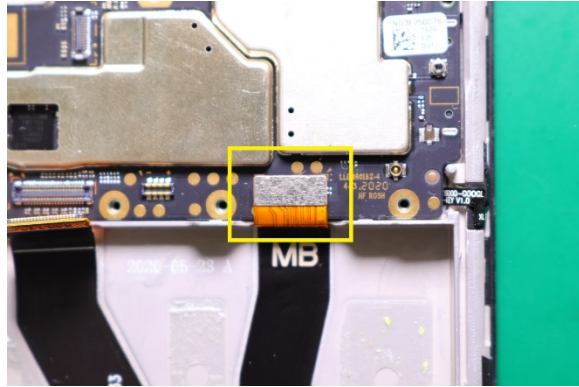
NOTE

Be sure the printing on the Main Flex is oriented in relation to the Front Housing as shown.

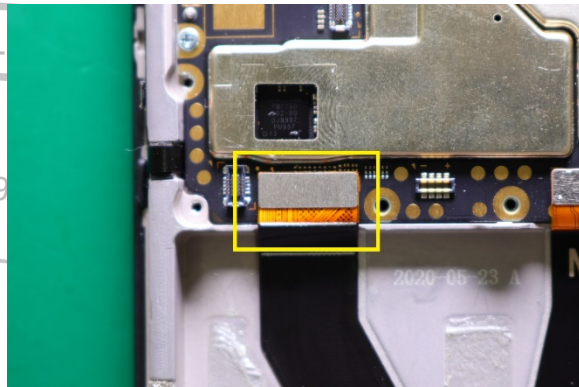
Lenovo Moto G9 Plus

Level 2 – Service and Repair Manual

4. Use fingers to place and secure the Main Flex BtB Connector on the Main PCB as shown.



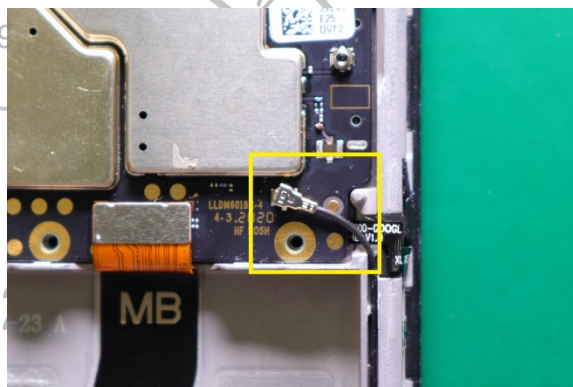
5. Use fingers to place the Display BtB Connector on the Main PCB as shown.



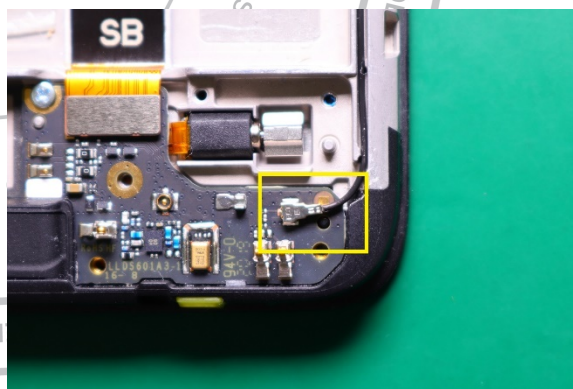
Coax Cable Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
XT2087-1 MB 4+128G DS CS XT2087-1 MB 4+128G SS CS XT2087-2 MB 4+128G DS CS XT2087-2 MB 6+128G DS CS ASSY,BD KIT,Odessa20 BR 4+128 MB PCBA	4
Odessa Sub Baord CS ASSY,DTR BRD,Odessa20 BR 4+128 SUB PCBA	9
LC5234AA000053+M600__RFmid-lowfrequencyc	24
LC5234AA000052+M600__RFcoaxialcable_coax	25

1. Use the Plastic Tweezer to place the black Coax Cable on the Main PCB as shown.



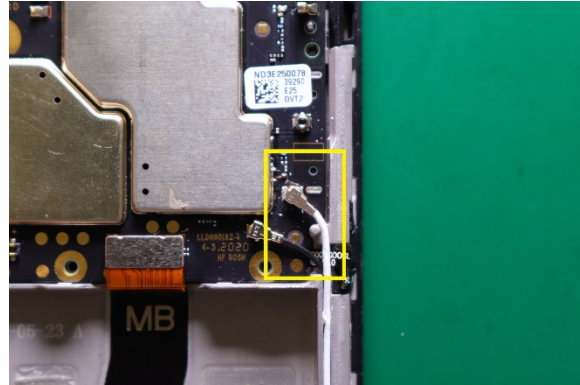
2. Use the Plastic Tweezer to place the black Coax Cable on the Sub Board as shown.



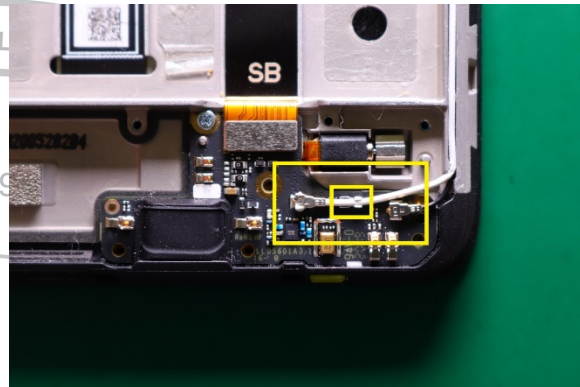
Lenovo Moto G9 Plus

Level 2 – Service and Repair Manual

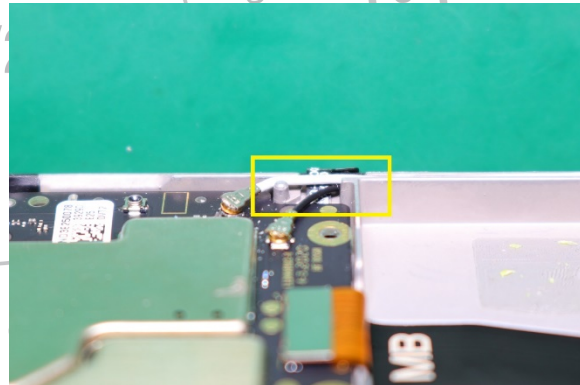
3. Use the Plastic Tweezer to place the white Coax Cable on the Main PCB as shown.



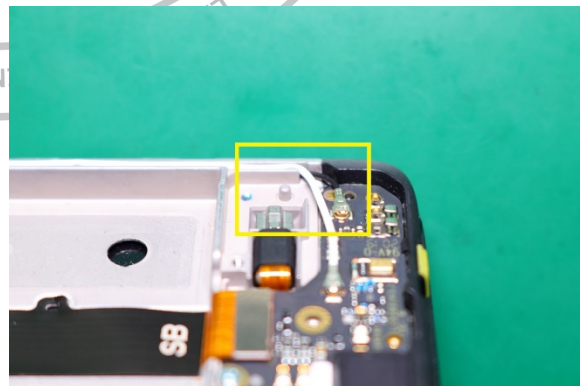
4. Use the Plastic Tweezer to place the white Coax Cable on the Sub Board as shown.



5. Use fingers to place and secure the Coax Cable in the pocket on the recess of the phone.



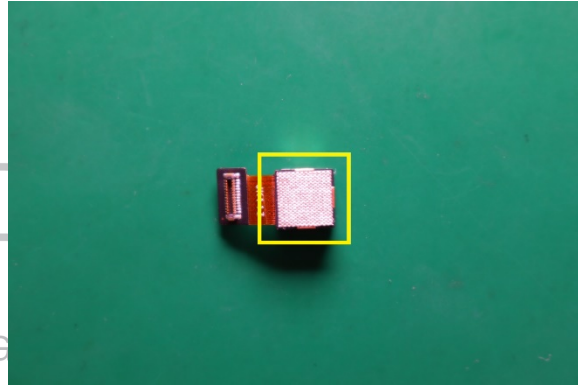
6. Use fingers to place and secure the Coax Cable in the pocket on the recess of the phone.



Rear-Facing Camera Conductive Cloth Assembly

<i>Description</i>	<i>Reference #</i>
MDL,CAMR,2MAF,Guam/Borneo/Odessa,L02BA00	8
LC5450AA000616+LXF_M600_A01_macroreargro	-

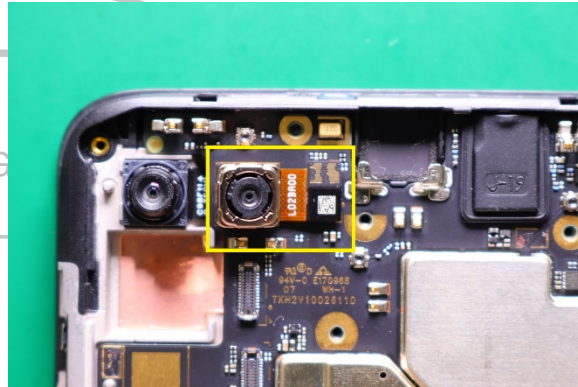
Use the Plastic Tweezer to place and secure the Rear-Facing Camera Conductive Cloth on the location as shown.



2M Macro Rear-Facing Camera Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
XT2087-1 MB 4+128G DS CS XT2087-1 MB 4+128G SS CS XT2087-2 MB 4+128G DS CS XT2087-2 MB 6+128G DS CS ASSY,BD KIT,Odessa20 BR 4+128 MB PCBA	4
MDL,CAMR,2MAF,Guam/Borneo/Odessa,L02BA00	8

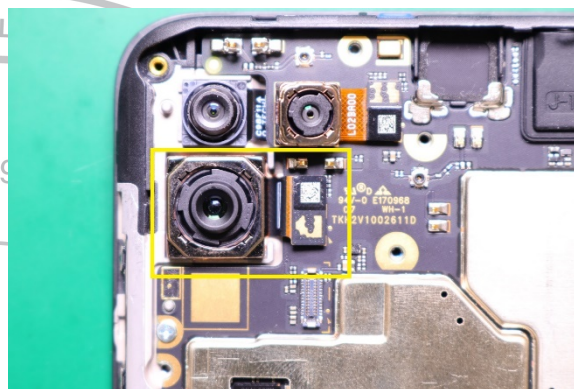
Use fingers to place and secure the 2M Macro Rear-Facing Camera on the Main PCB as shown.



64M Rear-Facing Camera Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
XT2087-1 MB 4+128G DS CS XT2087-1 MB 4+128G SS CS XT2087-2 MB 4+128G DS CS XT2087-2 MB 6+128G DS CS ASSY,BD KIT,Odessa20 BR 4+128 MB PCBA	4
MDL,CAMR,64M,AF,Liberty;AF64M;LGW1A20;Of	11

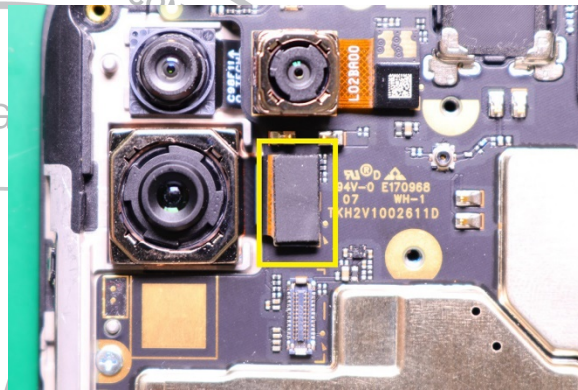
Use fingers to place the 64M Rear-Facing Camera on the Main PCB as shown.



Rear-Facing Camera Conductive Foam Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
XT2087-1 MB 4+128G DS CS XT2087-1 MB 4+128G SS CS XT2087-2 MB 4+128G DS CS XT2087-2 MB 6+128G DS CS ASSY,BD KIT,Odessa20 BR 4+128 MB PCBA	4
MDL,CAMR,64M,AF,Liberty;AF64M;LGW1A20;Of	11
LC5430AA000835+LXF_M600_A01_rearconnecto	-

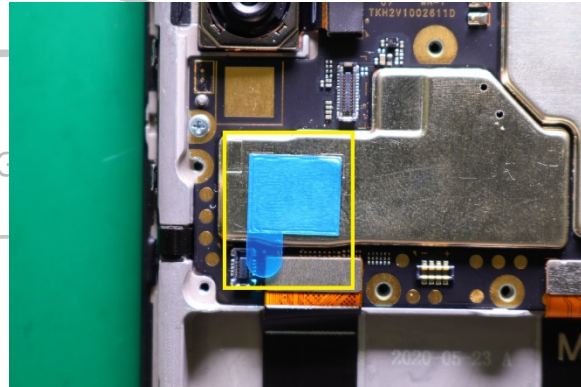
Use the Plastic Tweezer to place and secure the Rear-Facing Camera Conductive Foam on the location as shown.



PMU Copper Foil Assembly

<i>Description</i>	<i>Reference #</i>
Odessa FRU TP LCM Assy CS	1
XT2087-1 MB 4+128G DS CS XT2087-1 MB 4+128G SS CS XT2087-2 MB 4+128G DS CS XT2087-2 MB 6+128G DS CS ASSY,BD KIT,Odessa20 BR 4+128 MB PCBA	4
LC5470AA000163+LXF_M600_A01,PMUshielding	-

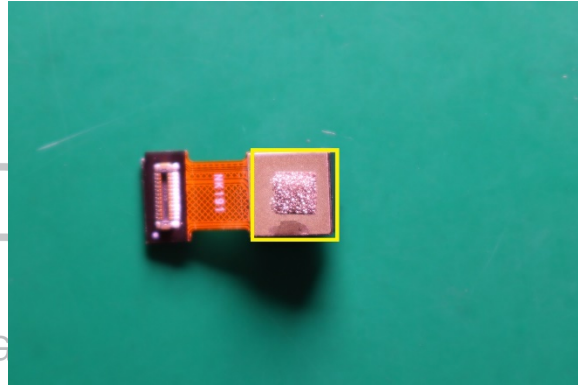
Use the Plastic Tweezer to place the PMU Copper Foil on the location as shown.



Rear-Facing Camera Conductive Sponge Assembly

<i>Description</i>	<i>Reference #</i>
MDL,CAMR,2M,Rear,Nairobi,tacoma;FOV;Ofil	12
LC5450AA000615+LXF_M600_A01_Depthoffield	-

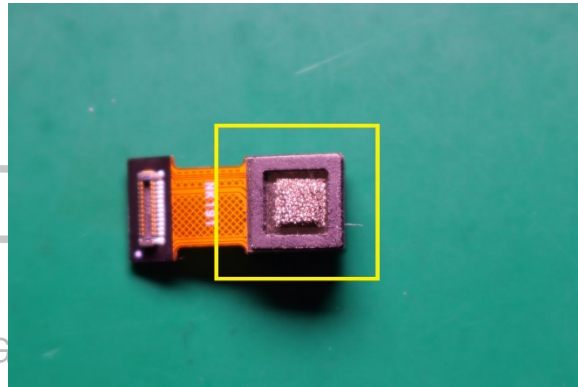
Use fingers to place and secure the Rear-Facing Camera Conductive Sponge on the location as shown.



Rear-Facing Camera Support Pad Assembly

<i>Description</i>	<i>Reference #</i>
MDL,CAMR,2M,Rear,Nairobi,tacoma;FOV;Ofil	12
LC5430AA000840+LXF_M600_A01_Depth-of-fie	-

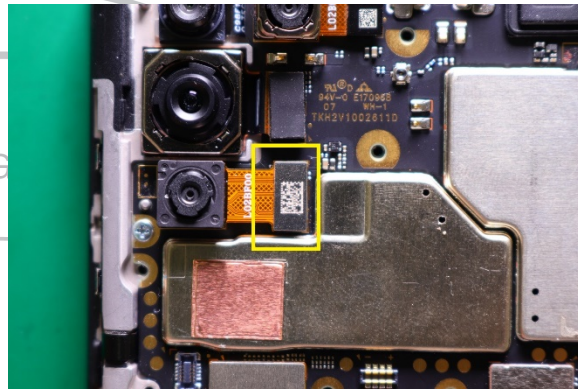
Use fingers to place and secure the Rear-Facing Camera Support Pad on the location as shown.



2M Depth Rear-Facing Camera Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
XT2087-1 MB 4+128G DS CS XT2087-1 MB 4+128G SS CS XT2087-2 MB 4+128G DS CS XT2087-2 MB 6+128G DS CS ASSY,BD KIT,Odessa20 BR 4+128 MB PCBA	4
MDL,CAMR,2M,Rear,Nairobi;tacoma;FOV;Ofil	12

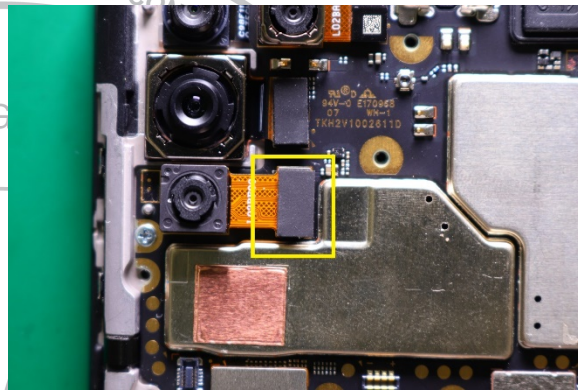
Use fingers to place and secure the 2M Depth
Rear-Facing Camera on the Main PCB as shown.



Rear-Facing Camera Conductive Foam Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
XT2087-1 MB 4+128G DS CS XT2087-1 MB 4+128G SS CS XT2087-2 MB 4+128G DS CS XT2087-2 MB 6+128G DS CS ASSY,BD KIT,Odessa20 BR 4+128 MB PCBA	4
MDL,CAMR,2M,Rear,Nairobi,tacoma;FOV;Ofil	12
LC5430AA000835+LXF_M600_A01_rearconnecto	-

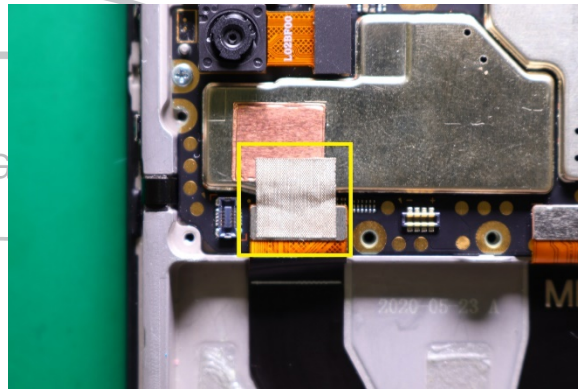
Use the Plastic Tweezer to place and secure the Rear-Facing Camera Conductive Foam on the location as shown.



LCD Connector Conductive Cloth Assembly

<i>Description</i>	<i>Reference #</i>
Odessa FRU TP LCM Assy CS	19
XT2087-1 MB 4+128G DS CS XT2087-1 MB 4+128G SS CS XT2087-2 MB 4+128G DS CS XT2087-2 MB 6+128G DS CS ASSY,BD KIT,Odessa20 BR 4+128 MB PCBA	4
LC5450AA000614+LXF_M600_A01_LCDconnector	-

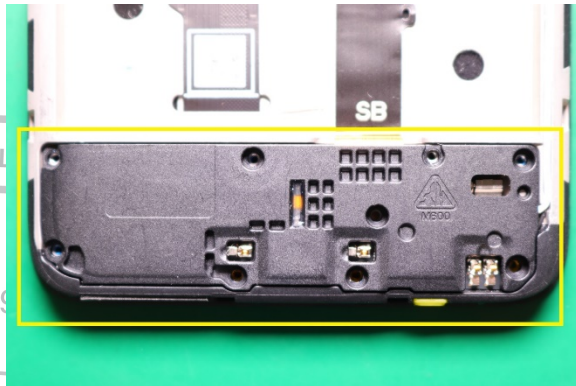
Use the Plastic Tweezer to place and secure the LCD Connector Conductive Cloth on the location as shown.



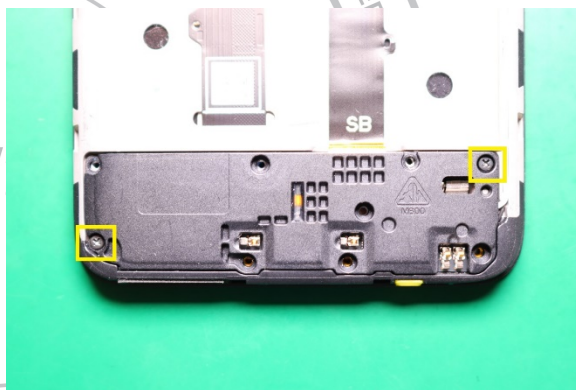
Speaker Box Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
LC4610AA000282+cross screw 1.4x3.5M600	15
LC5016AA000163+LXF_M600_A01_SpeakerBOXco	21

1. Use fingers to place the Speaker Box on the location as shown.



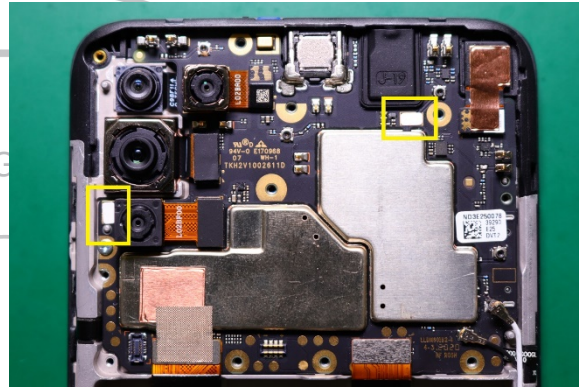
2. Use Torque Driver with PH00 Bit to tighten the two screws as shown (0.8 ± 0.05 kgf-cm).



Water Detection Label Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
XT2087-1 MB 4+128G DS CS XT2087-1 MB 4+128G SS CS XT2087-2 MB 4+128G DS CS XT2087-2 MB 6+128G DS CS ASSY,BD KIT,Odessa20 BR 4+128 MB PCBA	4
LC7015AA001336_Waterproof labeling_DGJL	-

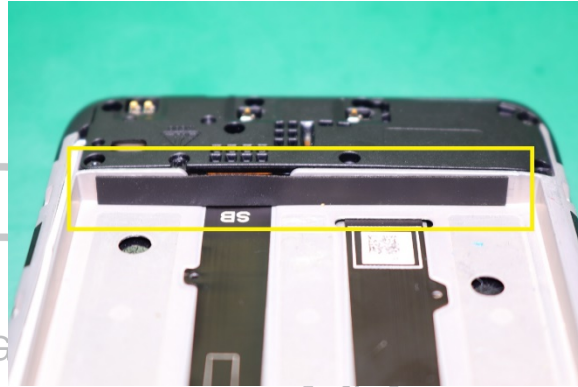
Use the Plastic Tweezer to place the Water Detection Label (WDL) on the Main PCB as shown.



Battery Foam Assembly

<i>Description</i>	<i>Reference #</i>
Odessa FRU TP LCM Assy CS	1
LC5430AA000837+LXF_M600_A01_Batter foam	-

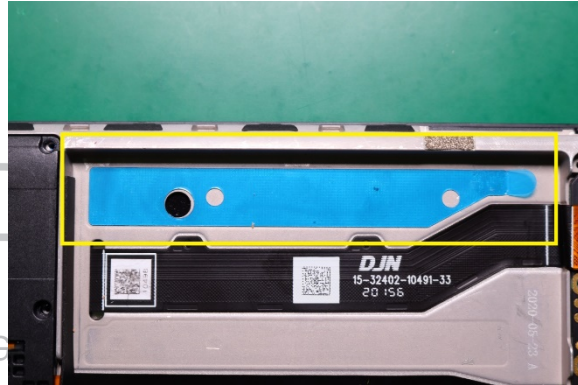
Use the Plastic Tweezer to place and secure the Battery Foam on the location as shown.



Battery Adhesive 1 Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
Odessa Battery Adhesive1&5420AA000820	-

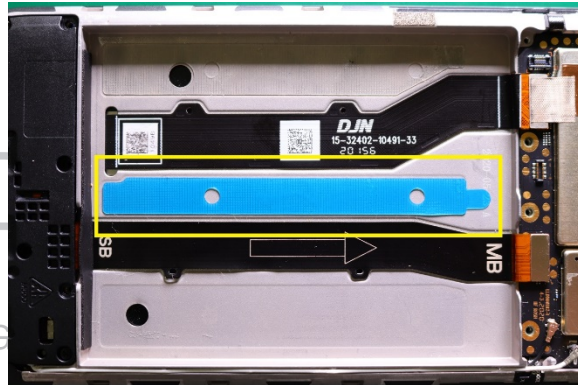
Use the Plastic Tweezer to place the Battery Adhesive 1 on the location as shown and peel off the liner.



Battery Adhesive 2 Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
Odessa Battery Adhesive2&5420AA000825	-

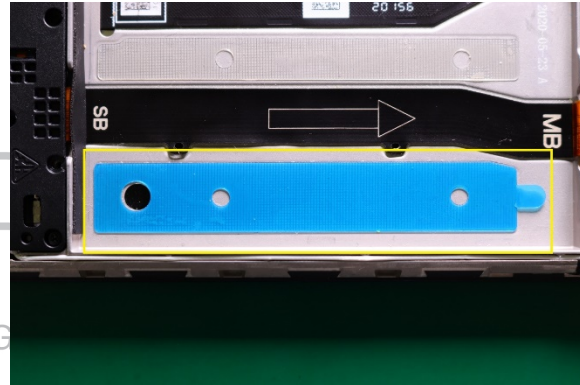
Use the Plastic Tweezer to place the Battery Adhesive 2 on the location as shown and peel off the liner.



Battery Adhesive 3 Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
Odessa Battery Adhesive3&5420AA000824	-

Use the Plastic Tweezer to place the Battery Adhesive 3 on the location as shown and peel off the liner.



Battery Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
XT2087-1 MB 4+128G DS CS XT2087-1 MB 4+128G SS CS XT2087-2 MB 4+128G DS CS XT2087-2 MB 6+128G DS CS ASSY,BD KIT,Odessa20 BR 4+128 MB PCBA	4
ACCY,BAT,5000mAh,MG50;ATL526582;ATL;WW	19

/// WARNING ///

Failure to adhere to Safety Critical Note(s) may increase risk of rupture, burning, or failure to function safely when used by the customer. Refer to the Battery Safety Guidelines.

// CAUTION //

Handle the Battery Pack with care. If dropped to the floor, it may be internally damaged and must be scrapped.

Ensure Battery and its insulation are not damaged (e.g. scratched, dented, punctured) prior to and throughout assembly.

Prior to assembly, ensure Battery edges and surfaces are not dented or deformed, and that fixtures and parts that will contact the Battery are free of foreign material.

Ensure screws and screwdrivers do not contact the Battery.

1. Use fingers to place the Battery on the Front Housing as shown.



Lenovo Moto G9 Plus

Level 2 – Service and Repair Manual

2. Use the Manual Roller to roll over the Battery for 10 cycles to secure the Battery as shown.

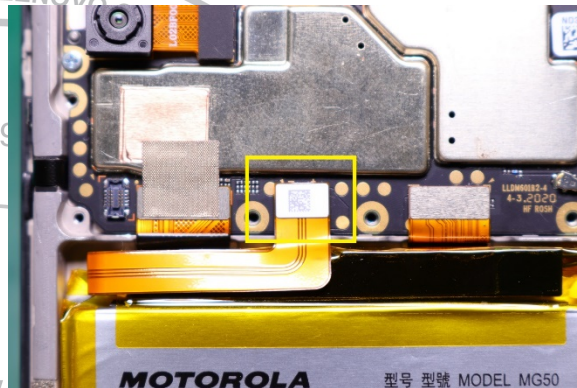
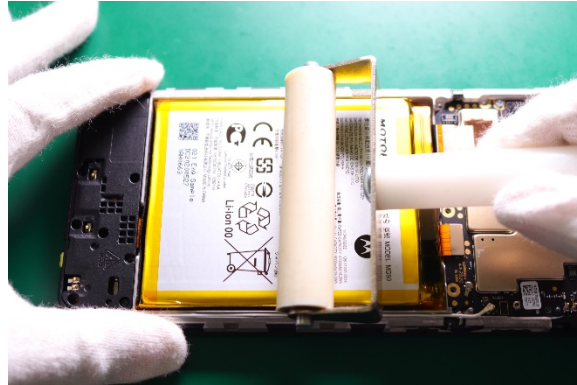
NOTE

For each cycle roll the Manual Roller from top to bottom of the Battery and vice versa.

NOTE

Each cycle should last for 3 seconds.

3. Use fingers to press and secure the Battery BtB Connector as shown.



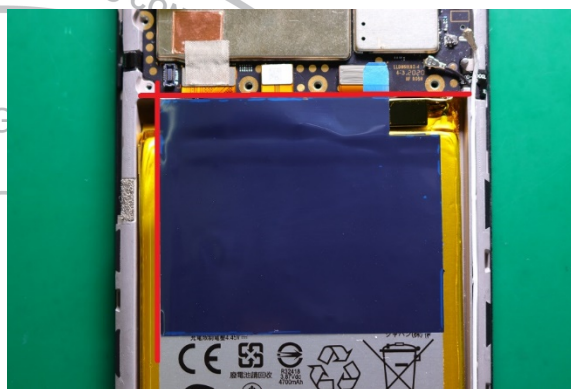
Battery Heat Spreader Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	19
XT2087-1 MB 4+128G DS CS XT2087-1 MB 4+128G SS CS XT2087-2 MB 4+128G DS CS XT2087-2 MB 6+128G DS CS ASSY,BD KIT,Odessa20 BR 4+128 MB PCBA	4
LC5470AA000155+LXF_M600_A01_Batterycool	-
ACCY,BAT,5000mAh,MG50;ATL526582;ATL;VW	19

Use the Plastic Tweezer to place and secure the Battery Heat Spreader on the location as shown.

NOTE

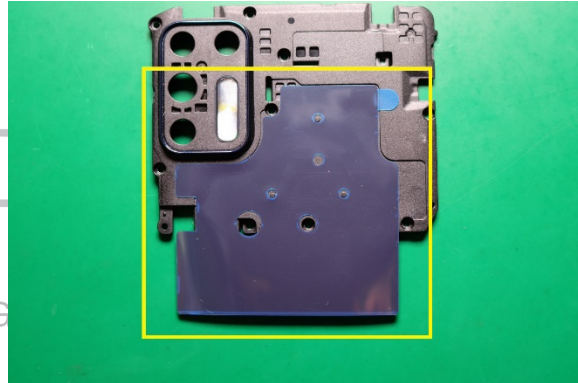
Place the Battery Heat Spreader along the yellow edge on the Battery.



Battery Cover Heat Spreader Assembly

Description	Reference #
Odessa Rear Housing DD CS Odessa Rear Housing CR CS	14
LC5470AA000160+LXF_M600_A01_Rearshellhea	-

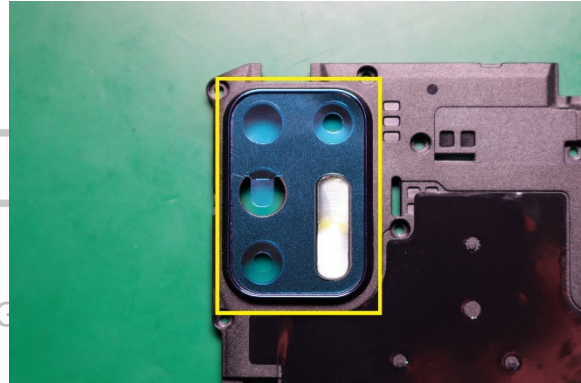
Use the Plastic Tweezer to place and secure the Battery Cover Heat Spreader on the location as shown.



Rear-Facing Camera Lens Adhesive Assembly

Description	Reference #
Odessa Rear Housing DD CS Odessa Rear Housing CR CS	14
Odessa Cameralens Adhesive&5420AA000821	-

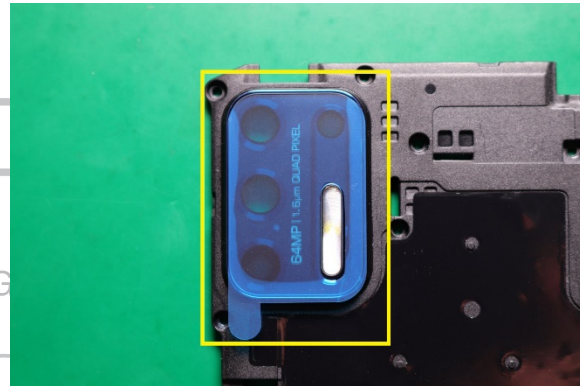
Use the Plastic Tweezer to place the Rear-Facing Camera Lens Adhesive on the location as shown and peel off the liner.



Rear-Facing Camera Lens Assembly

<i>Description</i>	<i>Reference #</i>
Odessa Rear Housing DD CS Odessa Rear Housing CR CS	14
LC5030AA000168+LXF_M600_A01_Lesu_darkblu LC5030AA000166+LXF_M600_A01_Lesu_Rosegol	26

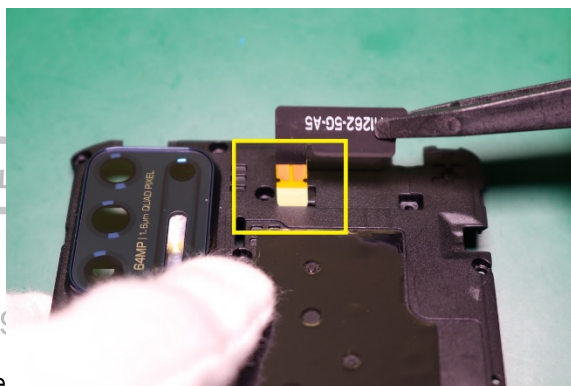
Use the Plastic Tweezer to place the Rear-Facing Camera Lens on the location as shown and peel off the liner.



5G WIFI Antenna Assembly

Description	Reference #
Odessa Rear Housing DD CS Odessa Rear Housing CR CS	14
LC5022AA000168+LXF_M600_A01_5G_WIFlanten	-

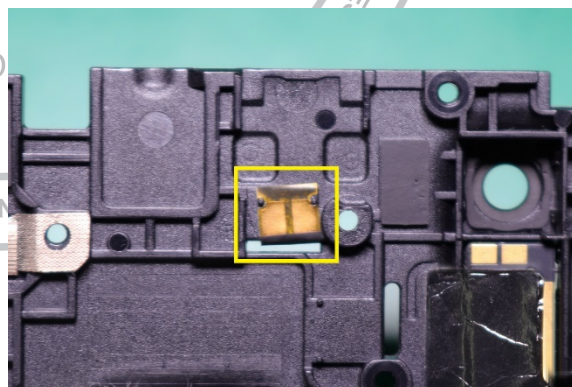
1. Use the Plastic Tweezer to slide the FPC small ear through the hole on the rear shell as shown.



2. Use fingers to place and secure the FPC according to the shape of the positioning groove of the rear shell.



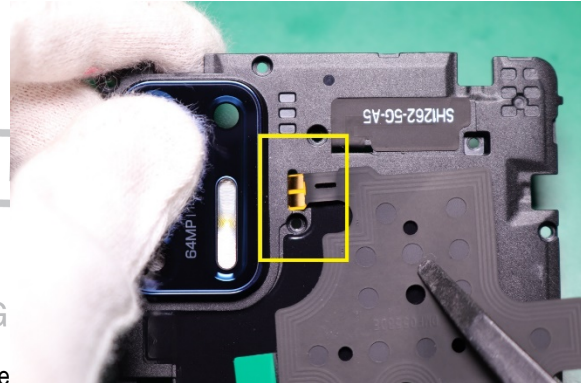
3. Use fingers to place and secure the FPC small ear according to the positioning column on the back of the back shell to the FPC positioning hole.



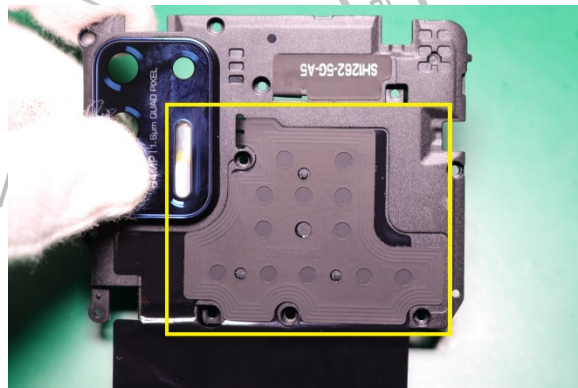
NFC Antenna Assembly

Description	Reference #
Odessa Rear Housing DD CS Odessa Rear Housing CR CS	14
LC5022AA000165+LXF_M600_A01_NFCantenna_	-

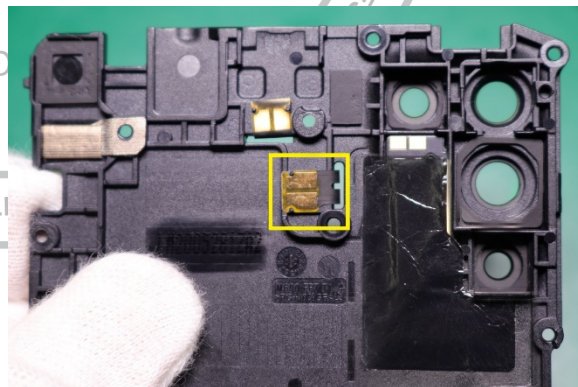
1. Use the Plastic Tweezer to slide the FPC small ear through the hole on the rear shell as shown.



2. Use fingers to place and secure the FPC according to the shape of the positioning groove of the rear shell.



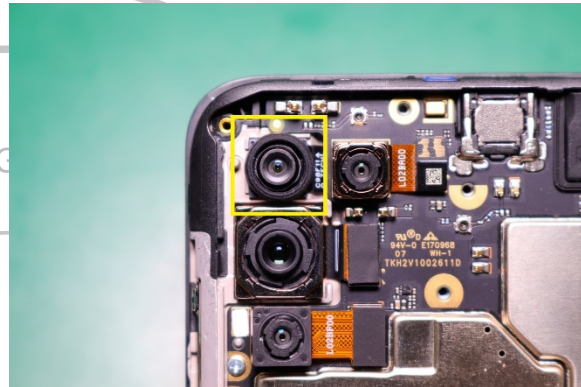
3. Use fingers to place and secure the FPC small ear according to the positioning column on the back of the back shell to the FPC positioning hole.



8M Rear-Facing Camera Grommet Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
XT2087-1 MB 4+128G DS CS XT2087-1 MB 4+128G SS CS XT2087-2 MB 4+128G DS CS XT2087-2 MB 6+128G DS CS ASSY,BD KIT,Odessa20 BR 4+128 MB PCBA	4
LC5061AA000488+LXF_M600_A01_Wideanglerea	27

Use the Plastic Tweezer to place the 8M Rear-Facing Camera Grommet on the location as shown.



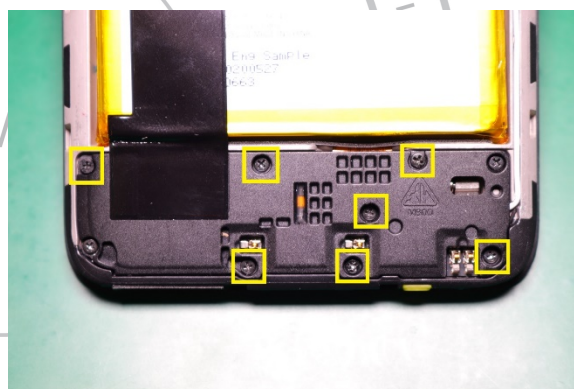
Speaker Box Heat Spreader Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
LC4610AA000282+cross screw 1.4x3.5M600	15
LC5016AA000163+LXF_M600_A01_SpeakerBOXco	21
LC5470AA000154+LXF_M600_A01_SpeakerBOXco	-

1. Use the Plastic Tweezer to place and secure the Speaker Box Heat Spreader on the location as shown and peel off the liner.



2. Use the Torque Driver with PH00 Bit to tighten the seven screws as shown (0.8 ± 0.05 kgf-cm).



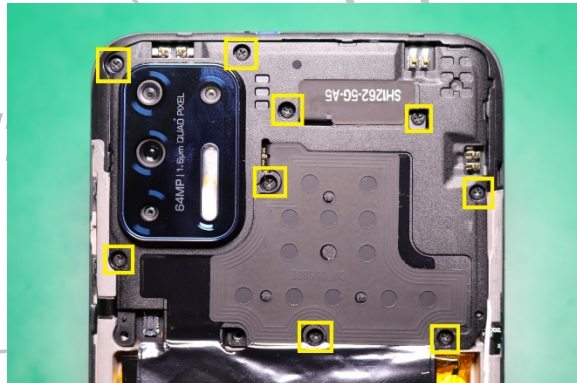
Top Bracket Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
Odessa Rear Housing DD CS Odessa Rear Housing CR CS	14
LC4610AA000282+cross screw 1.4x3.5M600	15

1. Use fingers to place the Top Bracket on the location as shown.



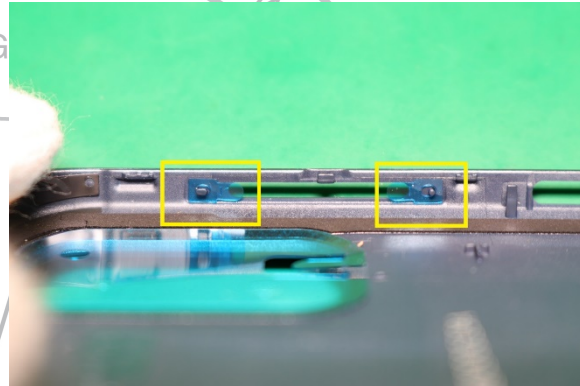
2. Use the Torque Driver with PH00 Bit to tighten the nine screws as shown (0.8 ± 0.05 kgf-cm).



Volume key Adhesive Assembly

Description	Reference #
LC5011AA000893+LXF_M600_A01,Battery cove LC5011AA000778+LXF_M600_A01_Batterycover HSG,BAT CVR,Odessa BR BATCVR KIT DD LM HSG,BAT CVR,Odessa BR BATCVR KIT CR LM Odessa Etching Battery Cover LATAM DD CS Odessa Etching Battery Cover LATAM CR CS Odessa Etching Battery Cover EMEA DD CS Odessa Etching Battery Cover EMEA CR CS Odessa Etching Battery Cover APAC DD CS Odessa Etching Battery Cover APAC CR CS	20
Odessa SideKey Adhesive&5420AA000819	-

Use fingers to place and secure the Volume Key Adhesive on the location as shown and peel off the liner.



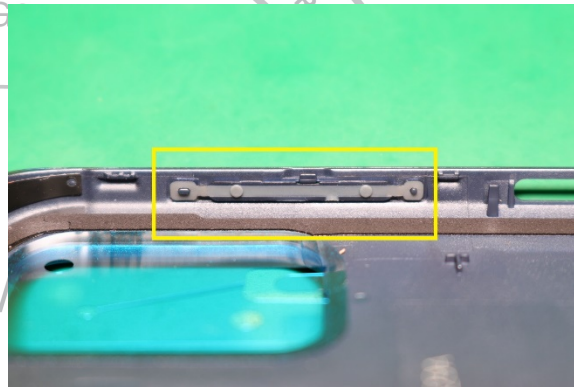
Volume Key Assembly

Description	Reference #
LC5011AA000893+LXF_M600_A01,Battery cove LC5011AA000778+LXF_M600_A01_Batterycover HSG,BAT CVR,Odessa BR BATCVR KIT DD LM HSG,BAT CVR,Odessa BR BATCVR KIT CR LM Odessa Etching Battery Cover LATAM DD CS Odessa Etching Battery Cover LATAM CR CS Odessa Etching Battery Cover EMEA DD CS Odessa Etching Battery Cover EMEA CR CS Odessa Etching Battery Cover APAC DD CS Odessa Etching Battery Cover APAC CR CS	20
LC5040AA000297+LXF_M600_A01_Volumekey_Di LC5040AA000302+LXF_M600_A01_VolumeKey_Di	29

Use the Plastic Tweezer to place and secure the Volume Key on the location as shown.

NOTE

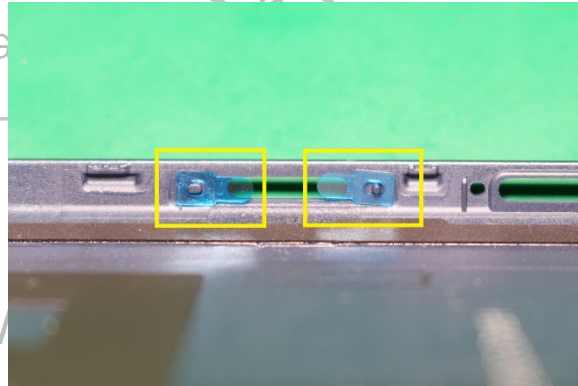
The Power button has a round hole on the cylinder and an elliptical hole on the elliptical cylinder.



Google Key Adhesive Assembly

Description	Reference #
LC5011AA000893+LXF_M600_A01,Battery cove LC5011AA000778+LXF_M600_A01_Batterycover HSG,BAT CVR,Odessa BR BATCVR KIT DD LM HSG,BAT CVR,Odessa BR BATCVR KIT CR LM Odessa Etching Battery Cover LATAM DD CS Odessa Etching Battery Cover LATAM CR CS Odessa Etching Battery Cover EMEA DD CS Odessa Etching Battery Cover EMEA CR CS Odessa Etching Battery Cover APAC DD CS Odessa Etching Battery Cover APAC CR CS	20
Odessa SideKey Adhesive&5420AA000819	-

Use fingers to place and secure the Google Key Adhesive on the location as shown and peel off the liner.



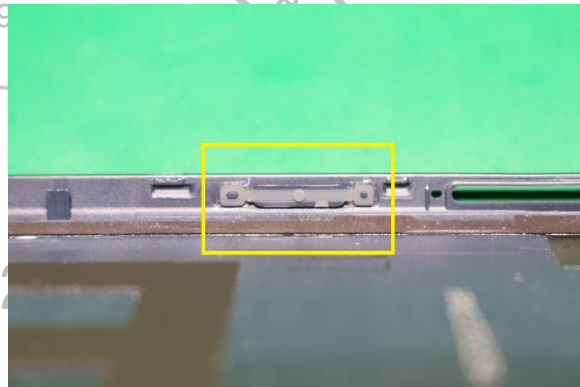
Google Key Assembly

Description	Reference #
LC5011AA000893+LXF_M600_A01,Battery cove LC5011AA000778+LXF_M600_A01_Batterycover HSG,BAT CVR,Odessa BR BATCVR KIT DD LM HSG,BAT CVR,Odessa BR BATCVR KIT CR LM Odessa Etching Battery Cover LATAM DD CS Odessa Etching Battery Cover LATAM CR CS Odessa Etching Battery Cover EMEA DD CS Odessa Etching Battery Cover EMEA CR CS Odessa Etching Battery Cover APAC DD CS Odessa Etching Battery Cover APAC CR CS	20
LC5040AA000298+LXF_M600_A01_GoogleKey_Di LC5040AA000301+LXF_M600_A01_Googlekey_tr	28

Use the Plastic Tweezer to place and secure the Google Key on the location as shown.

NOTE

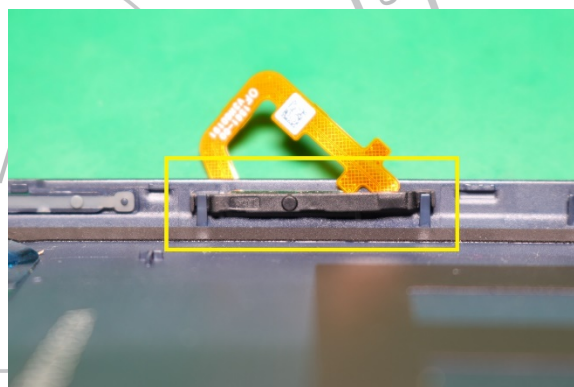
The Power button has a round hole on the cylinder and an elliptical hole on the elliptical cylinder.



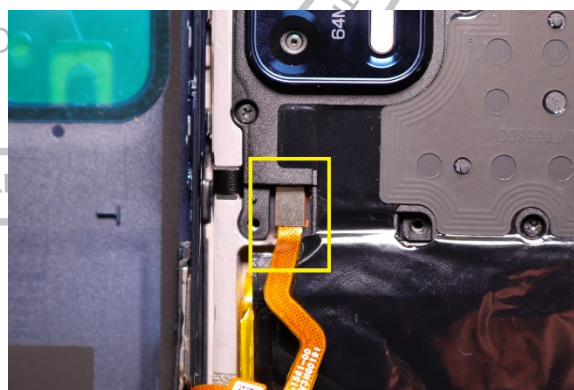
FPS Flex Assembly

<i>Description</i>	<i>Reference #</i>
Odessa FRU TP LCM Assy CS	1
Odessa Rear Housing DD CS Odessa Rear Housing CR CS	14
LC5011AA000893+LXF_M600_A01,Battery cove LC5011AA000778+LXF_M600_A01_Batterycover HSG,BAT CVR,Odessa BR BATCVR KIT DD LM HSG,BAT CVR,Odessa BR BATCVR KIT CR LM Odessa Etching Battery Cover LATAM DD CS Odessa Etching Battery Cover LATAM CR CS Odessa Etching Battery Cover EMEA DD CS Odessa Etching Battery Cover EMEA CR CS Odessa Etching Battery Cover APAC DD CS Odessa Etching Battery Cover APAC CR CS	20
ASSY,BND,Odessa20 FPS KIT DD ASSY,BND,Odessa20 FPS KIT CR	22

1. Use the Plastic Tweezer to place and secure the FPS on the location as shown.



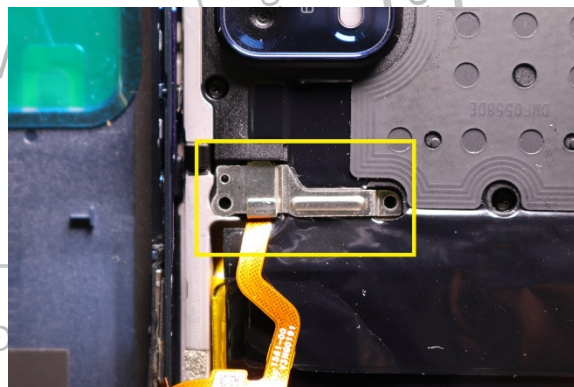
2. Use fingers to place the FPS Flex on the location as shown.



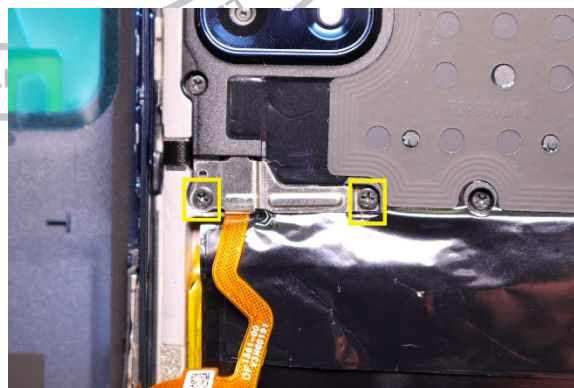
FPS Connector Bracket Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
Odessa Rear Housing DD CS Odessa Rear Housing CR CS	14
LC4610AA000282+cross screw 1.4x3.5M600	15
LC5011AA000893+LXF_M600_A01,Battery cove LC5011AA000778+LXF_M600_A01_Batterycover HSG,BAT CVR,Odessa BR BATCVR KIT DD LM HSG,BAT CVR,Odessa BR BATCVR KIT CR LM Odessa Etching Battery Cover LATAM DD CS Odessa Etching Battery Cover LATAM CR CS Odessa Etching Battery Cover EMEA DD CS Odessa Etching Battery Cover EMEA CR CS Odessa Etching Battery Cover APAC DD CS Odessa Etching Battery Cover APAC CR CS	20
ASSY,BND,Odessa20 FPS KIT DD ASSY,BND,Odessa20 FPS KIT CR	22
LC5090AA000261+LXF_M600_A01_Fingerprintrc	-

1. Use the Plastic Tweezer to place the FPS Connector Bracket on the location as shown.



2. Use the Torque Driver with PH00 Bit to tighten the two screws as shown (0.8 ± 0.05 kgf-cm).



Battery Inlay Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
Odessa Rear Housing DD CS Odessa Rear Housing CR CS	14
LC5011AA000893+LXF_M600_A01,Battery cove LC5011AA000778+LXF_M600_A01_Batterycover HSG,BAT CVR,Odessa BR BATCVR KIT DD LM HSG,BAT CVR,Odessa BR BATCVR KIT CR LM Odessa Etching Battery Cover LATAM DD CS Odessa Etching Battery Cover LATAM CR CS Odessa Etching Battery Cover EMEA DD CS Odessa Etching Battery Cover EMEA CR CS Odessa Etching Battery Cover APAC DD CS Odessa Etching Battery Cover APAC CR CS	20

1. Use the Plastic Tweezer to peel off the liner from the location as shown.



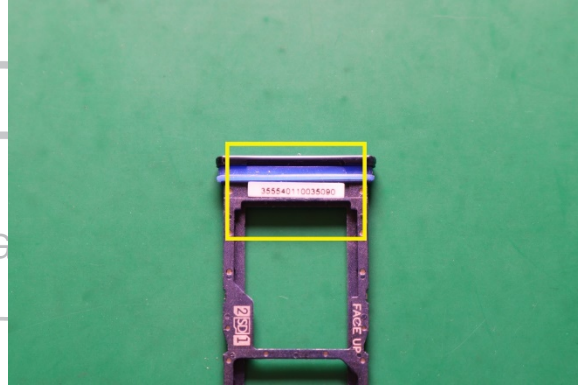
2. Use fingers to place the Battery Cover on the Front Housing and press it on all side to secure it as shown.



IMEI Label Assembly

Description	Reference #
LC5017AA000443+LXFM600_DD_SIM Tray_DS DP LC5017AA000442+LXFM600_CR_SIM Tray_DS DP LC5017AA000474+LXFM600_DD_SIM Tray_SS DP LC5017AA000473+LXFM600_CR_SIM Tray_SS DP	17
LC7015AA000352+SIM Card Tray Label	-

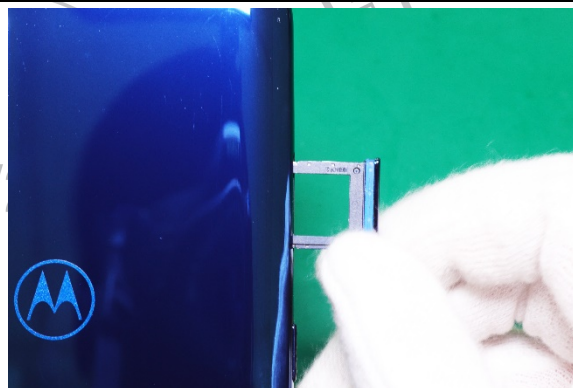
Use the Plastic Tweezer to place and secure the IMEI Label on the location as shown.



Sim Tray Assembly

Description	Reference #
Odessa FRU TP LCM Assy CS	1
LC5017AA000443+LXFM600_DD_SIM Tray_DS DP LC5017AA000442+LXFM600_CR_SIM Tray_DS DP LC5017AA000474+LXFM600_DD_SIM Tray_SS DP LC5017AA000473+LXFM600_CR_SIM Tray_SS DP	17
LC5011AA000893+LXF_M600_A01,Battery cove LC5011AA000778+LXF_M600_A01_Batterycover HSG,BAT CVR,Odessa BR BATCVR KIT DD LM HSG,BAT CVR,Odessa BR BATCVR KIT CR LM Odessa Etching Battery Cover LATAM DD CS Odessa Etching Battery Cover LATAM CR CS Odessa Etching Battery Cover EMEA DD CS Odessa Etching Battery Cover EMEA CR CS Odessa Etching Battery Cover APAC DD CS Odessa Etching Battery Cover APAC CR CS	20

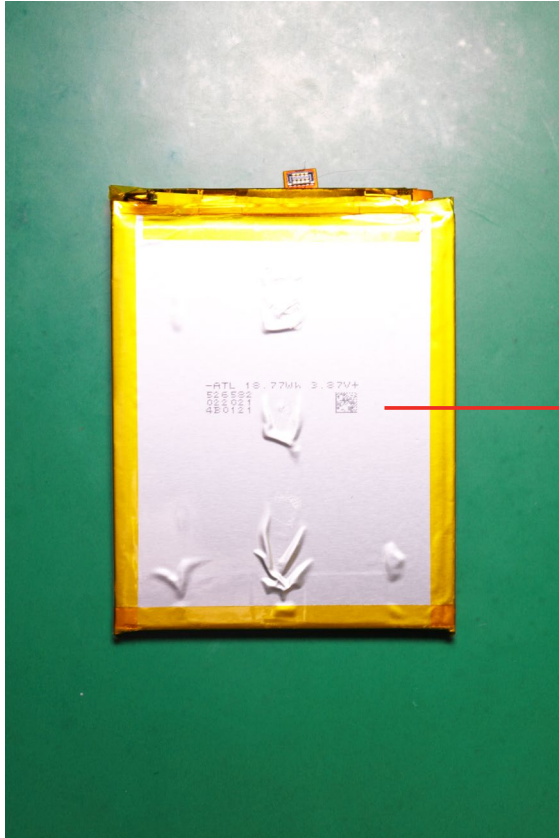
1. Insert the SIM Tray in the transceiver as shown.



2. Use finger to press the SIM Tray until it secures into the transceiver.

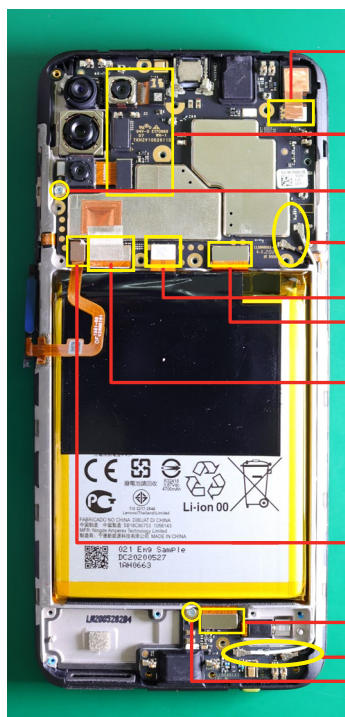
ASSEMBLY QUALITY CHECK

Battery Quality Check



Check if battery adhesive is reused
or with residue from the old one.

Grommet and Connectors Quality Check



Check if the front facing camera connector is well-seated and there's no sign of damage.

Check if the rear facing camera connectors are well-seated and there's no sign of damage.

Check if any missing/unseated screws.

Check if the battery connector is well-seated and there's no sign of damage.

Check if the display flex connector is well-seated and there's no sign of damage.

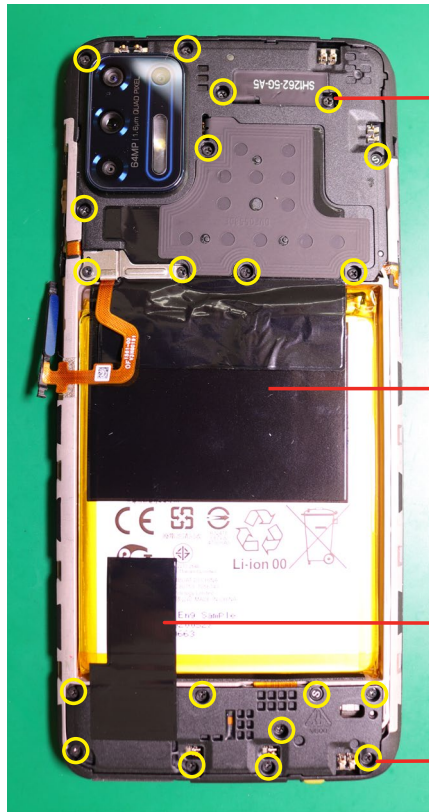
Check if the main flex connector is well-seated and there's no sign of damage.

Check if the FPS connector is well-seated and there's no sign of damage.

Check if the coax cable connector is well-seated and there's no sign of damage.

Check if any missing/unseated screws.

Screws Graphite Quality Check



Check if any missing/unseated screws.

Check if any missing/misaligned graphite tape.

Check if any missing/misaligned graphite tape.

Check if any missing/unseated screws.